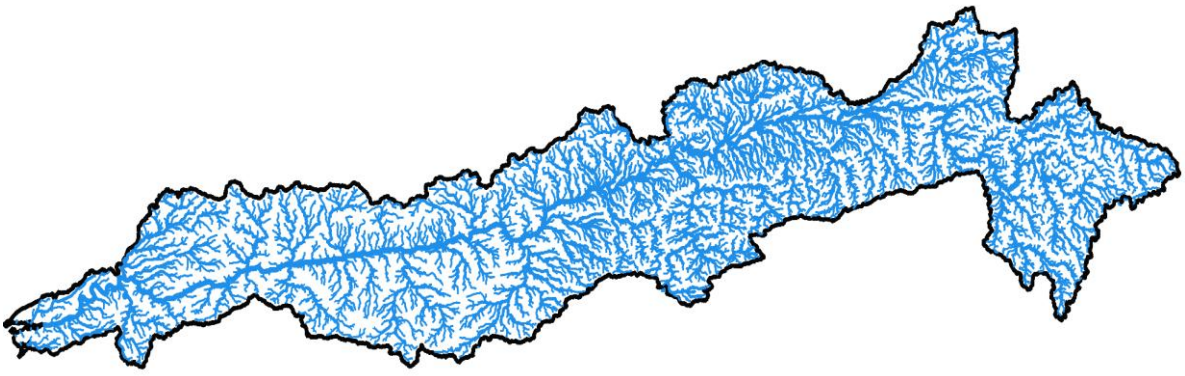
National River Conservation Directorate
Ministry of Jal Shakti, Department of Water Resources,
River Development & Ganga Rejuvenation
Government of India

Physico-Chemical and Biological Profile- Upper and Middle Basin



November 2025

© cNarmada, cGanga and NRCD, 2025

National River Conservation Directorate (NRCD)

The National River Conservation Directorate, functioning under the Department of Water Resources, River Development & Ganga Rejuvenation, and the Ministry of Jal Shakti, provides financial assistance to the State Government for the conservation of rivers under the Centrally Sponsored Schemes of 'National River Conservation Plan (NRCP)'. National River Conservation Plan to the State Governments/ local bodies to set up infrastructure for pollution abatement of rivers in identified polluted river stretches based on proposals received from the State Governments/ local bodies. (www.nrcd.nic.in)

Centers for Narmada River Basin Management and Studies (cNarmada)

The Center for Narmada River Basin Management and Studies (cNarmada) is a Brain Trust dedicated to River Science and River Basin Management. Established in 2024 by IIT Gandhinagar and IIT Indore, under the supervision of cGanga at IIT Kanpur, the center serves as a knowledge wing of the National River Conservation Directorate (NRCD). cNarmada is committed to restoring and conserving the Narmada River and its resources through the collation of information and knowledge, research and development, planning, monitoring, education, advocacy, and stakeholder engagement. (www.cnarmada.org)

Centers for Ganga River Basin Management and Studies (cGanga)

cGanga is a think tank formed under the aegis of NMCG, and one of its stated objectives is to make India a world leader in river and water science. The Centre is headquartered at IIT Kanpur and has representation from most leading science and technological institutes of the country. cGanga's mandate is to serve as a think-tank in the implementation and dynamic evolution of the Ganga River Basin Management Plan (GRBMP) prepared by the Consortium of 7 IITs. In addition to this, it is also responsible for introducing new technologies, innovations, and solutions into India. (www.cganga.org)

Acknowledgments

This report is a comprehensive outcome of the project jointly executed by IIT Gandhinagar (Lead Institute) and IIT Indore (Fellow Institute) under the supervision of cGanga at IIT Kanpur. It was submitted to the National River Conservation Directorate (NRCD) in 2025. We gratefully acknowledge the individuals who provided information for this report.

Team Members

Kiran Bala, cNarmada, IIT Indore
Mayur Shirish Jain, cNarmada, IIT Indore
Ritu Kothari, cNarmada, IIT Indore
Adarsh Singh, cNarmada, IIT Indore
Shweta Hiralkar, cNarmada, IIT Indore

PREFACE

The Narmada River, often referred to as the lifeline of central and western India, holds unparalleled significance for the region's water resources, ecosystems, and communities. Its hydrological complexity, combined with its ecological and cultural heritage, underscores the critical need for its preservation and sustainable management.

This report, focusing on physico-chemical and biological analysis studies of the Narmada River Basin, aims to provide an in-depth analysis of surface water, groundwater, rock, and soil parameters of the Narmada River. The collection and compilation of data on these parameters of both the Upper and Middle Basins will provide valuable insights into the water, soil, and rock quality status of the Narmada Basin.

The findings and recommendations presented in this report are intended to serve as a guiding framework for policymakers, researchers, and practitioners committed to sustainable water management in the Narmada Basin. By identifying trends and challenges, it aspires to support the development of policies that balance physico-chemical and biological quality perspectives, which help in ecological preservation with socioeconomic progress.

We are deeply grateful to all individuals, organizations, and institutions that contributed to the preparation of this report. Their dedication, expertise, and support have been instrumental in shaping this study. It is our hope that this report will inspire collaborative efforts and meaningful action toward the long-term health and sustainability of the Narmada River and its Basin.

Centre for Narmada River Basin Management and Studies (cNarmada)

IIT Gandhinagar, IIT Indore

Table of Contents

List of Tables 1
List of Figures 2
Abbreviations & Acronyms 4
1. 5
2. 7
3. 8
3.1 Upper Narmada Basin 14
3.1.1. Physical Parameters: 14
i) 14
ii) Total Dissolved Solids 15
3.1.2. 16
i) 16
ii) 17
iii) 19
iv) 19
v) 20
vi) 21
vii) 22
3.1.3 23
i) 23
3.2 24
3.2.1. Physical Parameters 24
i) Turbidity 24
ii) 25
i) pH 26
ii) Total Hardness 28
iii) 27
iv) 29
v) 30
vi) 30
vii) 31
3.2.3 Biological Parameter 34
i) Total Coliforms 34

4 33
4.1 37
i) 38
ii) 40
iii) 41
iv) 42
v) 43
vi) 45
vii) 46
viii) 46
4.2 46
i) 47
ii) 49
iii) 50
iv) 51
v) 53
vi) 53
vii) 54
viii) 55
Data Interpretation and Discussion 59
5. Soil Profile of the Narmada Basin..... 61
5.1 57
5.2 63
5.3 67
i) 67
ii) 68
iii) 69
6. 70
6.1. Chemical composition of the Rocks of the Upper Narmada Basin 77
i) 72
ii) 73
iii) 73
6.2 74
i) 74
ii) 75
iii) 76

7. 76
- 76
8. 76
9. 78

List of Tables

Table 1: Summary of the parameters analysed in the report. 6
Table 2: Drinking Water quality standards as per Bureau of Indian Standards BIS IS 10500: 2012. 7
Table 3: Water Quality Criteria 7
Table 4: Districts covered under MPPCB sampling stations in the Upper and Middle Narmada Basin. 9
Table 5: Station codes and Names of sampling stations of Upper and Middle Narmada Basin. 12
Table 6: Summarizes the yearly average of nine years of all sampling stations in the Upper Narmada Basin 24
Table 7: Summarizes the yearly average of COD of all the Middle Narmada Stations. 34
Table 8: Station codes and Names of Sampling stations of the Upper Narmada Basin 36
Table 9: Station codes and Names of sampling stations of the Middle Narmada Basin 40

List of Figures

Figure 1: District-wise distribution of sampling station numbers covered in the Upper Narmada Basin. 10

Figure 2: District-wise distribution of sampling station numbers covered in the Middle Narmada Basin. 11
Figure 3: Mapping of different sampling stations (1-24) of districts in the Upper Narmada Basin. 14
Figure 4: Mapping of different sampling stations (25-50) of districts in the Middle Narmada Basin. 15
Figure 5: Average, Maximum, and Minimum values over the last 9 years for Total Dissolved Solids in surface water across different sampling stations of Upper Narmada Basin. 17
Figure 6: pH yearly average values of surface water across all sampling stations of the Upper Narmada Basin. 18
Figure 7: Average, Maximum, and Minimum values over the last 9 years of Total Alkalinity in surface water across different sampling stations of the Upper Narmada Basin. 19
Figure 8: Average, Maximum, and Minimum values over the last 9 years of Total Hardness in surface water across different sampling stations of the Upper Narmada Basin. 20
Figure 9: Average, Maximum, and Minimum values over the last 9 years of Potassium in surface water across different sampling stations of the Upper Narmada Basin. 21
Figure 10: Average, Maximum, and Minimum values over the last 9 years of Dissolved Oxygen in surface water across different sampling stations of the Upper Narmada Basin. 22
Figure 11: Average, Maximum, and Minimum values over the last 9 years of Biochemical Oxygen Demand in surface water across different sampling stations of the Upper Narmada Basin. 23
Figure 12: Average, Maximum, and Minimum values over the last 9 years of Total Dissolved Solids in surface water across different sampling stations of the Middle Narmada Basin. 27
Figure 13: pH of the yearly average at all sampling stations of the Middle Narmada Basin. 28
Figure 14: Average, Maximum, and Minimum values over the last 9 years of Total Hardness surface water across different sampling stations of the Middle Narmada Basin. 29
Figure 15: Average, Maximum, and Minimum values over the last 9 years of Total Alkalinity in surface water across different sampling stations of the Middle Narmada Basin. 30
Figure 16: Average, Maximum, and Minimum values over the last 9 years of Potassium in surface water across different sampling stations of the Middle Narmada Basin. 31
Figure 17: Average, Maximum, and Minimum values over the last 9 years of Dissolved Oxygen in surface water across different sampling stations of the Middle Narmada Basin. 30
Figure 18: Average, Maximum, and Minimum values over the last 9 years of Biological Oxygen Demand across different sampling stations of the Middle Narmada Basin. 33
Figure 19: District-wise distribution of groundwater sampling station numbers covered in the Upper Narmada Basin. 41
Figure 20: pH yearly average values of groundwater of all sampling stations of 23 years of the Upper Narmada Basin. 44
Figure 21: Average, Maximum, and Minimum values of the 22 years of Chloride in groundwater across different sampling stations of the Upper Narmada Basin. 41
Figure 22: Average, Maximum, and Minimum values of the 20 years of Sulphate in groundwater across different sampling stations of the Upper Narmada Basin. 46
Figure 23: Average, Maximum, and Minimum values of 23 years of Fluoride in groundwater across different sampling stations of the Upper Narmada Basin. 47
Figure 24 : Average, Maximum, and Minimum values of 21 years of Total Hardness in groundwater across different sampling stations of the Upper Narmada Basin. 44
Figure 25 : Average, Maximum, and Minimum values of all 12 years of Total Alkalinity in groundwater across different sampling stations of the Upper Narmada Basin. 45
Figure 26: District-wise distribution of sampling stations in the Middle Narmada Basin. 52
Figure 27: pH yearly average of groundwater across all sampling stations of 23 years of the Middle Narmada Basin. 55

Figure 28: Average, Maximum, and Minimum values of 22 years of Chloride in groundwater across different sampling stations of the Middle Narmada Basin. 56
Figure 29: Average, Maximum, and Minimum values of all 20 years of Sulphate in groundwater across different sampling stations of the Middle Narmada Basin. 57
Figure 30: Average, Maximum, and Minimum values of all 23 years of fluoride in groundwater across different sampling stations of the Middle Narmada Basin. 52
Figure 31: Average, Maximum, and Minimum values of all 21 years of Total Hardness in groundwater across different sampling stations of the Middle Narmada Basin. 53
Figure 32: Average, Maximum, and Minimum values of all 12 years of Total Alkalinity in groundwater across different sampling stations of the Middle Narmada Basin. 60
Figure 33: Soil Series distribution for the different surveyed areas in the Basin 68
Figure 34: Land capability class of the surveyed area 67

Abbreviations & Acronyms

BIS	Bureau of Indian Standards
CWC	Central Water Commission
CGWB	Central Ground Water Board
BOD	Biological Oxygen Demand
COD	Chemical Oxygen Demand
CPCB	Central Pollution Control Board
DO	Dissolved Oxygen
GSI	Geological Survey of India
MPPCB	Madhya Pradesh Pollution Control Board
NWIC	National Water Informatics Centre
NRCD	National River Conservation Directorate
I-WRIS	Indian Water Resource Information System
SLUSI	Soil and Land Use Survey of India
TDS	Total Dissolved Solids
T. Col.	Total Coliform
d/s	Downstream
u/s	Upstream

1. Introduction

The Narmada River Basin extends from 21°40'12" to 23°41'24" N latitudes and 72°48'36" to 81°45'36" E longitudes, encompassing a total area of 98,796 sq. km, which constitutes about 3% of India's total geographical area (3,297,427.32 sq. km). Bounded by the Vindhyan ranges to the north, the Maikala hills to the east, the Satpura ranges to the south, and the Arabian Sea to the west, the basin lies at the northern extremity of the Deccan Plateau and covers significant portions of Madhya Pradesh and Gujarat, with smaller parts extending into Chhattisgarh and Maharashtra. The basin spans 40 districts, comprising 27 districts of Madhya Pradesh, 07districts of Gujarat, 04 districts of Chhattisgarh, and 02 districts of Maharashtra. The Narmada Basin exhibits an elongated shape, stretching approximately 915.65 km east to west and about 236 km north to south. The basin experiences a tropical monsoon climate characterized by distinct wet and dry seasons, with an average annual rainfall ranging from 800 to 1,200 mm. The Narmada River, the longest west-flowing river in India, drains into the Arabian Sea near Bharuch in Gujarat and is further subdivided into Upper, Middle, and Lower Narmada River basins.

There is considerable variation among the various districts within the Narmada Basin in terms of infrastructure, industrial development, and agricultural development. These developments result in multiple types of water resource uses, which can generate various kinds of solid-liquid waste and other pollutants that can impact the physicochemical and biological parameters of ground and surface water, soil, and rock in terms of physicochemical quality. The Basin encompasses river valleys, fertile plains, and hilly regions. Due to these geographical differences, which can ultimately lead to environmental challenges such as pollution, water availability and quality issues, and soil erosion, these issues may affect the physicochemical and biological properties of water, soil, and rock in the Narmada Basin.

The interpretation of physicochemical and biological parameters from the collected data is the primary focus of this report, which analyzes the quality parameters of surface water, groundwater, soil, and rock, aiming to identify the water quality status, potential pollution levels, and the various factors affecting the Narmada River's water quality.

We will also focus on identifying hotspots that require more attention, regular sampling, and further analysis due to poor water quality data in terms of physical, chemical, and biological

parameters. The possible reasons for these hotspots may be due to climatic conditions, such as heavy rainfall, which can ultimately increase pollutant runoff and sediments in rivers, as well as anthropogenic activities. In this report, we focused on the major parameters listed in Table 1.

Table 1: Summary of the parameters analyzed in the report.

Types	**Parameters**
Physical parameters	Turbidity, Total Dissolved Solids
Chemical Parameters	pH, Total Alkalinity, Total Hardness, Potassium, Sulphate, Chloride, Nitrate Nitrogen, Silicon Dioxide, Dissolved Oxygen, Biochemical Oxygen Demand
Biological Parameters	Total Coliforms

The report is divided into four sections: -

- Surface Water Physico-chemical Analysis
- Ground Water Physico-chemical Analysis
- Soil Physico-chemical Analysis
- Rock Physico-chemical Analysis

2. Water Quality Standards and Criteria

Water quality data is compared with the Bureau of Indian Standards (BIS) IS 10500:2012 limits set for drinking purposes. The BIS values of parameters are listed in Table 2.

The classification of Water Quality is based on its designated use. Table 3 provides the five categories (A to E), which are based on the different parameters.

Table 2: Drinking Water quality standards as per Bureau of Indian Standards BIS IS 10500: 2012.

Parameters	Acceptable Limit	Permissible Limit in Absence of an Alternate Source	Unit
pH	6.5-8.5	-	-
Total Dissolved Solids	500	2000	mg/L
Turbidity	1	5	NTU
Chloride	250	1000	mg/L
Sulphate	200	400	mg/L
Total hardness	200	600	mg/L
Total Alkalinity	200	600	mg/L
Dissolved Oxygen	6 or more		mg/L
Biochemical Oxygen Demand	3 or less		mg/L

Source: https://cpcb.nic.in/wqm/BIS_Drinking_Water_Specification.pdf

Table 3: Water Quality Criteria

Designated-Best-Use	Class of water	Criteria
Drinking Water Source without conventional treatment, but after disinfection	A	• Total Coliforms Organism MPN/100ml shall be 50 or less • pH between 6.5 and 8.5 • Dissolved Oxygen 6mg/l or more • Biochemical Oxygen Demand 5 days 20C 2mg/l or less
Outdoor bathing (Organized)	B	• Total Coliforms Organism MPN/100ml shall be 500 or less, pH between 6.5 and 8.5, Dissolved

Designated-Best-Use	Class of water	Criteria
		Oxygen 5mg/l or more ● Biochemical Oxygen Demand 5 days 20C 3mg/l or less
Drinking water source after conventional treatment and disinfection	C	● Total Coliforms Organism MPN/100ml shall be 5000 or less, pH between 6 and 9, Dissolved Oxygen 4mg/l or more ● Biochemical Oxygen Demand 5 days 20C 3mg/l or less
Propagation of Wildlife and Fisheries	D	● pH between 6.5 and 8.5, Dissolved Oxygen 4mg/l or more ● Free Ammonia (as N) 1.2 mg/l or less
Irrigation, Industrial Cooling, Controlled Waste Disposal	E	● pH between 6.0 and 8.5 ● Electrical Conductivity at 25 °C micro mhos/cm Max 2250 ● Sodium absorption Ratio Max 26 ● Boron Max. 2mg/l
	Below-E	Not Meeting A, B, C, D & E Criteria

Source: https://cpcb.nic.in/water-quality-criteria/

3. Surface Water Physico-Chemical and Biological Analysis

Data has been collected from MPPCB for surface water analysis, which includes 50 sampling stations across the 13 districts of the Narmada Basin. Districts and names of sampling stations for both the Upper and Middle Basins are listed in Tables 4 and 5.

Table 4: Districts covered under MPPCB sampling stations in the Upper and Middle Narmada Basin.

S. No.	Districts Covered by MPPCB
1	Anuppur
2	Dindori
3	Mandla
4	Jabalpur
5	Narsinghpur
6	Narmadapuram
7	Sehore
8	Dewas
9	Dhar
10	Khandwa
11	Khargone
12	Badwani
13	Alirajpur

Source: Narmada: River at a Glance, cNarmada (2024) and MPPCB.

Each district of the Upper and Middle Basin has a different set of sampling stations, as shown in Figures 1& 2.

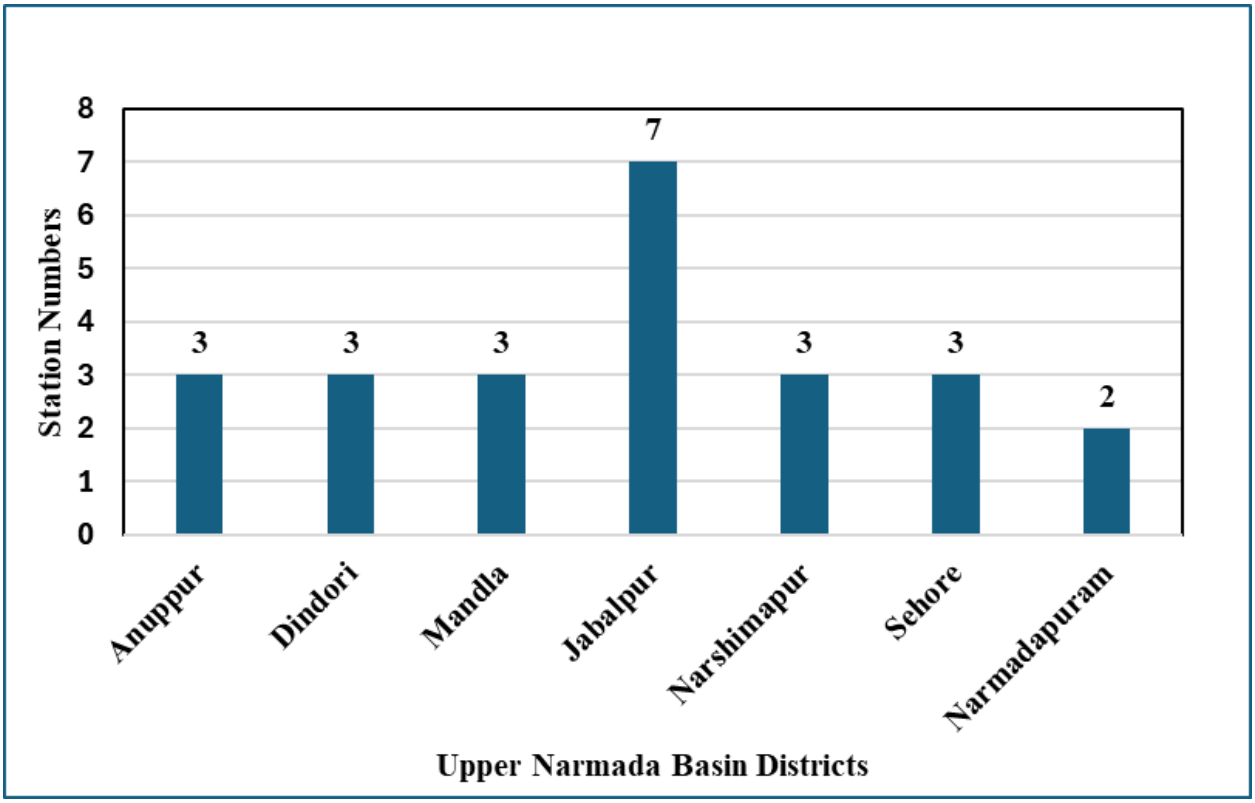


Figure 1: District-wise distribution of sampling station numbers covered in the Upper Narmada Basin.

Source: Narmada: River at a Glance (2024), cNarmada and MPPCB

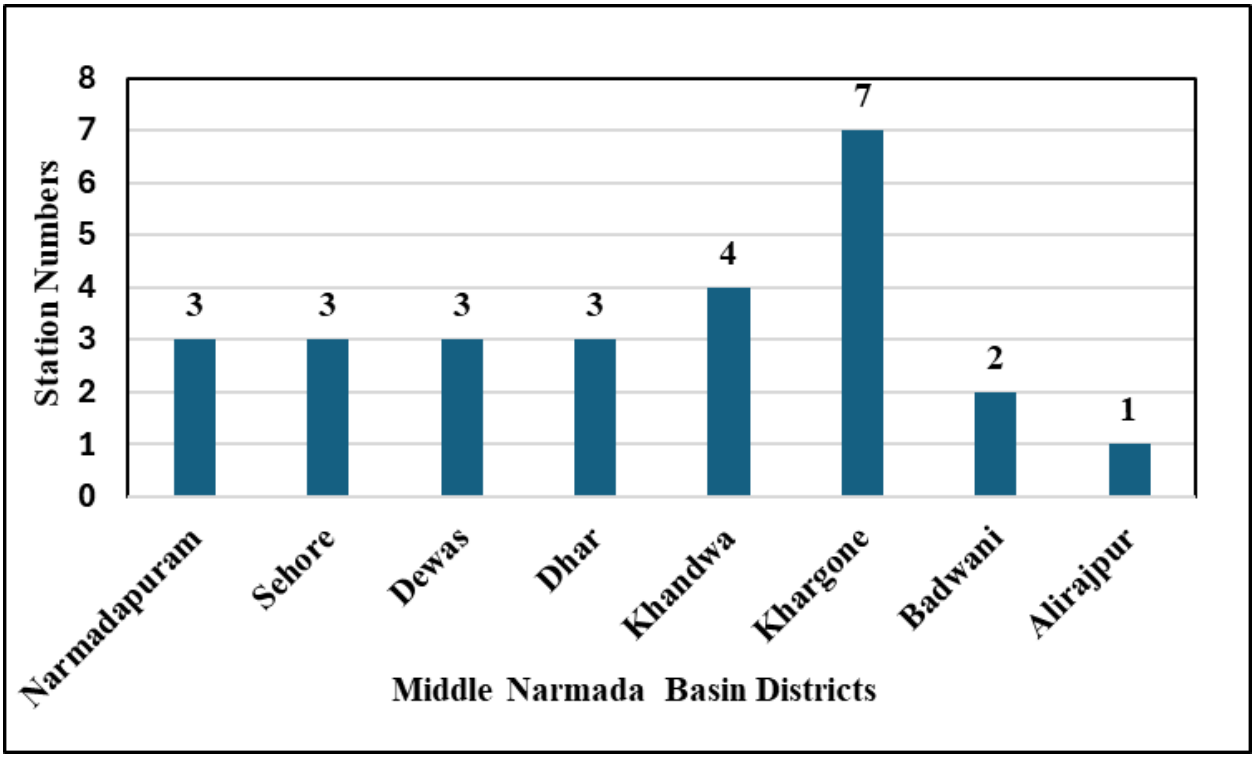


Figure 2: District-wise distribution of sampling station numbers covered in the Middle Narmada Basin.

Source: Narmada: River at a Glance (2024) cNarmada and MPPCB

Table 5 : Station codes and Names of sampling stations of Upper and Middle Narmada Basin.

Station code	Station Names of the Upper Narmada Basin
SU1	Narmada at Amarkantak Origin point
SU2	Narmada at Pushkar dam, Amarkantak
SU3	Narmada at Kapildhara, Amarkantak
SU4	Narmada at Chandanghat, Dindori
SU5	Narmada at Dindori u/s, Dindori
SU6	Narmada at Dindori d/s Dindori Jogotikara Ghat
SU7	Narmada at Mandla near Shamshanghat, Mandla
SU8	Narmada at Road bridge, Raftaghat, Mandla
SU9	Narmada at Bhairav temple, Shahastradhara, Mandla
SU10	Narmada at road bridge d/s of Bargi dam
SU11	Narmada at Jamtara Jabalpur
SU12	Narmada at Lalpur, near the water supply intake well point
SU13	Narmada at Tilwaraghat, Jabalpur
SU14	Narmada at Panchawatighat, before mixing with the Bawanganga River
SU15	Narmada at Saraswatighat after mixing with the Bawanganga River
SU16	Narmada near road bridge, Jhansighat, Jabalpur
SU17	Narmada NH-44, Road bridge Barmanghat Narsinghpur
SU18	Narmada at 100 mt d/s Barmanghat, Narsinghpur.
SU19	Narmada SH-44, Jhikoli village, Narsinghpur

Station code	Station Names of the Upper Narmada Basin
SU20	Narmada at the road bridge Sandia, Pipariya
SU21	Narmada u/s of Jait village, Sehore.
SU22	Narmada D/s of Jait village, Sehore.
SU23	Narmada at Shahganj rest house
SU24	Narmada u/s before confluence of Tawa River at Village Bandrabhan

Station Codes and Names of the Middle Narmada Basin	
SM25	Narmada d/s after confluence of the Tawa River at Ramnagar Village, Bandrabhan
SM26	Narmada at Sethanighat
SM27	Narmada near Budhnighat, Budni
SM28	Narmada, d/s textile unit, Village Holipura, Budni
SM29	Narmada at Korighat
SM30	Narmada 100 mts. d/s after confluence of SPM Nallah
SM31	Narmada before the confluence of the river Jammar, Nemawar
SM32	Narmada at Nemawar
SM33	Narmada 500 mts. d/s of Jain Mandir, Nemawar
SM34	Narmada River at Hanuwantia, Khandwa
SM35	Narmada river at Punasa dam
SM36	Narmada at U/s Omkareshwar Dam
SM37	Narmada at D/s Omkareshwar Dam
SM38	Narmada near Mortaka bridge, Badwah, Khargone
SM39	Narmada at Dhareshwar, Khargone
SM40	Narmada at U/s Mandleshwar near Jalood intake well point, Khargone
SM41	Narmada at Mandleshwar d/s Khargone
SM42	Narmada at Maheshwar, Khargone
SM43	Narmada at Shahastradhara, Jalkoti
SM44	Narmada at Khalghat, Khargone
SM45	Narmada at Dharampuri, before mixing the Domestic nalla
SM46	Narmada at Dharampuri d/s
SM47	Narmada at Semalda u/s Badwani
SM48	Narmada at Rajghat, Badwani
SM49	Narmada at Koteshwar, Nisarpur, Dhar
SM50	Narmada at Kakrana, Alirajpur

Source:https://www.mppcb.mp.gov.in/narmada-report16-17-17-18eng.aspx

There are three new stations in the MPPCB data, which we named N1, N2, and N3. N1 is d/s of Mella Area, Near Darogaghat, Jabalpur, added in the Year 2024-2025 with significantly less data, and Stations N2 and N3, Narmada, Barmanghat, Mainghat, Narsinghpur, and Narmada River at Narmada- Kshipra Sangam are the new stations added in the sampling

stations. No data is available for these stations. Figures 3 & 4 show the locations of sampling stations in the Upper Narmada Basin (1-24) and Middle Narmada Basin (25-50).

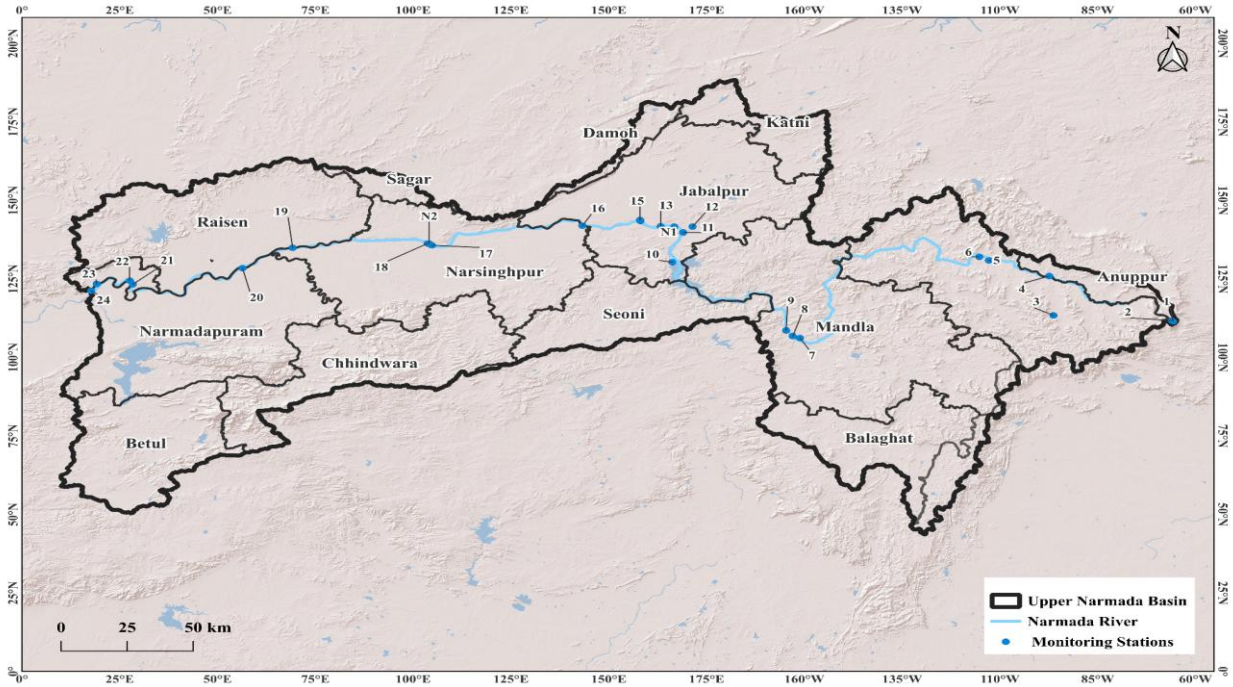


Figure 3: Mapping of different sampling stations (1-24) of districts in the Upper Narmada Basin.

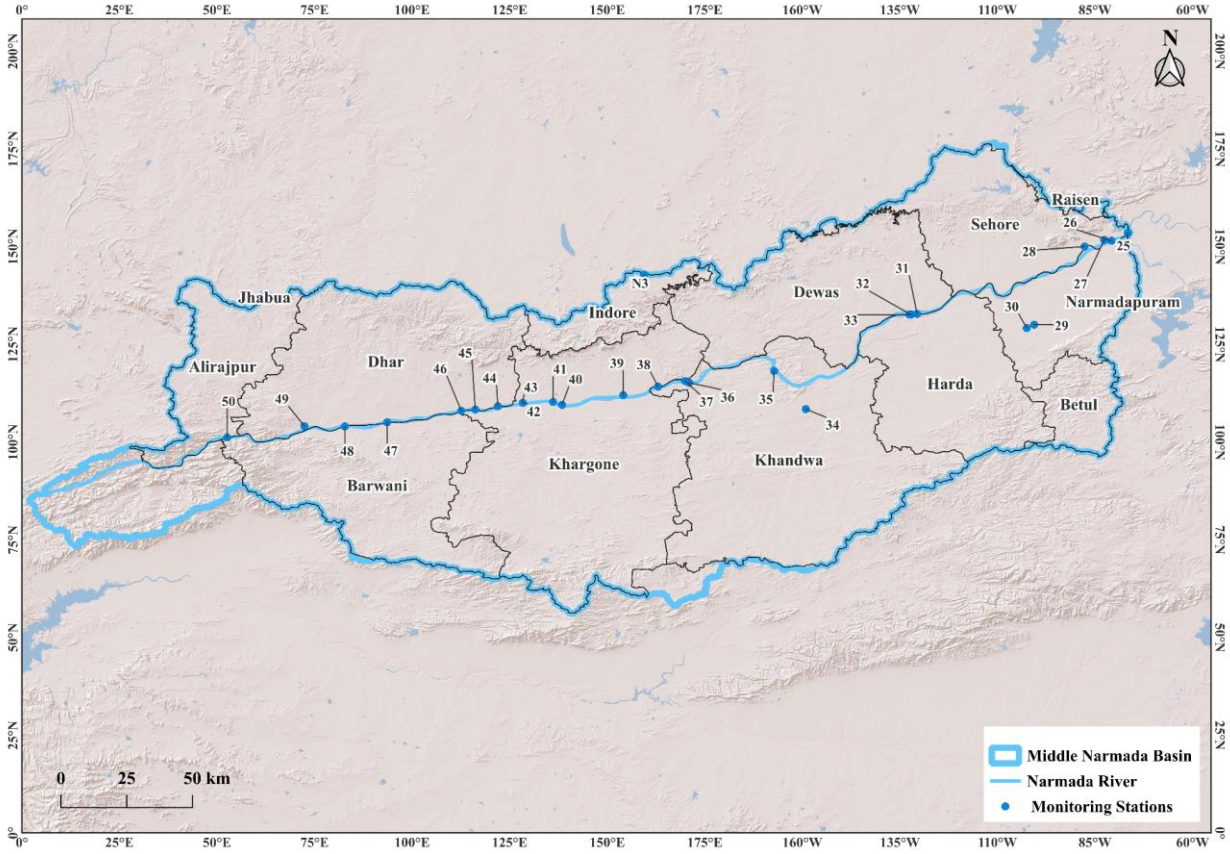


Figure 4: Mapping of different sampling stations (25-50) of districts in the Middle Narmada Basin.

3.1 Upper Narmada Basin

There are a total of 24 stations in the Upper Narmada Basin. The data covers a period of 9 years, from 2016 to 2025 (April – March) for all parameters. The collected data is grouped into three main categories: Physical, Chemical, and Biological Parameters.

3.1.1. Physical Parameters:

Two major physical parameters, Turbidity and Total Dissolved Solids data, are analyzed to assess the water quality of the Upper Narmada Basin.

i) Turbidity

As per the BIS standard, the limit for turbidity is 1-5 NTU. Here, we have analyzed the monthly data for each year (2016-2025) at all stations. According to the values of turbidity, the highest

yearly average was observed at 80.74 NTU at station SU24 (Narmada u/s before the confluence of River Tawa near Bandrabhan). At this same station, the maximum values of turbidity observed were 424, 390, and 98 NTU in July, August, and September 2023-2024, which are typically monsoon months. Similar values are observed for station SU22, Narmada River, at the d/s of Jait Village, Sehore, and Station SU23, Narmada River, at Shahganj Guest House, in the same months and year. A similar pattern of extremely high turbidity values was also recorded at other stations during the monsoon season.

The maximum increase in turbidity values during the monsoon season can lead to sediment disturbance and is also correlated with agricultural mechanization and intensive farming in these areas. The figures for the seasonal variation of turbidity are provided for reference in Annexure 3A.

ii) Total Dissolved Solids

Another important parameter, evaluated to assess the water quality of the Narmada Basin, is the total dissolved solids.

According to the standard set by the Bureau of Indian Standards (BIS), the acceptable range of total dissolved solids is 500-2000mg/L. The data showed that the maximum value of total dissolved solids among all years was recorded as 610 mg/L at station SU15 Narmada, at Saraswatighat, after the Bawanganga River.

The average values of all years across all sampling stations fall within the range set by BIS. The highest average value among all stations was observed at 265 mg/L at Station SU17, located on Narmada NH-44, near the Road Bridge Barmanghat, Narsinghpur, as shown in Figure 5.

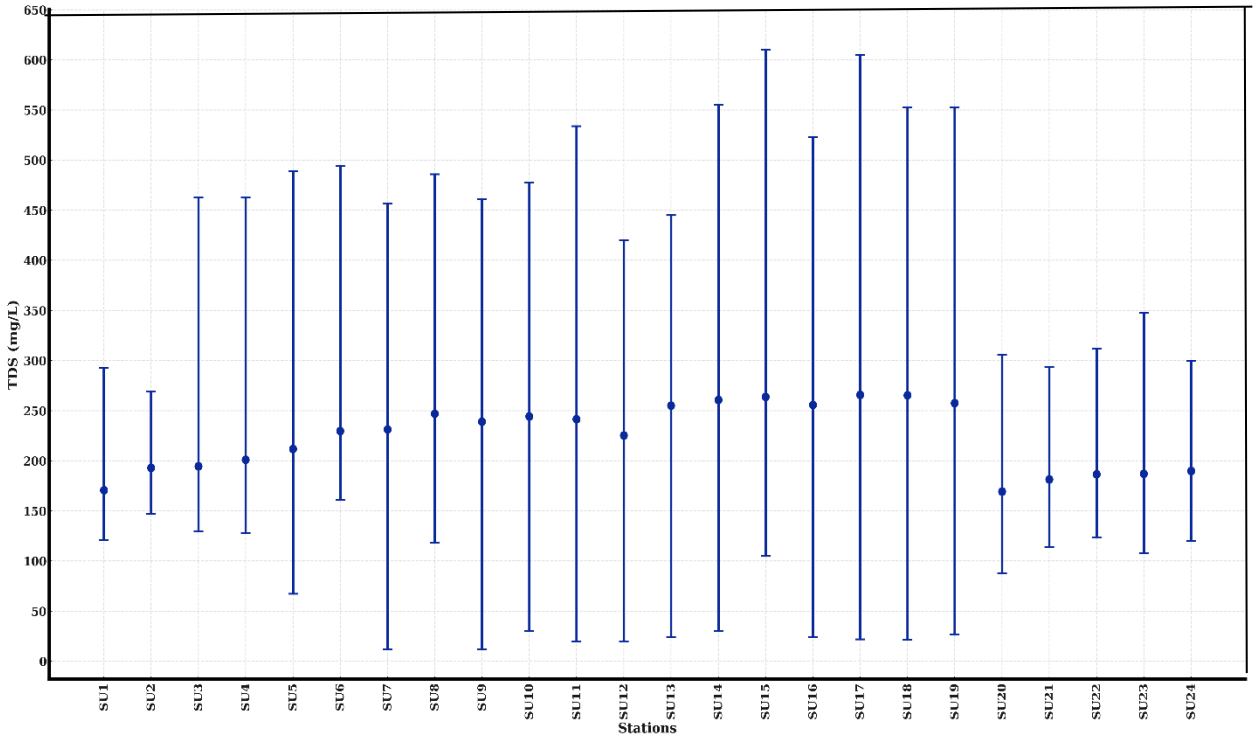


Figure 5: Average, Maximum, and Minimum values over the last 9 years for Total Dissolved Solids in surface water across different sampling stations of Upper Narmada Basin.

3.1.2. Chemical Parameters:

The data analyzed for the chemical parameters include pH, Total Alkalinity, Total Hardness, Potassium, Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), and Chemical Oxygen Demand (COD) across all years and sampling stations.

i) pH

The highest average pH value across all years and stations was observed as 8.3 at Station SU20, located on the Narmada River near Road Bridge Sandhiyaghat, which falls within the BIS permissible limit of 6.5 to 8.5. The pH yearly average values of all years at all sampling stations are shown in Heat Map Figure 6.

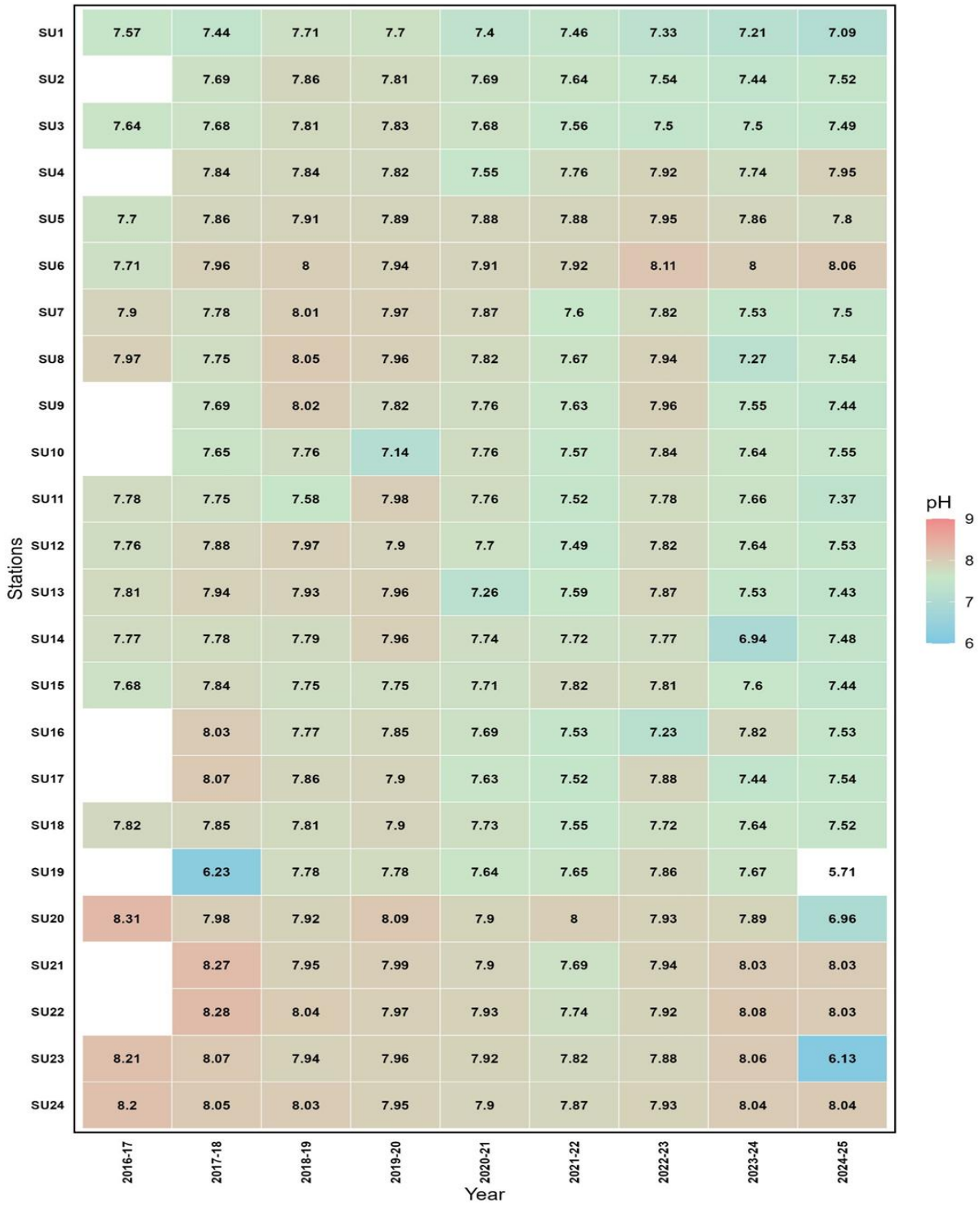


Figure 6: pH yearly average values of surface water across all sampling stations of the Upper Narmada Basin.

ii) Total Alkalinity

The BIS limit set for Total Alkalinity is 200-600mg/L. The highest average value among nine

years across all stations was observed at 128 mg/L at Station SU21, located in the Narmada River u/s of the village of Jait, Sehore. At Station 24, the Narmada u/s before the confluence of the Tawa River at Village Bandrabhan, the maximum value of 176 mg/L was recorded. The average values for all years and stations are within the limit, as shown in Figure 7.

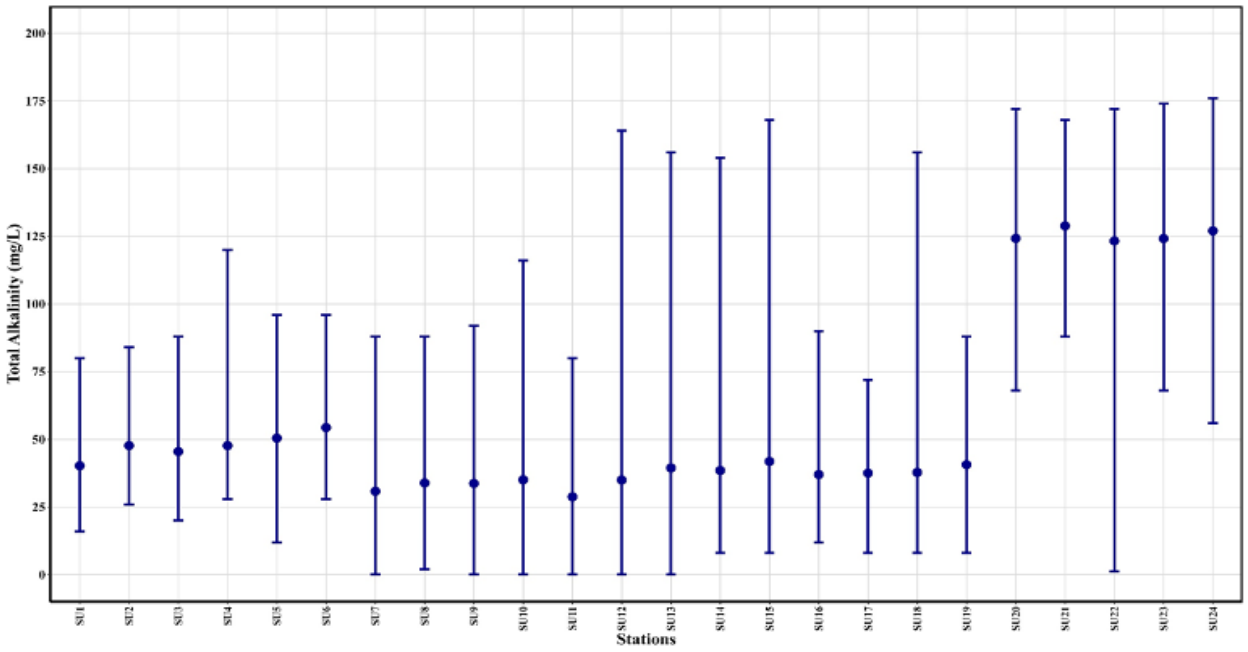


Figure 7: Average, Maximum, and Minimum values over the last 9 years of Total Alkalinity in surface water across different sampling stations of the Upper Narmada Basin.

iii) Total Hardness

The BIS limit of Total hardness is set to 200-600 mg/L. The highest average value among nine years across all stations was observed at 132 mg/L at station SU6, Narmada at Dindori d/s Dindori Jogotikara Ghat. The Maximum Value was observed at 287 mg/L at station SU12, Narmada, at Lalpur, near the water supply intake well point. The average values of all years across all stations for Total Hardness are within the limits, as shown in Figure 8.

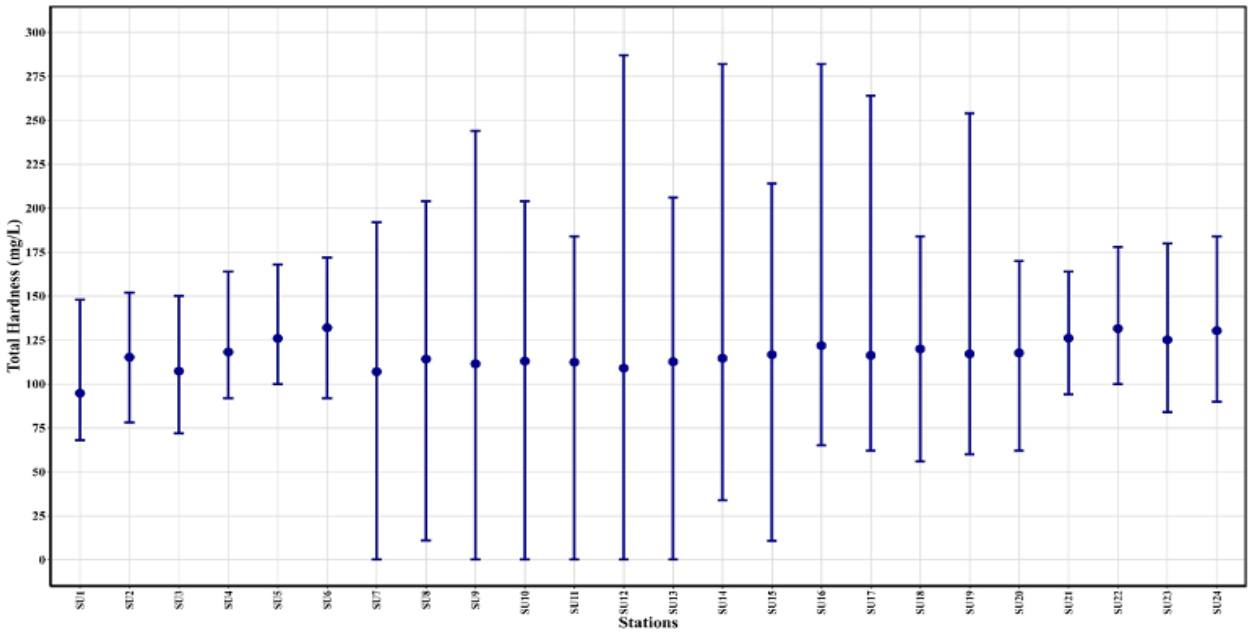


Figure 8: Average, Maximum, and Minimum values over the last 9 years of Total Hardness in surface water across different sampling stations of the Upper Narmada Basin.

iv) Potassium

The highest average values for all nine years at all stations range from 3.52 to 1.69 mg/L. The highest average value for Potassium was observed at 3.52 mg/L at Station 18 Narmada at 100

mt d/s Barmanghat, Narsinghpur, and the maximum value was 18 mg/L at the same station, SU13 Narmada, at Tilwaraghat, Jabalpur. The average values for all nine years at all stations are shown in Figure 9. There is no standard available for Potassium in BIS; however, according to WHO guidelines, a concentration of 10 mg/L is considered safe, with limits varying depending on different conditions (Ref. 7).

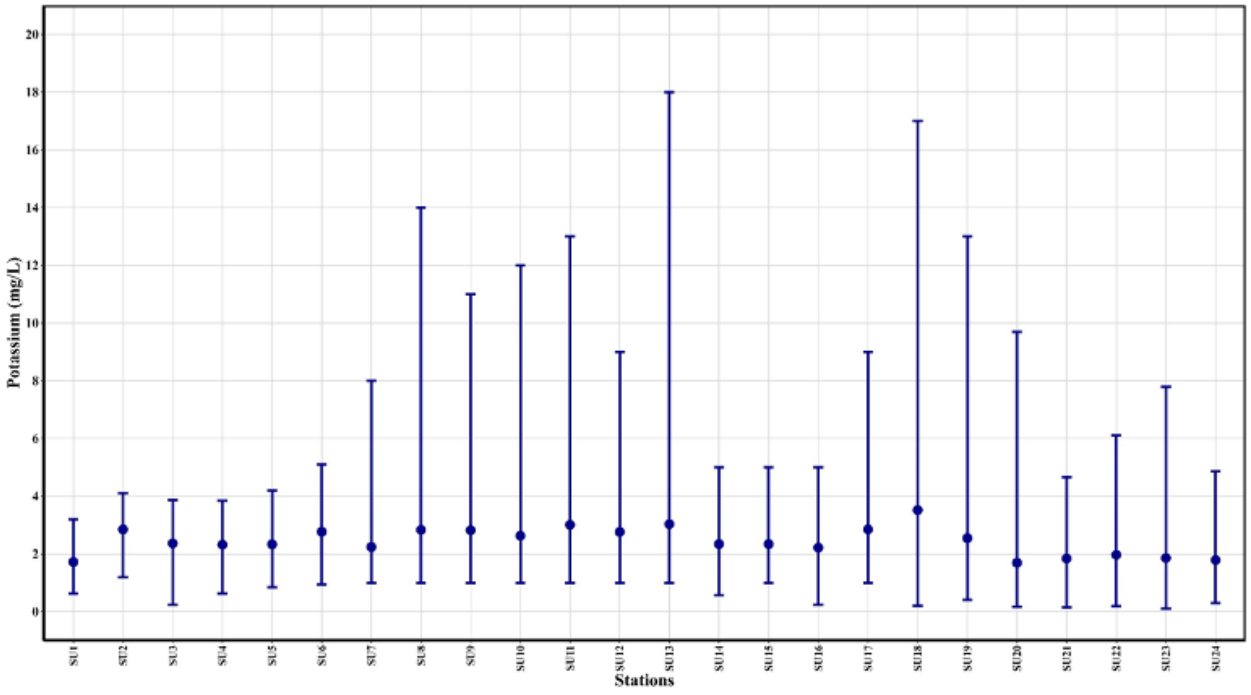


Figure 9: Average, Maximum, and Minimum values over the last 9 years of Potassium in surface water across different sampling stations of the Upper Narmada Basin.

v) Dissolved Oxygen (DO)

The highest average value of dissolved oxygen across all nine years and all sampling stations was recorded as 8.29 mg/L at Station SU 22 Narmada d/s of Jait village, and the maximum value of 14.5 mg/L was observed at the same station. All average values of all years across all stations are within the required limits, i.e., 6 or more, as shown in Figure 10.

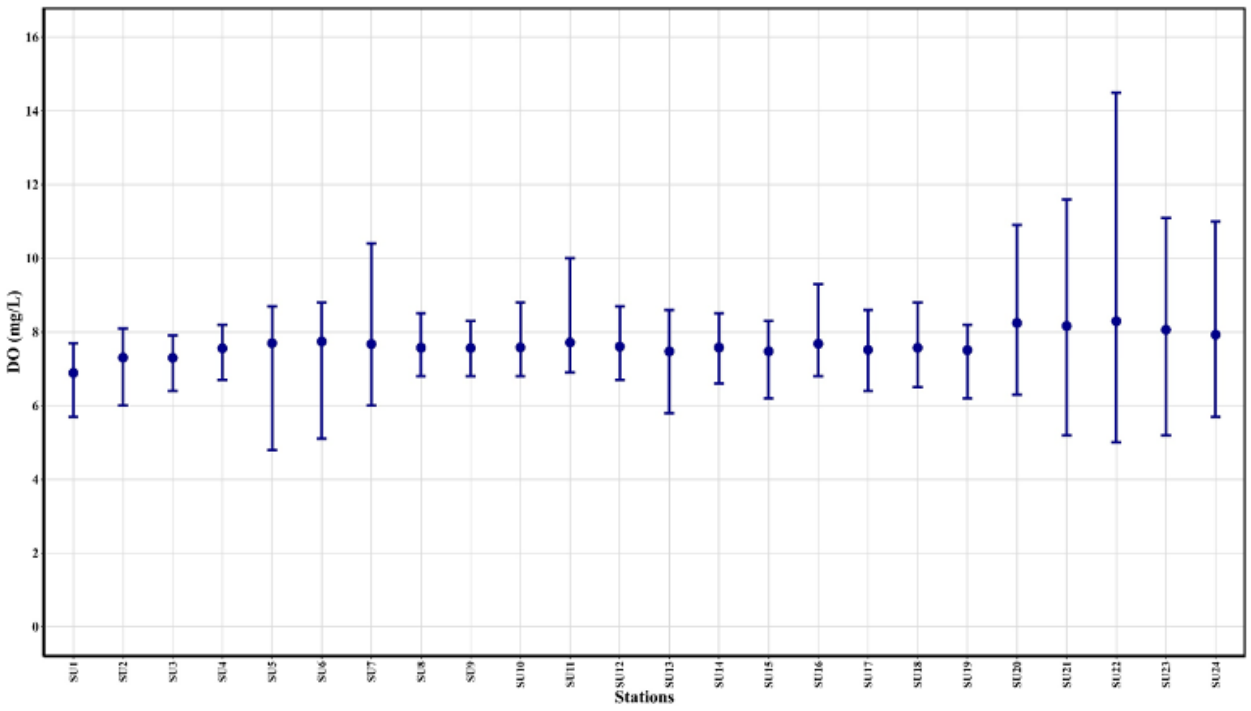


Figure 10: Average, Maximum, and Minimum values over the last 9 years of Dissolved Oxygen in surface water across different sampling stations of the Upper Narmada Basin.

vi) Biochemical Oxygen Demand (BOD)

The analysis indicates that the highest average value across all years and stations observed was 1.84 mg/L at Station SU2 Narmada at Pushkar Dam, Amarkantak, which is within the limits set by BIS, i.e., <3mg/L. The maximum value of 3.1 mg/L was observed at Station SU22 Narmada d/s of Jait village, Sehore. Narmada at the road bridge, Sandia, Pipariya. The average values for all nine years of sampling stations are shown in Figure 11.

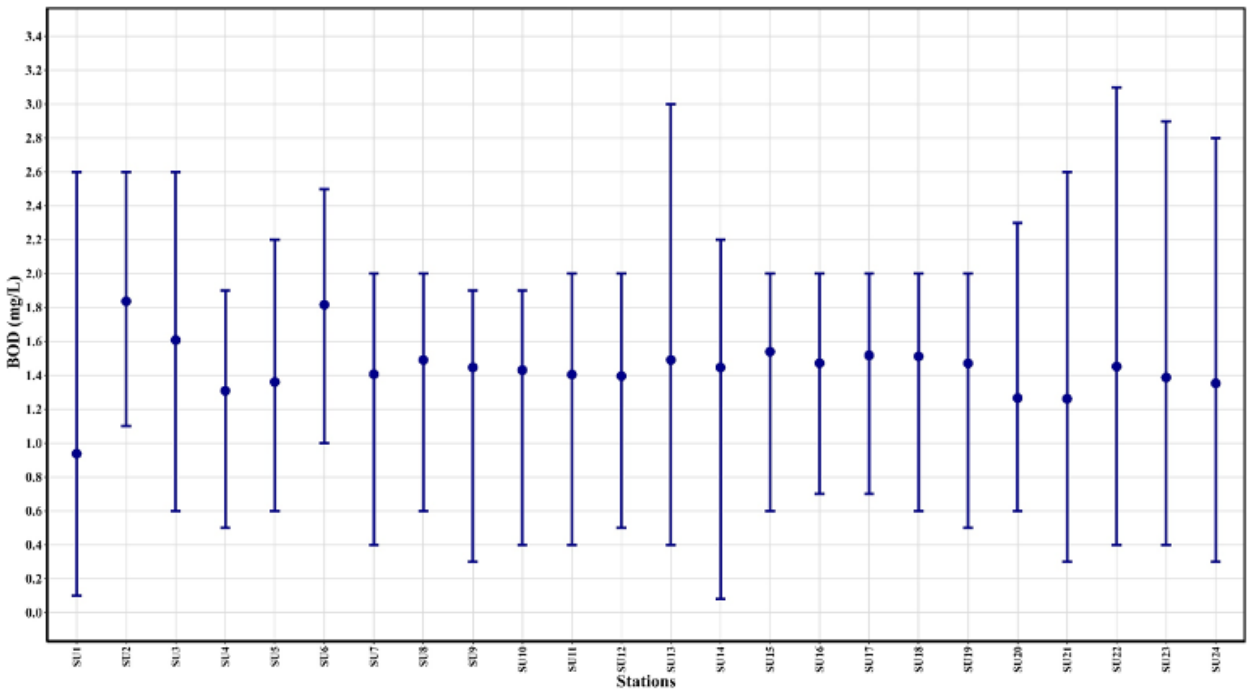


Figure 11: Average, Maximum, and Minimum values over the last 9 years of Biochemical Oxygen Demand in surface water across different sampling stations of the Upper Narmada Basin.

vii) Chemical Oxygen Demand (COD)

The analysis indicates a yearly average value that ranges from 1 to 25 mg/L across all stations over a nine-year period. Details of all stations' yearly averages are provided in Table 6.

Table 6: Summarizes the yearly average of nine years of all sampling stations in the Upper Narmada Basin

Chemical Oxygen Demand									
Station Codes	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
SU1	7.6	5.3	4.9	4.9	2.7	6.1	5.0	9.8	8.9
SU2	-	12.3	15.5	14.7	10.2	14.1	12.8	17.2	13.5
SU3	11.2	9.0	11.4	9.8	5.2	10.4	14.2	16.0	13.7
SU4	-	8.1	8.2	6.1	3.1	6.5	9.4	13.1	12.4
SU5	20.167	9.4	7.4	6.13	3.13	6.50	10.4	10.2	13.9
SU6	21.0	16.3	15.1	13.88	8.62	13.13	19.1	17.6	17.7
SU7	12.17	12.0	13.7	20.0	13.67	11.8	12.3	16.1	11.6
SU8	12.08333	10.9	16.1	21.0	15.1	12.8	13.0	17.3	10.5
SU9	-	16.0	17.7	20.0	14.8	12.1	13.2	18.9	12.8
SU10	-	5.7	18.4	19.8	12.3	14.3	16.8	14.0	13.1
SU11	12.25	12.3	15.1	23.7	12.5	15.3	11.6	14.4	10.1
SU12	10.75	9.8	14.8	18.92	11.50	14.67	13.6	15.2	10.3
SU13	11.67	10.7	18.3	20.8	14.5	12.8	15.5	14.4	14.3
SU14	13.22	12.4	15.9	19.83	16.42	12.17	14.0	14.6	16.6
SU15	13.5	12.6	16.4	19.17	17.17	13.83	15.3	17.9	18.5
SU16	-	12.8	15.3	15.50	16.08	11.50	14.8	15.5	11.5
SU17	-	8.8	18.8	20.3	14.73	16.7	12.7	18.0	12.3
SU18	13.67	14.3	16.6	18.8	17.0	16.5	12.5	16.3	15.8
SU19	-	8.5	18.3	15.7	19.0	13.2	14.2	16.8	11.8
SU20	13.97	12.94	13.1	13.86	13.50	13.43	13.27	12.71	12.78
SU21	-	16.21	12.3	19.55	13.02	11.56	11.37	16.48	12.84
SU22	-	19.39	18.9	24.50	20.33	15.10	12.17	16.09	12.89
SU23	11.57	12.17	14.7	17.16	17.68	13.50	13.02	17.83	13.17
SU24	14.25	22.6	13.9	17.1	13.8	11.7	13.9	14.13	11.34

3.1.3 Biological Parameter

i) Total Coliform

The range of Total Coliforms in the Upper Narmada Basin, as recorded across nine years and at all sampling stations, varied between 1.8 and 54,000 MPN/100 mL, which is significantly higher than the limits of 50 MPN/100 mL for category A. Many stations (SM20-SM24) across the Upper Narmada Basin have recorded high values of Total Coliforms, primarily in the months of October to February and in the years 2018-2021, that is, post-monsoon months. However, the current year's data (2024-2025) shows a maximum average value of 181 MPN/100 ml at Station SU23, Narmada River, at Shahganj Guest House, which is better than

previous years' data, specifically for the above-mentioned stations.

According to the current situation of Total Coliforms in the river water, the water quality is categorized as B, as per the CPCB water quality criteria (Table 5), since the Total Coliform values are above 50 MPN/100ml and less than 500 MPN/100ml. This is an alarm of concern that requires further investigation, regular sampling analysis, and identification of the contamination source to improve the water quality for drinking purposes.

3.2 Middle Narmada Basin

There are a total of 26 stations in the Middle Narmada Basin. The data covers a period of 9 years, from 2016 to 2025 (April – March) for all parameters. The collected data is grouped into three main categories: Physical, Chemical, and Biological Parameters.

3.2.1. Physical Parameters

i) Turbidity

According to the BIS standard, the limit for turbidity is 1-5 NTU. Here, we have analyzed the monthly data for each year at all sampling stations. According to the values, the highest yearly average was observed at 81.94 NTU at Station SM32 Narmada, located at Nemawar, and at Station SM31 Narmada before the confluence of the River Jammar, also at Nemawar. The maximum values of turbidity observed were 510 NTU at Station SM33 Narmada 500 mts. d/s of Jain Mandir, Nemawar, and 498 NTU at Station SM31 in August 2019-2020, which are typically monsoon months. Similar high turbidity values are observed for many sampling stations and across all nine years from July to September. Graphs for seasonal variation are present in Annexure 3B for reference.

A similar pattern of extremely high turbidity values has been recorded for other sampling stations (SM25-SM33), especially during the monsoon season of the year (2018-2025). The maximum increase in turbidity values that occurs during the monsoon season can lead to sediment disturbance and is also correlated with agricultural mechanization and intensive farming in these areas.

ii) Total Dissolved Solids

According to the standard set by the Bureau of Indian Standards (BIS), the acceptable range is 500-2000mg/L for total dissolved solids.

In the analyzed data, the highest average value for nine years across all sampling stations was recorded as 221.2 mg/L at Station SM33, Narmada, 500 meters. d/s of Jain Mandir, Nemawar.

The average maximum value of dissolved solids was recorded as 434mg/L at the same station. Average values of nine years across all sampling stations are within the range and shown in Figure 12.

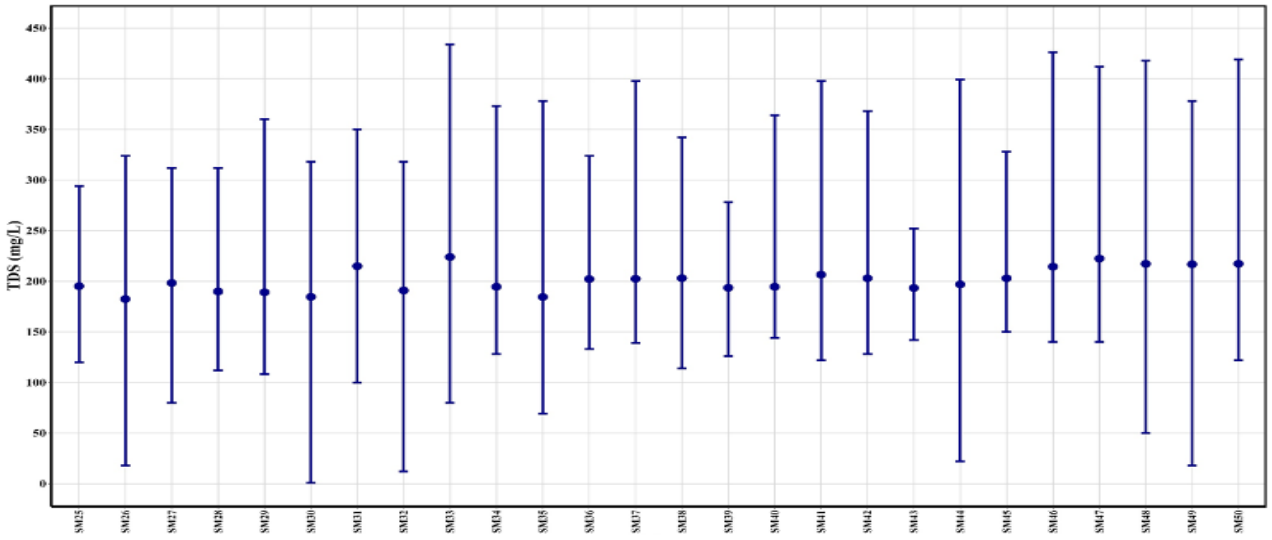


Figure 12: Average, Maximum, and Minimum values over the last 9 years of Total Dissolved Solids in surface water across different sampling stations of the Middle Narmada Basin.

3.2.2. Chemical Parameters

i) pH

The pH values were recorded over a nine-year period at all sampling stations. The maximum yearly average value observed was 8.69 at Station SM34, Narmada River, at Hanumantia, Khandwa, which falls within the BIS permissible limit of 6.5 to 8.5. Variations in pH, with yearly average values from nine years at all sampling stations, are shown in the Heat Map (Figure 13).

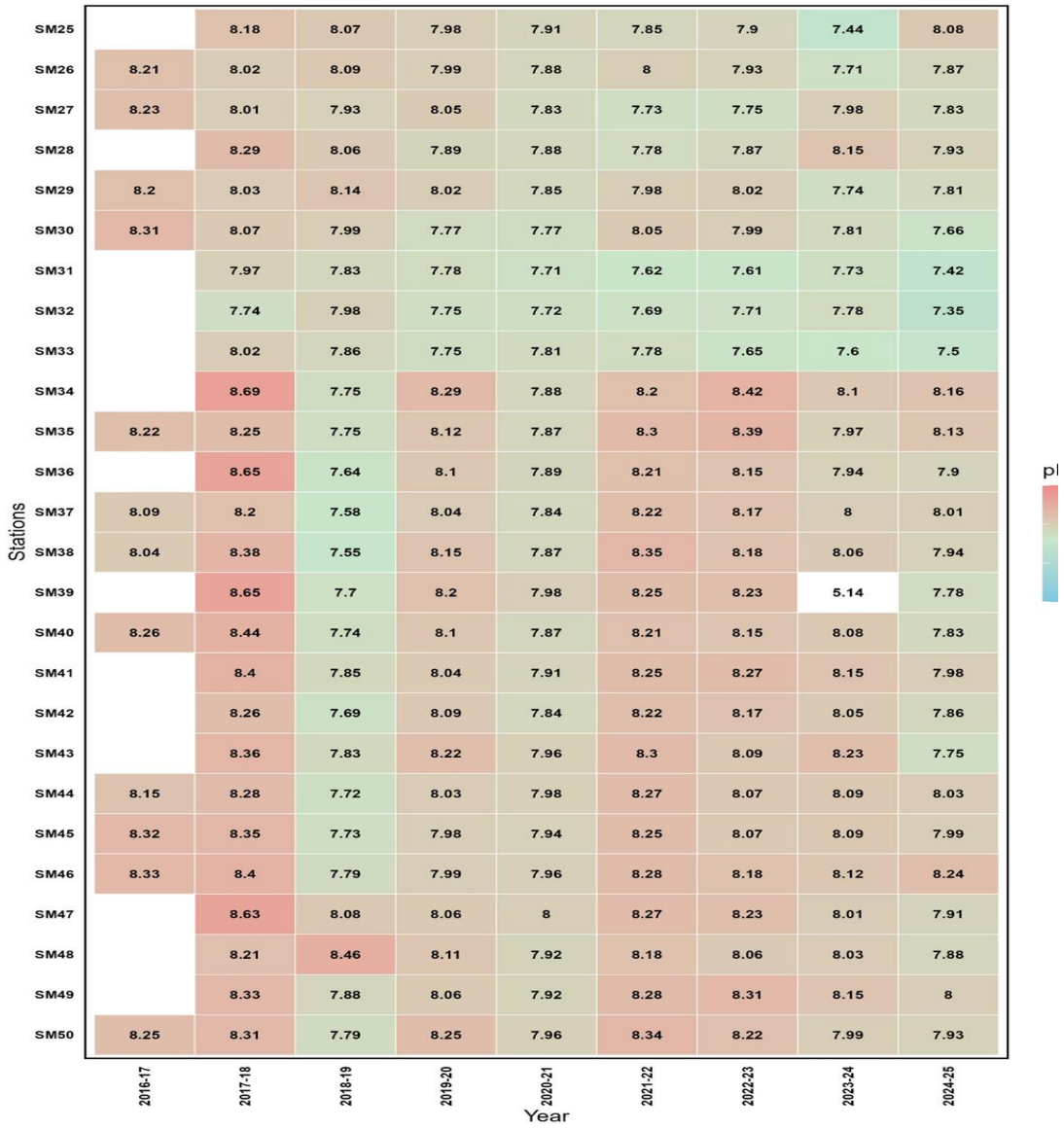


Figure 13: pH of the yearly average at all sampling stations of the Middle Narmada Basin.

ii) Total Hardness

The standard set by BIS for total hardness is 200-600mg/L. The highest average value of nine years across all sampling stations was 168.62 mg/L at Station SM33 Narmada, 500 m. d/s of

Jain Mandir, Nemawar. The maximum value of 410 mg/L was recorded at Station SM48 (Narmada at Rajghat). The average values for the nine years across all sampling stations are within the range, as shown in Figure 14.

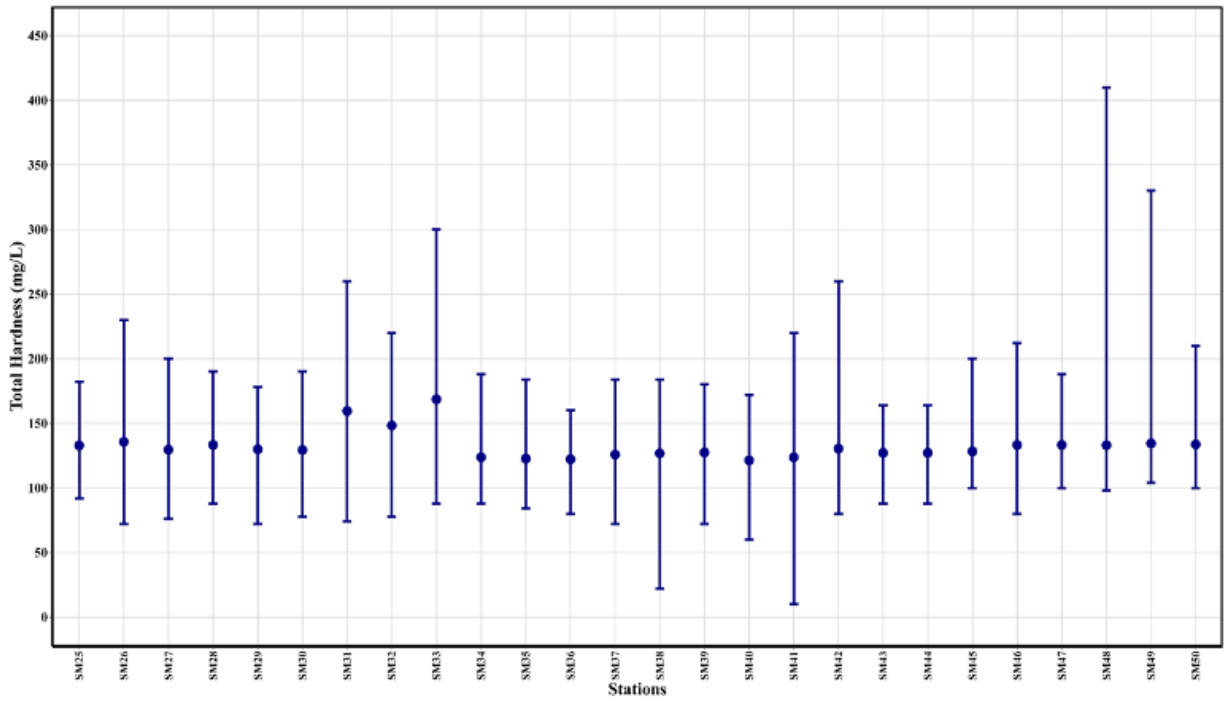


Figure 14: Average, Maximum, and Minimum values over the last 9 years of Total Hardness surface water across different sampling stations of the Middle Narmada Basin.

iii) Total Alkalinity

The standard set by BIS for total alkalinity is 200-600mg/L**.** The highest average value of nine years across all sampling stations was recorded at 149 mg/L at Station SM46 Narmada, d/s of Dharampuri. The maximum value of 238 mg/L was recorded at Station SM26 Narmada at Sethanighat, Narmadapuram. The average values for the nine years at all sampling stations fall within the specified range and are illustrated in Figure 15.

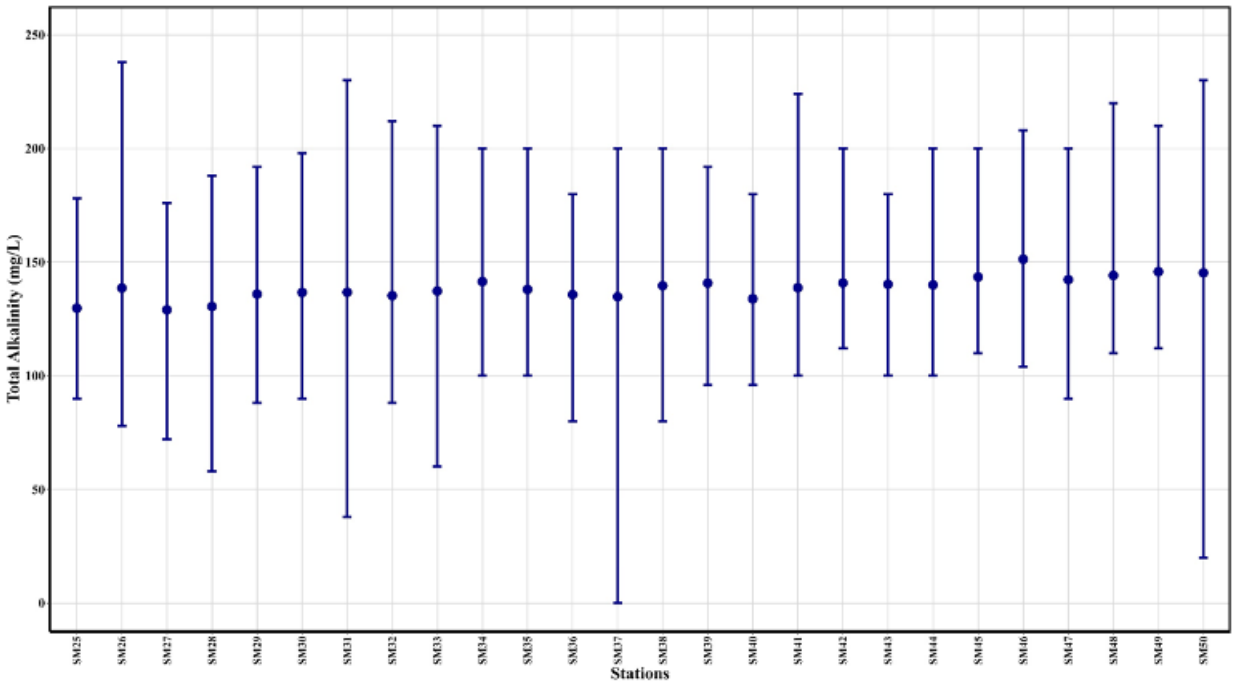


Figure 15: Average, Maximum, and Minimum values over the last 9 years of Total Alkalinity in surface water across different sampling stations of the Middle Narmada Basin.

iv) Potassium

The average values for nine years at all sampling stations range from 2.3 to 1.09 mg/L. The highest average value for Potassium was observed at Station SM25 Narmada d/s after the confluence of the Tawa River at Ramnagar Village. The maximum value of potassium was observed at 7.7 mg/L at Station SM48 Narmada, Rajghat, Barwani. The average values of all 9 years of all stations are shown in Figure 16. There are no standard limits available for Potassium in BIS; however, according to WHO guidelines, a safe limit of up to 10 mg/L is recommended, which varies depending on different conditions (Ref. 7).

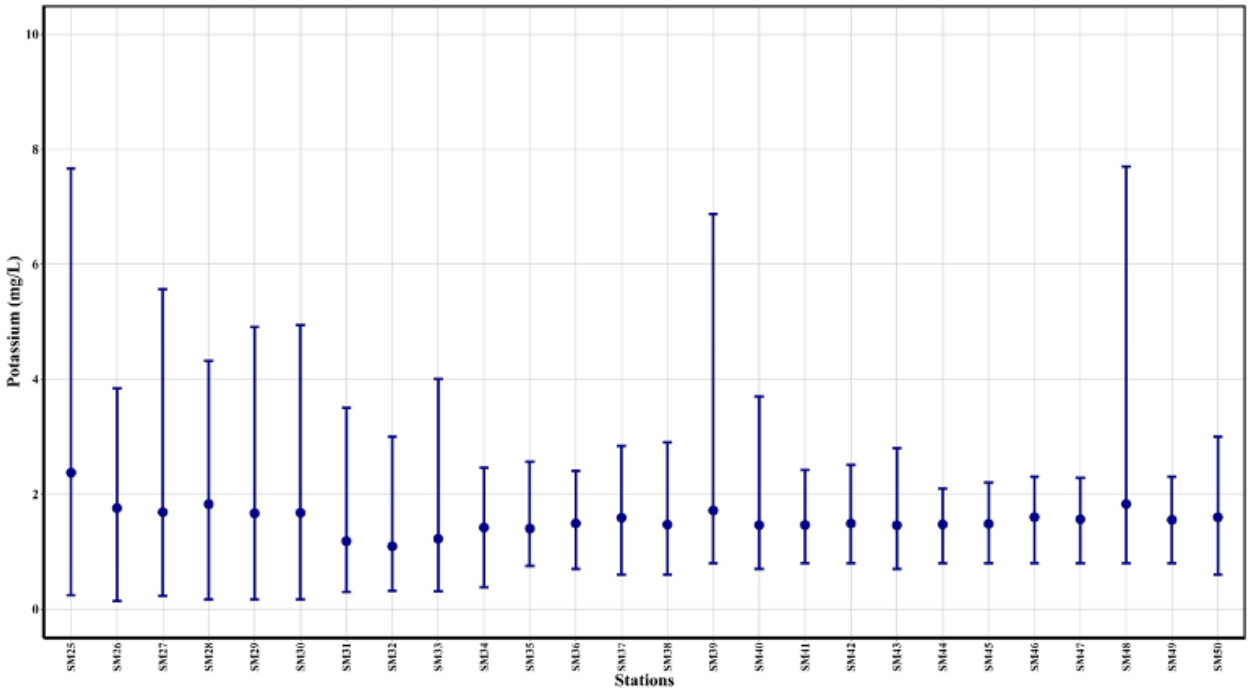


Figure 16: Average, Maximum, and Minimum values over the last 9 years of Potassium in surface water across different sampling stations of the Middle Narmada Basin.

v) Dissolved oxygen

The highest average value of dissolved oxygen across all years and stations was recorded at 8.17 mg/L at Station SM25, Narmada d/s of the confluence of the Tawa River at Ramnagar Village, Bandharban. The same station also recorded a maximum value of 14.10 mg/L. All the nine-year average values are within the limits of 6 or more, as shown in Figure 17.

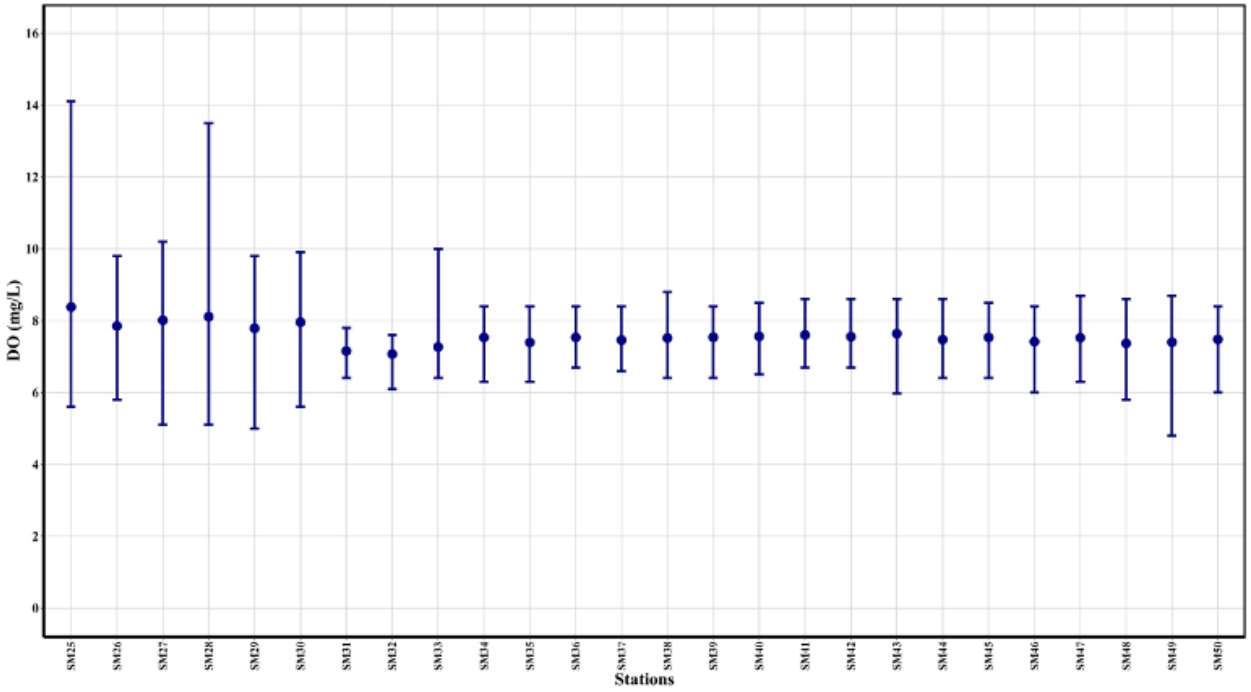


Figure 17: Average, Maximum, and Minimum values over the last 9 years of Dissolved Oxygen in surface water across different sampling stations of the Middle Narmada Basin.

vi) Biochemical Oxygen Demand

As per BIS limits, the Biochemical oxygen demand should not exceed 3 mg or less. Figure 18

shows that the average values for the nine years across all sampling stations fall within this range. The highest average value across all sampling stations was observed at 1.99 mg/L Station SM33, Narmada 500 mts. d/s of Jain Mandir, Nemawar. The maximum value of 3.2 mg/L was observed at Station SM 31, Narmada, before the confluence of the rivers Jammar and Nemawar.

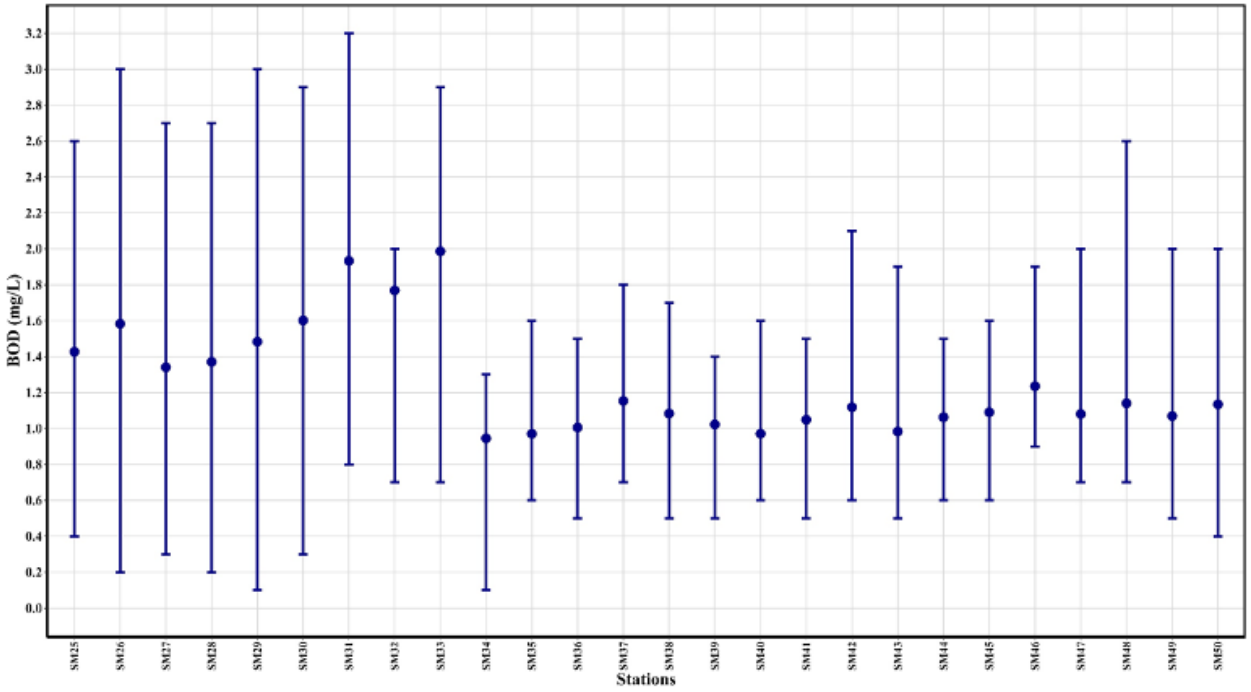


Figure 18: Average, Maximum, and Minimum values over the last 9 years of Biological Oxygen Demand across different sampling stations of the Middle Narmada Basin.

vii) Chemical Oxygen Demand

Chemical Oxygen Demand (COD) analysis indicates a yearly average value for all nine years, ranging from 7 to 23 mg/L across all stations in all years. Details of all stations' yearly averages are provided in Table 7.

Table 7: Summarizes the yearly average of COD of all the Middle Narmada Stations.

Chemical Oxygen Demand									
Station Codes	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25
SM25		19.39	15.57	21.19	17.82	17.56	17.16	13.30	14.41
SM26	17.33	15.54	21.33	21.30	19.58	21.68	18.27	18.87	19.73
SM27	13.42	12.99	19.53	14.65	15.32	12.73	13.92	18.11	12.02
SM28		22.56	17.25	16.97	19.33	11.78	18.77	17.72	12.03
SM29	14.91	14.73	21.33	19.70	17.76	18.03	15.27	17.12	14.15
SM30	18.96	17.17	21.35	22.99	23.20	18.45	15.10	22.65	19.67
SM31		13.33	11.75	18.42	12.64	14.56	14.17	14.00	11.50
SM32		12.04	10.83	11.26	10.74	13.23	12.33	12.67	10.67
SM33		12.67	10.55	13.00	11.73	14.39	15.50	14.57	13.00
SM34		9.40	11.60	10.60	7.99	8.30	7.68	7.67	9.00
SM35	9.93	11.25	11.55	11.56	8.63	9.15	7.81	7.50	7.83
SM36		12.99	12.49	10.70	8.86	8.48	7.89	7.97	8.24
SM37	14.23	15.53	14.81	12.60	10.18	9.46	10.89	10.12	9.57
SM38	12.37	13.50	10.90	12.19	11.60	10.87	9.58	9.45	10.39
SM39		8.83	8.99	10.72	9.01	10.20	9.55	18.21	10.54
SM40	10.59	11.03	9.64	10.76	8.34	8.06	8.27	8.18	8.63
SM41		12.21	11.89	10.16	8.46	8.72	9.15	8.86	10.87
SM42		14.71	13.60	11.32	10.41	10.45	10.15	13.18	10.13
SM43		10.80	10.32	9.45	8.62	9.07	8.56	7.79	10.40
SM44	11.04	12.41	12.55	11.34	9.76	10.36	8.93	9.37	8.15
SM45	11.61	13.63	13.20	11.86	10.32	9.95	9.27	21.53	9.30
SM46	14.38	15.92	15.92	14.79	11.46	11.26	10.39	11.24	11.13
SM47		14.61	13.41	12.13	8.68	9.86	8.80	8.99	11.31
SM48		16.75	12.63	13.95	10.59	11.10	10.55	10.90	11.09
SM49		12.40	12.13	11.90	9.36	12.97	9.40	9.54	13.29
SM50	9.66	13.12	13.58	11.39	9.93	13.81	9.96	10.70	8.47

3.2.3 Biological Parameter

i) Total Coliforms

The highest value of Total Coliforms in the Middle Narmada Basin, across all years and stations, was recorded at 92000 MPN/100 mL, which is significantly higher than the limit of 50 MPN/100 mL. Many stations (SM25-SM30) across the Middle Narmada Basin have recorded values with this high range of total coliforms, mainly in the months of October to March in years (2018-2024), post-monsoon and monsoon months. Although in the current year

data (2024-2025) the maximum average value is 221 MPN/100 ml at Station SM28, Narmada d/s textile unit, Village Holipura, Budni Dhar, which is better than the previous year's data specifically for these stations but still point of concern and requires further investigations, sample analysis to make the water quality better for drinking purposes.

4 Groundwater Physico-chemical Analysis

Data for the Upper and Middle Narmada Basin groundwater is collected from the National Water Informatics Centre (NWIC), Department of Water Resources, River Development & Ganga Rejuvenation of the Central Ground Water Board, covering the years 2000-2022, for physicochemical parameters. The Narmada Basin comprises a total of 313 sampling stations, located in the Upper and Middle Basins. The groundwater analysis is conducted in two sections: the Upper Basin and the Middle Basin. Station codes and Names of Sampling stations of the Upper and Middle Narmada Basin are in Tables 8 and 9.

Table 8: Station codes and Names of Sampling stations of the Upper Narmada Basin

Station Codes	Station Names	Station Codes	Station Names	Station Codes	Station Names	Station Code	Station Name
GU1	Amarkantak	GU55	Simariya	GU109	Manegaon2	GU163	Sonkhera
GU2	Karanjiya	GU56	Siloni	GU110	Karakbel New		
GU3	Gorakhpur	GU57	Bargi Pz	GU111	Bauchhar		
GU4	Gadasarai	GU58	Bargi1	GU112	Dangidhana		
GU5	Gadasarai-D	GU59	Manegaon1	GU113	Sundernagar		
GU6	Gadasarai-S	GU60	Bheraghat New	GU114	Bhainsa		
GU7	Sagar Tola	GU61	Jain Dharamshala	GU115	Nandwara		
GU8	Bijhauri	GU62	Panda Ki Madhia	GU116	Ramkhiria		
GU9	Dindori	GU63	Deotal	GU117	Gundrai2		
GU10	Shahpur	GU64	Bedi Nagar	GU118	Tendukhera		
GU11	Vikrampur1	GU65	Madan Mahal	GU119	Dobhi		
GU12	Salaiya	GU66	Saliwada	GU120	Betli		
GU13	Amarpur1	GU67	Barela	GU121	Kareli Basti		
GU14	Harra	GU68	Sadar Bazar	GU122	Koudiya		
GU15	Chabi	GU69	Panchpedi	GU123	Deoribadwani		
GU16	Indira	GU70	Nagar Nigam Complex	GU124	Salichauka		
GU17	Rehgaon	GU71	Datta Mandir	GU125	Kundali		
GU18	Devgaon	GU72	Cherital	GU126	HarraiDW		
GU19	Mahania patpara	GU73	Bus Stand	GU127	Sathiya		
GU20	Khari	GU74	Raddi Chowki	GU128	Surla		
GU21	Babaliya	GU75	Railway Station	GU129	Chhindi		
GU22	Khamher Kheda	GU76	Kanch Ghar	GU130	Delakhari		
GU23	Kalpi	GU77	Jabalpur	GU131	Renikhera		
GU24	Kudomali New	GU78	Jabalpur(D)	GU132	Mahaljhir		
GU25	Mangalganj	GU79	Jabalpur(S)	GU133	Sarni		
GU26	Gwari	GU80	Adhartal Naka	GU134	Ghoradongri		
GU27	Mandla1	GU81	Gokalpur	GU135	Thapa		

Station Codes	Station Names	Station Codes	Station Names	Station Codes	Station Names	Station Code	Station Name
GU28	Mahrajpur	GU82	Ranjhi	GU136	Betul1		
GU29	Pathiri patpara	GU83	Umariya	GU137	Khedi		
GU30	Ramnagar1	GU84	Padaria	GU138	Jogli		
GU31	Padmi choraha	GU85	Bishanpura	GU139	Nimpani		
GU32	Bamhni New	GU86	Kundam Pz	GU140	Bhonra		
GU33	Anjania	GU87	Kundam	GU141	Siarmau		
GU34	Ghughri	GU88	Ghughara	GU142	Deori		
GU35	Mudiya Rijhika	GU89	Sihora1	GU143	Udaipura		
GU36	Bichhia1	GU90	Sihora Pz	GU144	Khiria		
GU37	Sijhora	GU91	Kachupura	GU145	Dhangwan		
GU38	Motinala	GU92	Majholi	GU146	Silwani		
GU39	Mangli	GU93	Patan Pz	GU147	Sultanpur		
GU40	Saletekhri	GU94	Shahpura	GU148	Dam Dongri		
GU41	Damoh(S)	GU95	Banjari	GU149	Hathi palan		
GU42	Damoh2	GU96	Ghansor1	GU150	Bari		
GU43	Birsa	GU97	Mehta	GU151	Sandia		
GU44	Supkhar	GU98	Khamaria	GU152	Matkuli		
GU45	Jawaditula	GU99	Sahasna	GU153	Pachmarhi		
GU46	Bhaisanghat	GU100	Gharghatia	GU154	Suktawa		
GU47	Mukki	GU101	Makarjhir	GU155	Kesla		
GU48	Parsatola	GU102	Lakhnadon1	GU156	Gurra New		
GU49	Baihar(D)	GU103	Gaurabibi	GU157	BagratawaDW		
GU50	Baihar1	GU104	Dhuma	GU158	Sohagpur		
GU51	Khurmundi	GU105	Nagan Deori	GU159	Semri Harchand		
GU52	Bagholi	GU106	Dargada	GU160	Baharpur		
GU53	Paraswara	GU107	Jhoteshwar	GU161	BabaiDW		

Station Codes	Station Names	Station Codes	Station Names	Station Codes	Station Names	Station Code	Station Name
GU54	Umariapan	GU108	Gotegaon	GU162	Nimsadia		

Table 9: Station codes and Names of sampling stations of the Middle Narmada Basin.

Station Codes	Station Names	Station Codes	Station Names	Station Codes	Station Names
GM164	PathrautaDW	GM216	Udaipur	GM268	Lunera
GM165	Raisalpur	GM217	Bangarda	GM269	Dhar
GM166	Sanwalkhera	GM218	Mundi	GM270	Dhar (D)
GM167	DolariaDW	GM219	Dagad Khedi	GM271	Dhar (S)
GM168	Bhilatdeo	GM220	Bori saray	GM272	Amjhira
GM169	Seonimalwa	GM221	Bedia	GM273	Amjhera(D)
GM170	Chincholi(S)	GM222	Chanera	GM274	Tanda
GM171	Pathakhera	GM223	Kahlari	GM275	Bagh New
GM172	Chirapatala	GM224	Jawar	GM276	Bagh(D)
GM173	Khokharkheda	GM225	Roshiya	GM277	Bagh(S)
GM174	Budhni	GM226	Deshgaon New	GM278	Palasi
GM175	Bayan	GM227	Chhegaonmakhan	GM279	Kukshi1
GM176	Neelkachar	GM228	Khandwa	GM280	Pipalya
GM177	Malibayan	GM229	Jaswadi1	GM281	Dhulsar
GM178	Bordi	GM230	Rudhy Bhata	GM282	Singhana
GM179	Nadan	GM231	Khedi New	GM283	Dehari
GM180	Larkui New	GM232	Kalamkalan	GM284	Gandhwani
GM181	Rala	GM233	Khalwa1	GM285	Zeerabad
GM182	Chhidgaon	GM234	Gurhi	GM286	Manawar1
GM183	Timarni	GM235	Balwara1	GM287	Manawar(D)
GM184	Temagaon	GM236	Pandhana	GM288	Mandu
GM185	Mohanpur1	GM237	Kusumbiya	GM289	Gujri1
GM186	Chhuri Khal	GM238	Borgaon Buzurg	GM290	Dhamnod
GM187	Sonpura Colony	GM239	Jhiri	GM291	Tawlai
GM188	Chhipawad	GM240	Bhulwani	GM292	Dharampuri1
GM189	Khirkiya(D)	GM241	Dhulkot	GM293	Chachariya
GM190	Mandla	GM242	Bhagwanpura(D)	GM294	Sendhwa
GM191	Masangaon	GM243	Ghatti	GM295	Niwali1
GM192	HardaDW	GM244	Segaon	GM296	Palsud
GM193	Handia	GM245	Un	GM297	Balsamund
GM194	Handia(S)	GM246	Khargone	GM298	Julwania
GM195	Pipilianankar	GM247	Divalgaon	GM299	Baruphatak
GM196	Khategaon(S)	GM248	Bamnala New	GM300	Rajpur(D)
GM197	Khategaon(D)	GM249	Bhikangaon1	GM301	Rajpur
GM198	Dhayali	GM250	Bhikangaon(D)	GM302	Borlai
GM199	Satwas New	GM251	Daudwa	GM303	Fatta
GM200	Kantaphor	GM252	Sanawad New	GM304	Nanpur
GM201	Kannod	GM253	Sanawad(D)	GM305	Borkua
GM202	Kusumania	GM254	Amba	GM306	Aalirajpur(S)
GM203	Bijawad	GM255	Gogaon	GM307	Alirajpur
GM204	Bijawad(D)	GM256	Sawda	GM308	Khattali
GM205	Punjapura	GM257	Kasrawad1	GM309	Jobat New

Station Codes	Station Names	Station Codes	Station Names	Station Codes	Station Names
GM206	Bhikupura	GM258	Kasrawad(S)	GM310	Jobat(D)
GM207	Udainagar	GM259	Maheshwar	GM311	Badaguda
GM208	Pipri	GM260	Maheswar(S)	GM312	Ambua
GM209	Thapana	GM261	Dhargaon	GM313	Bhabra New
GM210	Ghosali	GM262	Piplyabuzrug		
GM211	Dhangaon	GM263	Baddiya		
GM212	Karoli	GM264	Barwah		
GM213	Gujar Khedi	GM265	Balwara		
GM214	Kelwa kalan	GM266	Nandpura		
GM215	Daulatpur	GM267	Manpur(D)		

4.1 Upper Narmada Basin

There are a total of 163 groundwater sampling stations that cover 12 districts of the Upper Narmada Basin. A detail of the number of sampling stations is illustrated in Figure 19.

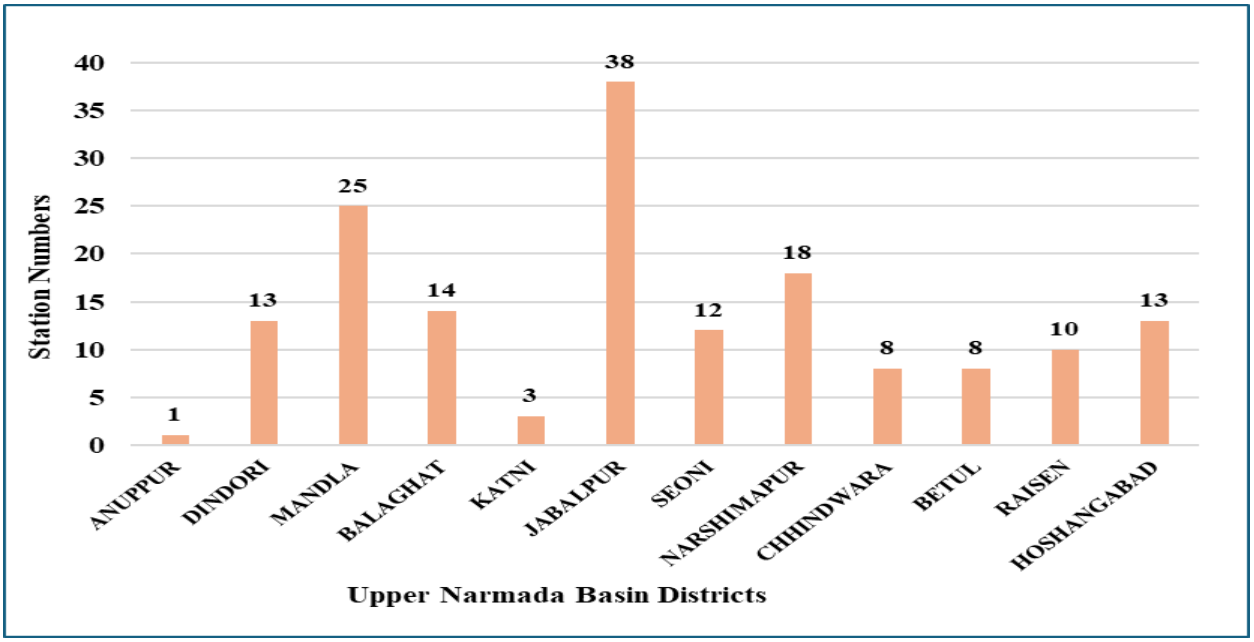


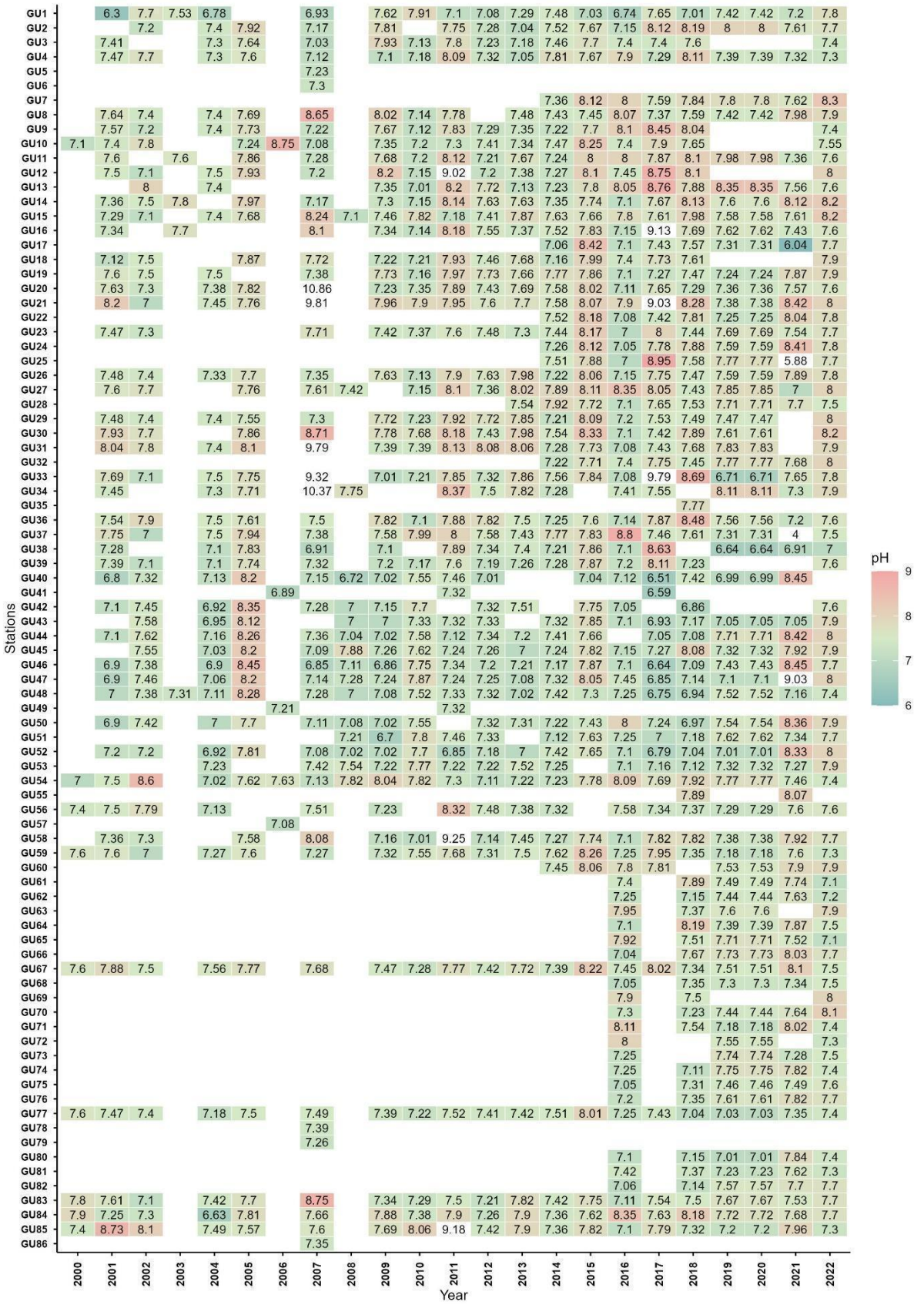
Figure 19: District-wise distribution of groundwater sampling station numbers covered in the Upper Narmada Basin.

The groundwater parameters analyzed in this report are crucial for assessing water quality and health. The following parameters have been measured:

i) pH

The pH data for all sampling stations in the Upper Narmada Basin are available for the period from 2000 to 2022, and yearly average values for all 23 years across different sampling stations are shown in the heatmap (Figure 20).

The highest average value of all 23 years for pH values across these stations ranges from 6.93 to 7.98, which fall within the BIS permissible limit of **6.5 to 8.5**. A noticeable increase in pH values was observed at several stations, exceeding the upper limit of 8.5. The maximum pH values were recorded at stations GU20 (10.86 Khari) and GU34 (10.37 Ghughri), as well as at several other stations, with maximum values ranging from 8.6 to 9.81 during the period. Apart from these values, the pH levels at all stations remained within the acceptable range, indicating overall stable quality conditions in the basin.


Stations
Year
pH
9
8
7
6
2000
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021
2022

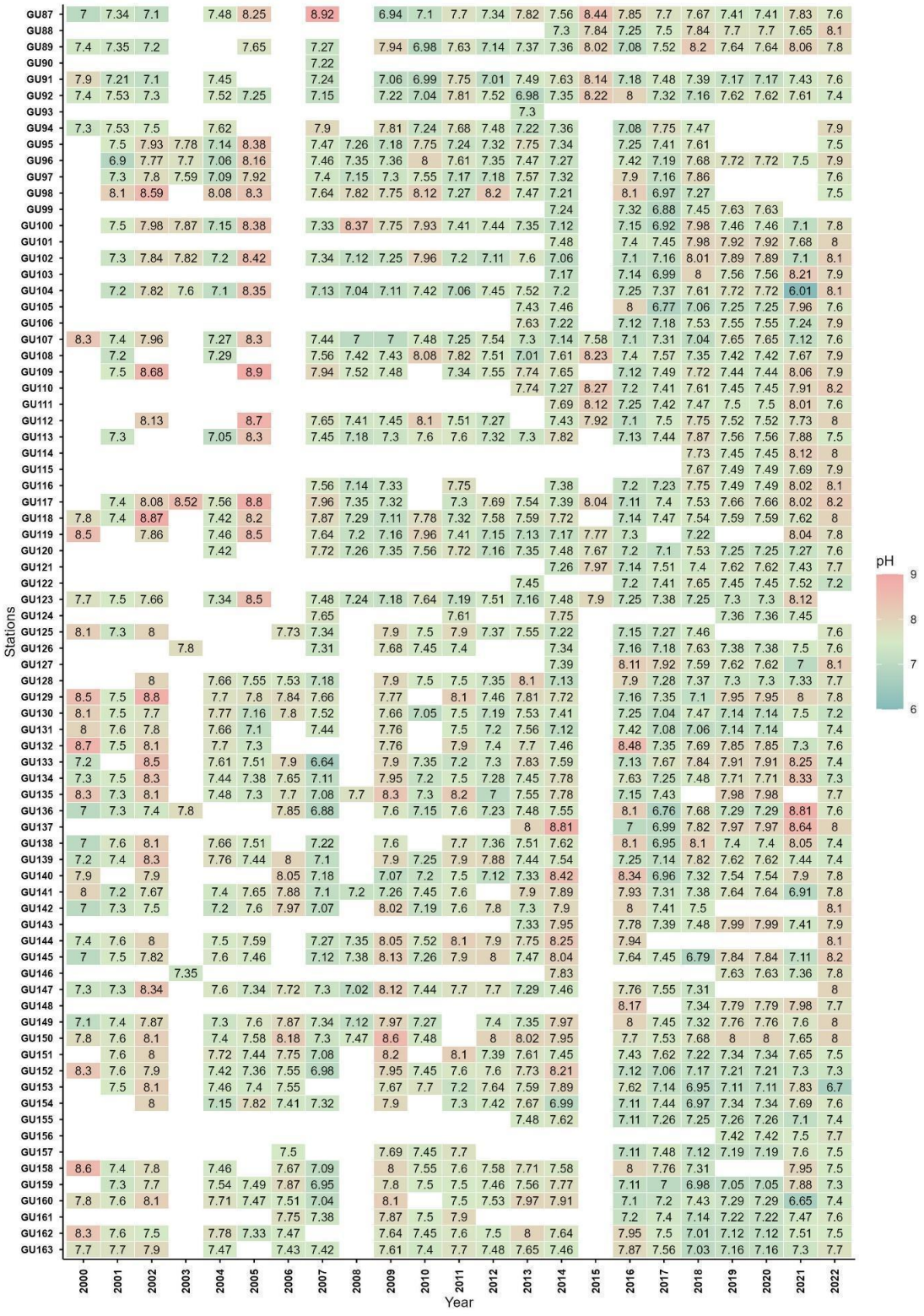


Figure 20: pH yearly average values of groundwater of all sampling stations of 23 years of the Upper Narmada Basin.

ii) Chloride

The BIS standard limit for Chloride is between 250 and 1000mg/L. Data collected over the

years 2000-2022 for all 163 sampling stations show that the highest average value for all years for chloride is 236.27 mg/L, recorded at Station GU156 (Gurra New), and 216.56 mg/L at Station GU50 (Baihar 1). The maximum chloride value, 849.7 mg/L, was observed at Station 56 (Siloni). All the chloride values for all years and stations are within the permissible limits for the Upper Narmada Basin, as shown in Figure 21.

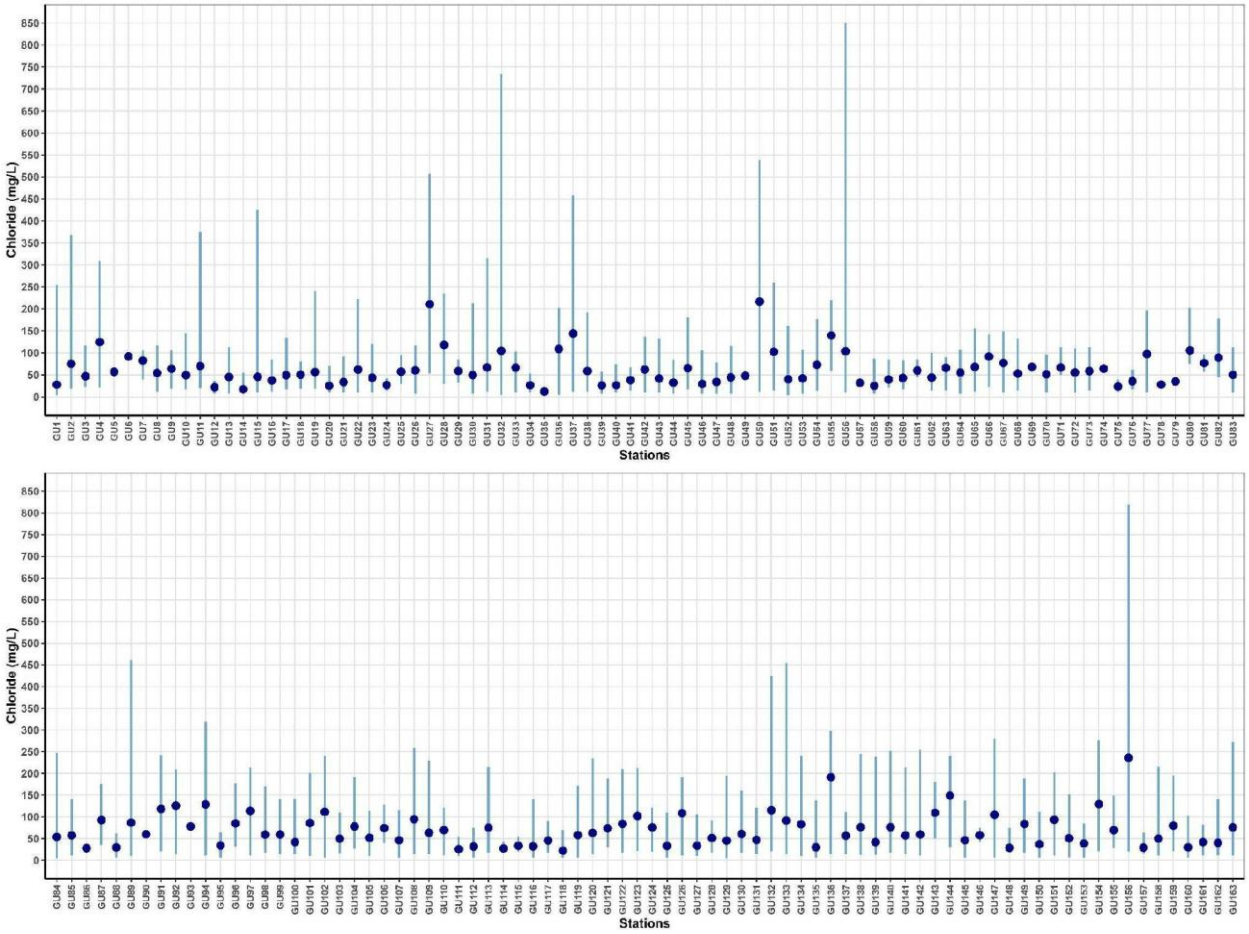


Figure 21: Average, Maximum, and Minimum values of the 22 years of Chloride in groundwater across different sampling stations of the Upper Narmada Basin.

iii) Sulphate

The BIS standard limit for sulphate is 200-400 mg/L. Data collected over the years (2000-2022) for all 163 sampling stations show that the highest average value for all years of sulphate, at 199.53 mg/L, was recorded at Station GU144 (Khiria). Additionally, at this same station, the maximum sulphate values reached 575 mg/L and 500 mg/L. In all other years and stations, the all-year average values of sulphate levels did not exceed the upper limit, as shown in Figure

22.

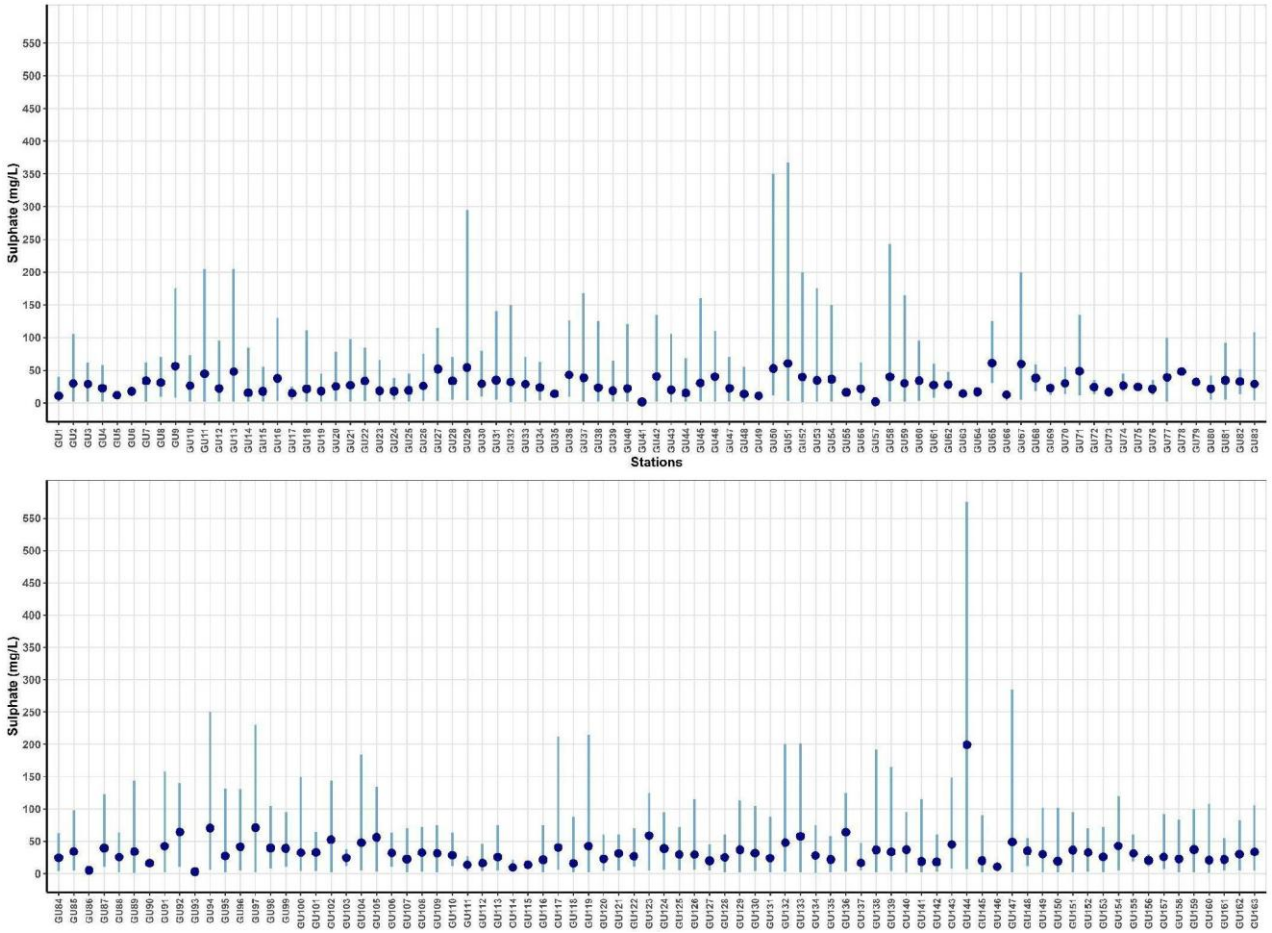


Figure 22: Average, Maximum, and Minimum values of the 20 years of Sulphate in groundwater across different sampling stations of the Upper Narmada Basin.

iv) Fluoride

The BIS standard limit for fluoride is 1-1.5 mg/L. The data for fluoride collected over (2000-2022) for all sampling stations. The highest average fluoride value for all years was observed at 1.31 mg/L at Station GU74 (Raddi Chowki), which falls within the standard limits. A maximum value of 2.8 mg/L was recorded at Station GU36 (Bicchai1). The overall average for all years across all stations is within the permissible limit, as shown in Figure 23.

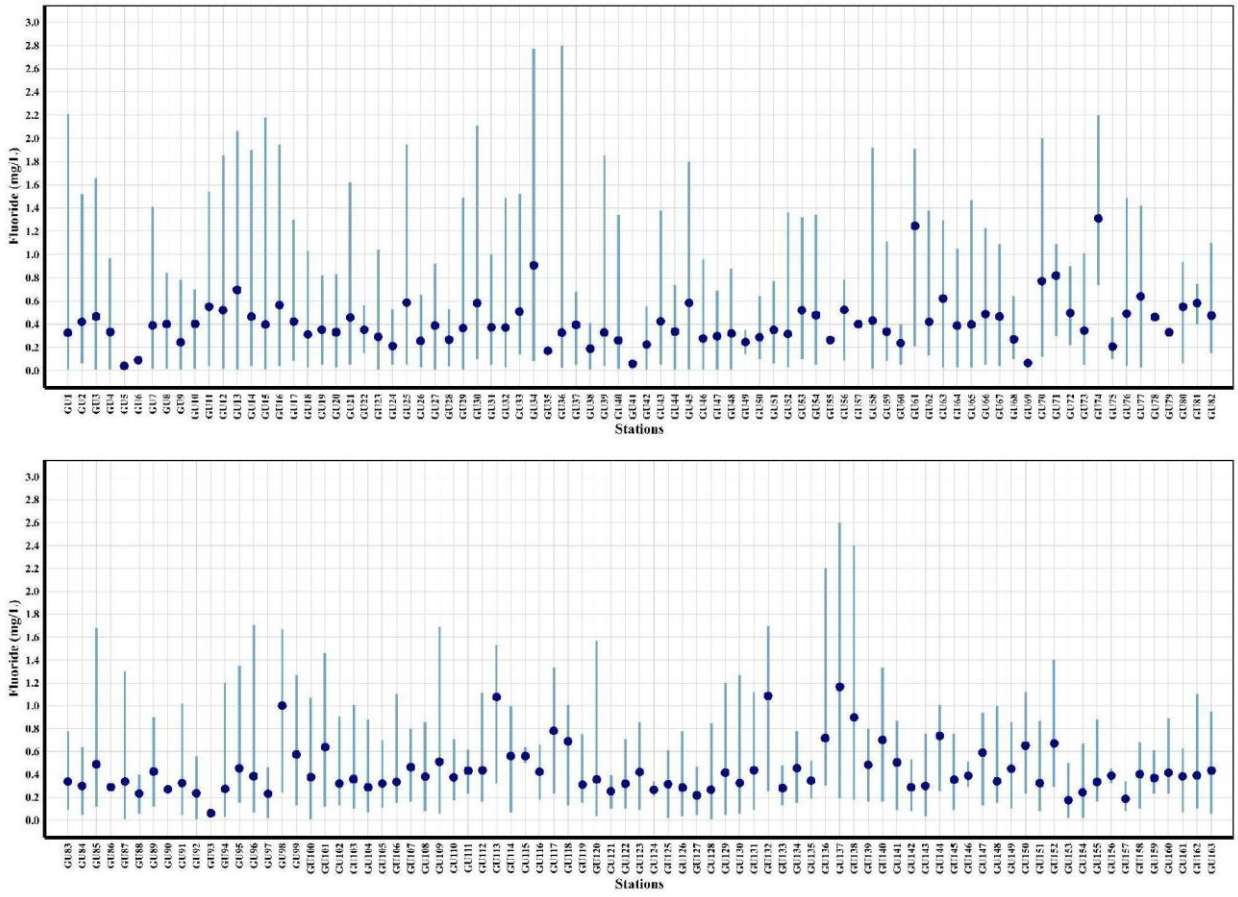


Figure 23: Average, Maximum, and Minimum values of 23 years of Fluoride in groundwater across different sampling stations of the Upper Narmada Basin.

v) Total Hardness

The BIS standard limit for Total Hardness is between 200 and 600mg/L. We have collected and analysed data over the years 2000-2022 for all stations, which shows that the highest overall average value of all years for total hardness is 536.67 mg/L, recorded at Station GU156 (Gurra New), and also 507 mg/L, recorded at Station GU94 (Shahpura), which are within the limits.

The maximum total hardness value, 1,250 mg/L, was observed at Station GU56 (Siloni) in 2021. Additionally, three other stations reported significant hardness values: Station GU94 (Shahpura) with 1015 mg/L in 2004, Station GU144 (Khiria) with 1000 mg/L in 2008, and

Station GU156 (Gurra New) with 1040 mg/L in 2022. Numerous sampling stations recorded total hardness values ranging from 610 mg/L to 955 mg/L, as shown in Figure 24, which exceeds the permissible limit. However, it is important to note that each of these stations exhibited high values only once or twice during the study period, while the total hardness values for all other years remained within the acceptable range.

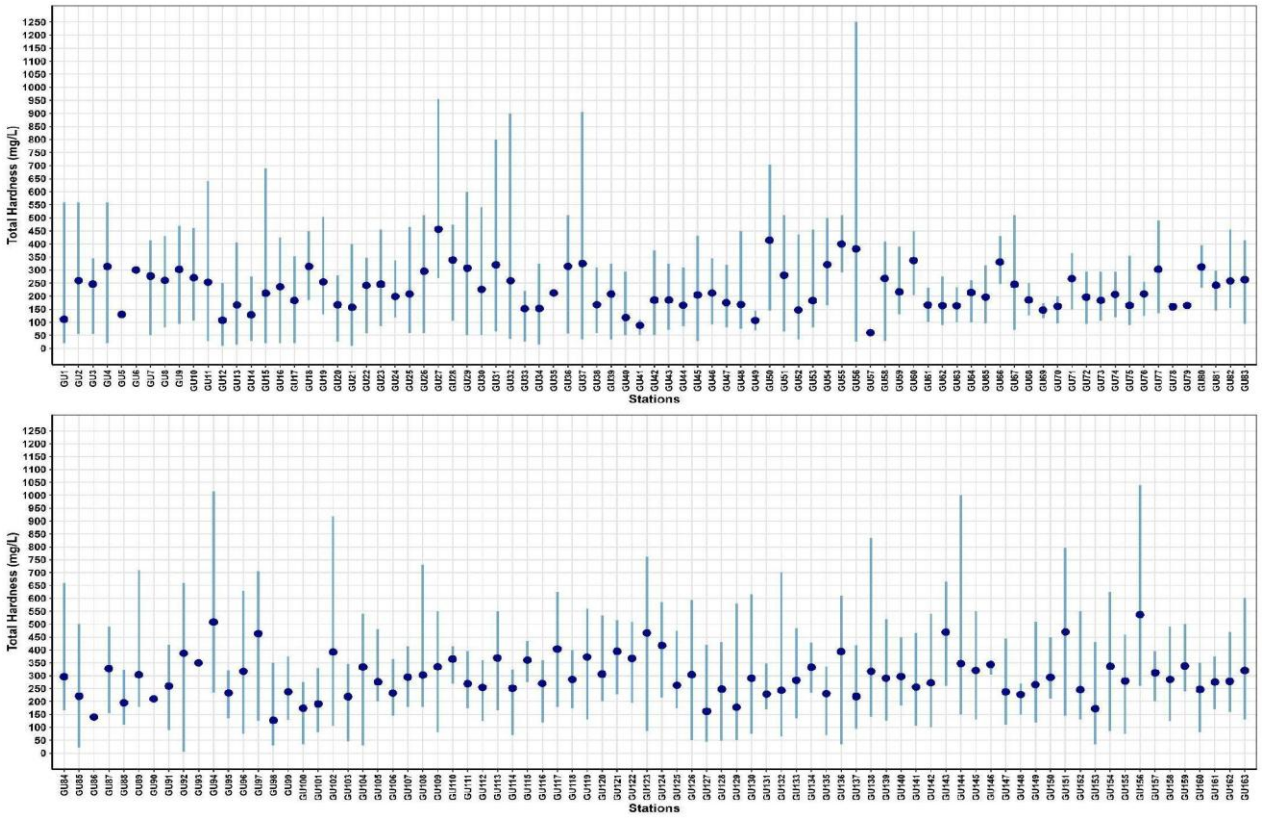


Figure 24: Average, Maximum, and Minimum values of 21 years of Total Hardness in groundwater across different sampling stations of the Upper Narmada Basin.

vi) Total Alkalinity

The BIS standard limit for Total Alkalinity is between 200 and 600mg/L. We have collected and analyzed data from 2004 to 2015 for all stations, which shows that the highest overall average value for total alkalinity across all years is 505 mg/L, recorded at Station GU144 (Khiria). The maximum total alkalinity value observed at this station was 665 mg/L. All other stations remained within the permissible limit during the entire period, as shown in Figure 25.

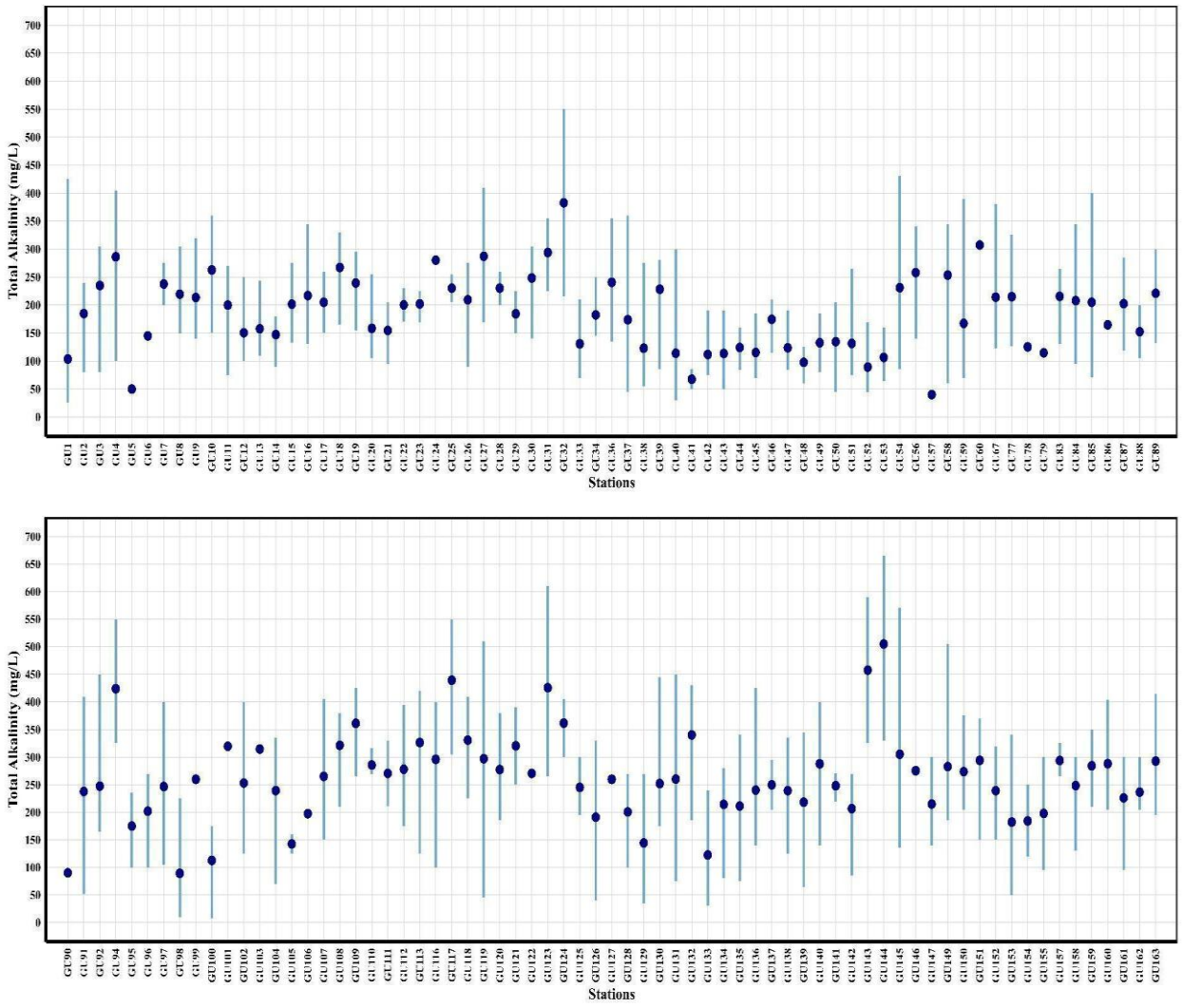


Figure 25: Average, Maximum, and Minimum values of all 12 years of Total Alkalinity in groundwater across different sampling stations of the Upper Narmada Basin.

vii) Silicon dioxide and Total Dissolved Solids

The SiO2 maximum value limit is 210mg/L(Ref.9), considered safe for drinking water, and the data collected from 2016 to 2022 for all sampling stations lie within these limits. The maximum value was observed at 93 mg/L at Station GU56 (Siloni). Similarly, TDS values across all stations for all years are within the limit, i.e., 500-2000mg/L.

viii) Nitrate

The BIS standard limit for Nitrate Nitrogen is set at < 45mg/L. The data collected and analysed over (2000-2022) for all sampling stations. The highest average value of all years was observed at 124 mg/L at Station GU97 (Mehta). The maximum Value of Nitrate was observed at 424 mg/L at Station GU84, Padaria. Many of the stations located in the Upper Narmada Basin exhibited nitrate levels that were significantly elevated and exceeded the established limits.

4.2 Middle Narmada Basin

There are a total of 150 sampling stations for groundwater (164-313) that cover 12 districts of

the Middle Narmada Basin. Details of the number of stations are illustrated in Figure 26.

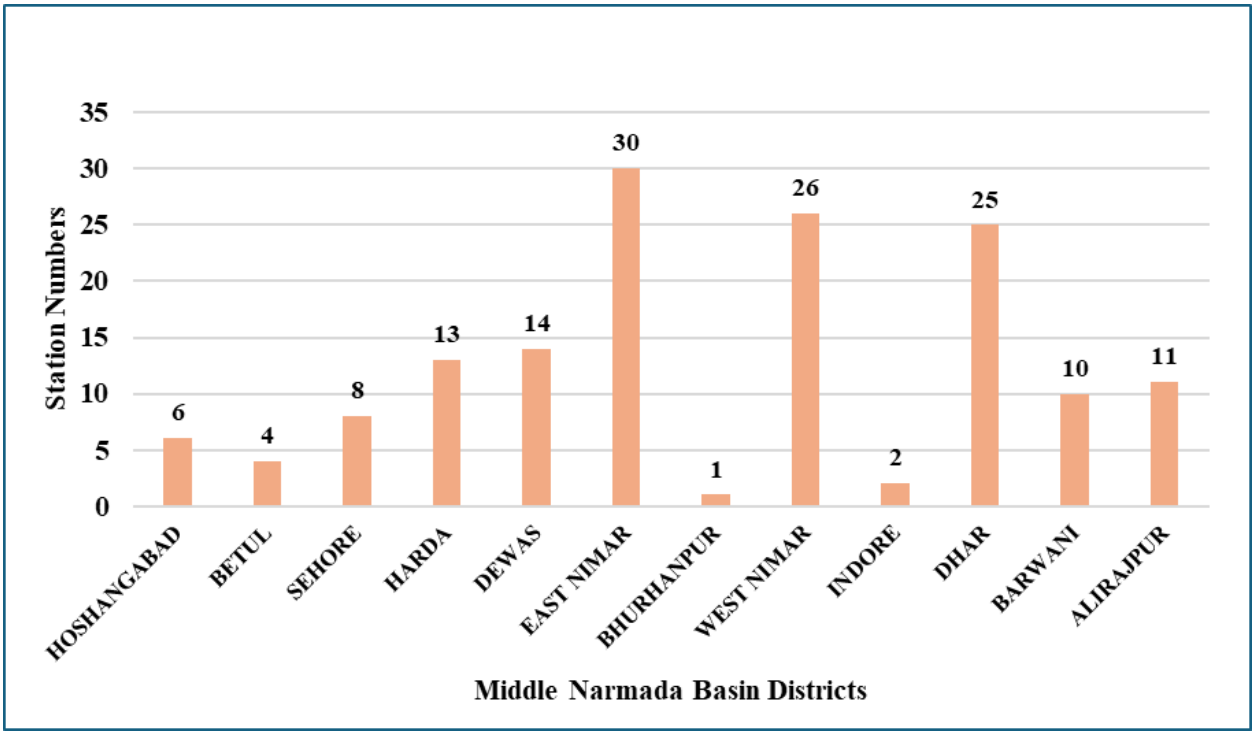
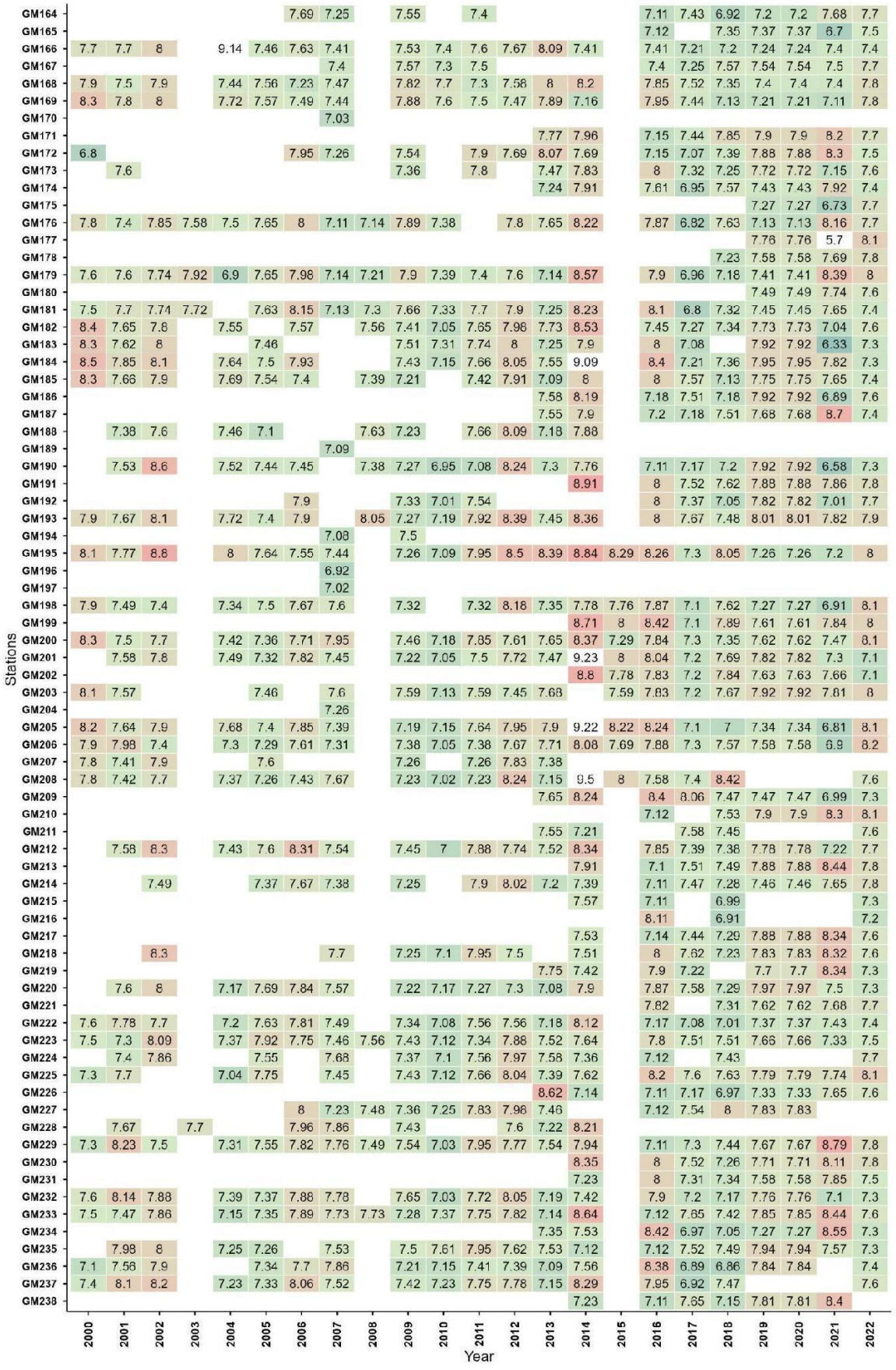


Figure 26: District-wise distribution of sampling stations in the Middle Narmada Basin.

i) pH

The pH data for all sampling stations in the Middle Narmada Basin are available for the period from 2000 to 2022, and yearly average values of all 23 years across different sampling stations are shown in the heat map.

The highest average value of all 22 years for pH values across these stations ranges from 6.92 to 8.12, which falls within the BIS permissible limit of **6.5 to 8.5**. A noticeable increase in pH values was observed at several stations, exceeding the upper limit of 8.5. The maximum pH values were recorded at Stations GM208 (9.5, Pipri) and GM256 (9.43, Sawda), as well as at several other stations, with maximum values ranging from 8.89 to 9.5 during the period. The pH yearly average values are shown in the Heat map Figure 27.


Stations
Year
pH
9
8
7
6
GM164 7.69 7.25 7.55 7.4 7.11 7.43 6.92 7.2 7.2 7.68 7.7
GM165 7.12 7.35 7.37 7.37 6.7 7.5
GM166 7.7 7.7 8 9.14 7.46 7.63 7.41 7.53 7.4 7.6 7.67 8.09 7.41 7.41 7.21 7.2 7.24 7.24 7.4 7.4
GM167 7.4 7.57 7.3 7.5 7.4 7.25 7.57 7.54 7.54 7.5 7.7
GM168 7.9 7.5 7.9 7.44 7.56 7.23 7.47 7.82 7.7 7.3 7.58 8 8.2 7.85 7.52 7.35 7.4 7.4 7.4 7.8
GM169 8.3 7.8 8 7.72 7.57 7.49 7.44 7.88 7.6 7.5 7.47 7.89 7.16 7.95 7.44 7.13 7.21 7.21 7.11 7.8
GM170 7.03
GM171 7.77 7.96 7.15 7.44 7.85 7.9 7.9 8.2 7.7
GM172 6.8 7.95 7.26 7.54 7.9 7.69 8.07 7.69 7.15 7.07 7.39 7.88 7.88 8.3 7.5
GM173 7.6 7.36 7.8 7.47 7.83 8 7.32 7.25 7.72 7.72 7.15 7.6
GM174 7.24 7.91 7.61 6.95 7.57 7.43 7.43 7.92 7.4
GM175 7.27 7.27 6.73 7.7
GM176 7.8 7.4 7.85 7.58 7.5 7.65 8 7.11 7.14 7.89 7.38 7.8 7.65 8.22 7.87 6.82 7.63 7.13 7.13 8.16 7.7
GM177 7.76 7.76 5.7 8.1
GM178 7.23 7.58 7.58 7.69 7.8
GM179 7.6 7.6 7.74 7.92 6.9 7.65 7.98 7.14 7.21 7.9 7.39 7.4 7.6 7.14 8.57 7.9 6.96 7.18 7.41 7.41 8.39 8
GM180 7.49 7.49 7.74 7.6
GM181 7.5 7.7 7.74 7.72 7.63 8.15 7.13 7.3 7.66 7.33 7.7 7.9 7.25 8.23 8.1 6.8 7.32 7.45 7.45 7.65 7.4
GM182 8.4 7.65 7.8 7.55 7.57 7.56 7.41 7.05 7.65 7.98 7.73 8.53 7.45 7.27 7.34 7.73 7.73 7.04 7.6
GM183 8.3 7.62 8 7.46 7.51 7.31 7.74 8 7.25 7.9 8 7.08 7.92 7.92 6.33 7.3
GM184 8.5 7.85 8.1 7.64 7.5 7.93 7.43 7.15 7.66 8.05 7.55 9.09 8.4 7.21 7.36 7.95 7.95 7.82 7.3
GM185 8.3 7.66 7.9 7.69 7.54 7.4 7.39 7.21 7.42 7.91 7.09 8 8 7.57 7.13 7.75 7.75 7.65 7.4
GM186 7.58 8.19 7.18 7.51 7.18 7.92 7.92 6.89 7.6
GM187 7.55 7.9 7.2 7.18 7.51 7.68 7.68 8.7 7.4
GM188 7.38 7.6 7.46 7.1 7.63 7.23 7.66 8.09 7.18 7.88
GM189 7.09
GM190 7.53 8.6 7.52 7.44 7.45 7.38 7.27 6.95 7.08 8.24 7.3 7.76 7.11 7.17 7.2 7.92 7.92 6.58 7.3
GM191 8.91 8 7.52 7.62 7.88 7.88 7.86 7.8
GM192 7.9 7.33 7.01 7.54 8 7.37 7.05 7.82 7.82 7.01 7.7
GM193 7.9 7.67 8.1 7.72 7.4 7.9 8.05 7.27 7.19 7.92 8.39 7.45 8.36 8 7.67 7.48 8.01 8.01 7.82 7.9
GM194 7.08 7.5
GM195 8.1 7.77 8.8 8 7.64 7.55 7.44 7.26 7.09 7.95 8.5 8.39 8.84 8.29 8.26 7.3 8.05 7.26 7.26 7.2 8
GM196 6.92
GM197 7.02
GM198 7.9 7.49 7.4 7.34 7.5 7.67 7.6 7.32 7.32 8.18 7.35 7.78 7.76 7.87 7.1 7.62 7.27 7.27 6.91 8.1
GM199 8.71 8 8.42 7.1 7.89 7.61 7.61 7.84 8
GM200 8.3 7.5 7.7 7.42 7.36 7.71 7.95 7.46 7.18 7.85 7.61 7.65 8.37 7.29 7.84 7.3 7.35 7.62 7.62 7.47 8.1
GM201 7.58 7.8 7.49 7.32 7.82 7.45 7.22 7.05 7.5 7.72 7.47 9.23 8 8.04 7.2 7.69 7.82 7.82 7.3 7.1
GM202 8.8 7.78 7.83 7.2 7.84 7.63 7.63 7.66 7.1
GM203 8.1 7.57 7.46 7.6 7.59 7.13 7.59 7.45 7.68 7.59 7.83 7.2 7.67 7.92 7.92 7.81 8
GM204 7.26
GM205 8.2 7.64 7.9 7.68 7.4 7.85 7.39 7.19 7.15 7.64 7.95 7.9 9.22 8.22 8.24 7.1 7 7.34 7.34 6.81 8.1
GM206 7.9 7.98 7.4 7.3 7.29 7.61 7.31 7.38 7.05 7.38 7.67 7.71 8.08 7.69 7.88 7.3 7.57 7.58 7.58 6.9 8.2
GM207 7.8 7.41 7.9 7.6 7.26 7.26 7.83 7.38
GM208 7.8 7.42 7.7 7.37 7.26 7.43 7.67 7.23 7.02 7.23 8.24 7.15 9.5 8 7.58 7.4 8.42 7.6
GM209 7.65 8.24 8.4 8.06 7.47 7.47 7.47 6.99 7.3
GM210 7.12 7.53 7.9 7.9 8.3 8.1
GM211 7.55 7.21 7.58 7.45 7.6
GM212 7.58 8.3 7.43 7.6 8.31 7.54 7.45 7 7.88 7.74 7.52 8.34 7.85 7.39 7.36 7.78 7.78 7.22 7.7
GM213 7.91 7.1 7.51 7.49 7.88 7.88 8.44 7.8
GM214 7.49 7.37 7.67 7.38 7.25 7.9 8.02 7.2 7.39 7.11 7.47 7.28 7.46 7.46 7.65 7.8
GM215 7.57 7.11 6.99 7.3
GM216 8.11 6.91 7.2
GM217 7.53 7.14 7.44 7.29 7.88 7.88 8.34 7.6
GM218 8.3 7.7 7.25 7.1 7.95 7.5 7.51 8 7.62 7.23 7.83 7.83 8.32 7.6
GM219 7.75 7.42 7.9 7.22 7.7 7.7 8.34 7.3
GM220 7.6 8 7.17 7.69 7.84 7.57 7.22 7.17 7.27 7.3 7.08 7.9 7.87 7.58 7.29 7.97 7.97 7.5 7.3
GM221 7.82 7.31 7.62 7.62 7.68 7.7
GM222 7.6 7.78 7.7 7.2 7.63 7.81 7.49 7.34 7.08 7.56 7.56 7.18 8.12 7.17 7.08 7.01 7.37 7.37 7.43 7.4
GM223 7.5 7.3 8.09 7.37 7.92 7.75 7.46 7.56 7.43 7.12 7.34 7.88 7.52 7.64 7.8 7.51 7.51 7.66 7.66 7.33 7.5
GM224 7.4 7.86 7.55 7.68 7.37 7.1 7.56 7.97 7.58 7.36 7.12 7.43 7.7
GM225 7.3 7.7 7.04 7.75 7.45 7.43 7.12 7.66 8.04 7.39 7.62 8.2 7.6 7.63 7.79 7.79 7.74 8.1
GM226 8.62 7.14 7.11 7.17 6.97 7.33 7.33 7.65 7.6
GM227 8 7.23 7.48 7.36 7.25 7.83 7.98 7.46 7.12 7.54 8 7.83 7.83
GM228 7.67 7.7 7.96 7.86 7.43 7.6 7.22 8.21
GM229 7.3 8.23 7.5 7.31 7.55 7.82 7.76 7.49 7.54 7.03 7.95 7.77 7.54 7.94 7.11 7.3 7.44 7.67 7.67 8.79 7.8
GM230 8.35 8 7.52 7.26 7.71 7.71 8.11 7.8
GM231 7.23 8 7.31 7.34 7.58 7.58 7.85 7.5
GM232 7.6 8.14 7.88 7.39 7.37 7.88 7.78 7.65 7.03 7.72 8.05 7.19 7.42 7.9 7.2 7.17 7.76 7.76 7.1 7.3
GM233 7.5 7.47 7.86 7.15 7.35 7.89 7.73 7.73 7.28 7.37 7.75 7.82 7.14 8.64 7.12 7.65 7.42 7.85 7.85 8.44 7.6
GM234 7.35 7.53 8.42 6.97 7.05 7.27 7.27 8.55 7.3
GM235 7.98 8 7.25 7.26 7.53 7.5 7.61 7.95 7.62 7.53 7.12 7.12 7.52 7.49 7.94 7.94 7.57 7.3
GM236 7.1 7.56 7.9 7.34 7.7 7.86 7.21 7.15 7.41 7.39 7.09 7.56 8.38 6.89 6.86 7.84 7.84 7.4
GM237 7.4 8.1 8.2 7.23 7.33 8.06 7.52 7.42 7.23 7.75 7.78 7.15 8.29 7.95 6.92 7.47 7.6
GM238 7.23 7.11 7.65 7.15 7.81 7.81 8.4
2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022
Year

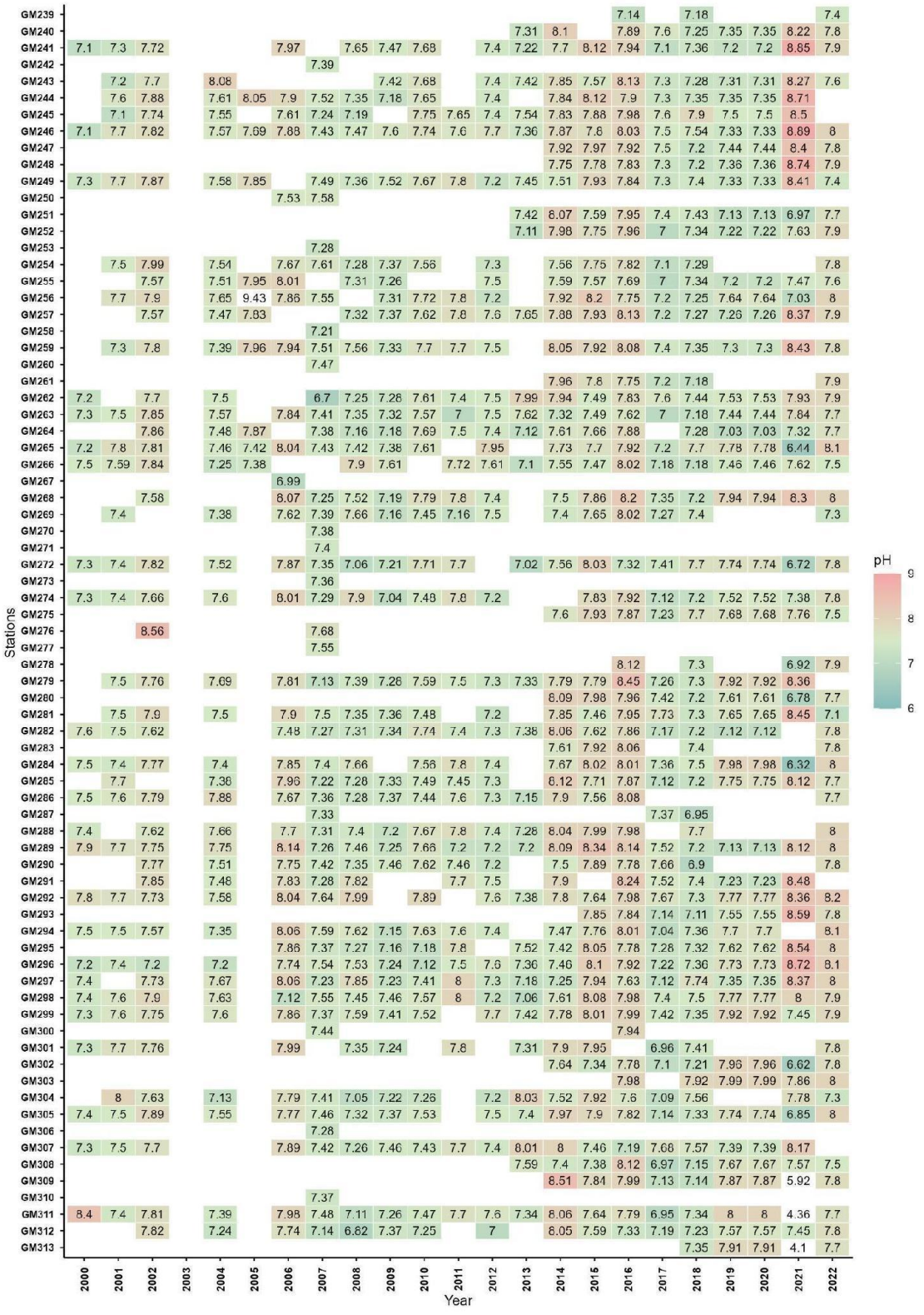


Figure 27: pH yearly average of groundwater across all sampling stations of 23 years of the Middle Narmada Basin.

ii) Chloride

The BIS standard limit for chloride is between 250 and 1000mg/L. We have collected and analyzed data over the years (2001-2022) for all sampling stations, showing that the highest average value for all years for chloride is 500 mg/L, which was recorded at Station GM304 (Nanpur). The maximum chloride value, 1026 mg//L, was observed at Station GM238 (Borgaon Buzurg). All the chloride average values for all years and stations are within the permissible limits for the Middle Narmada Basin, as shown in Figure 28.

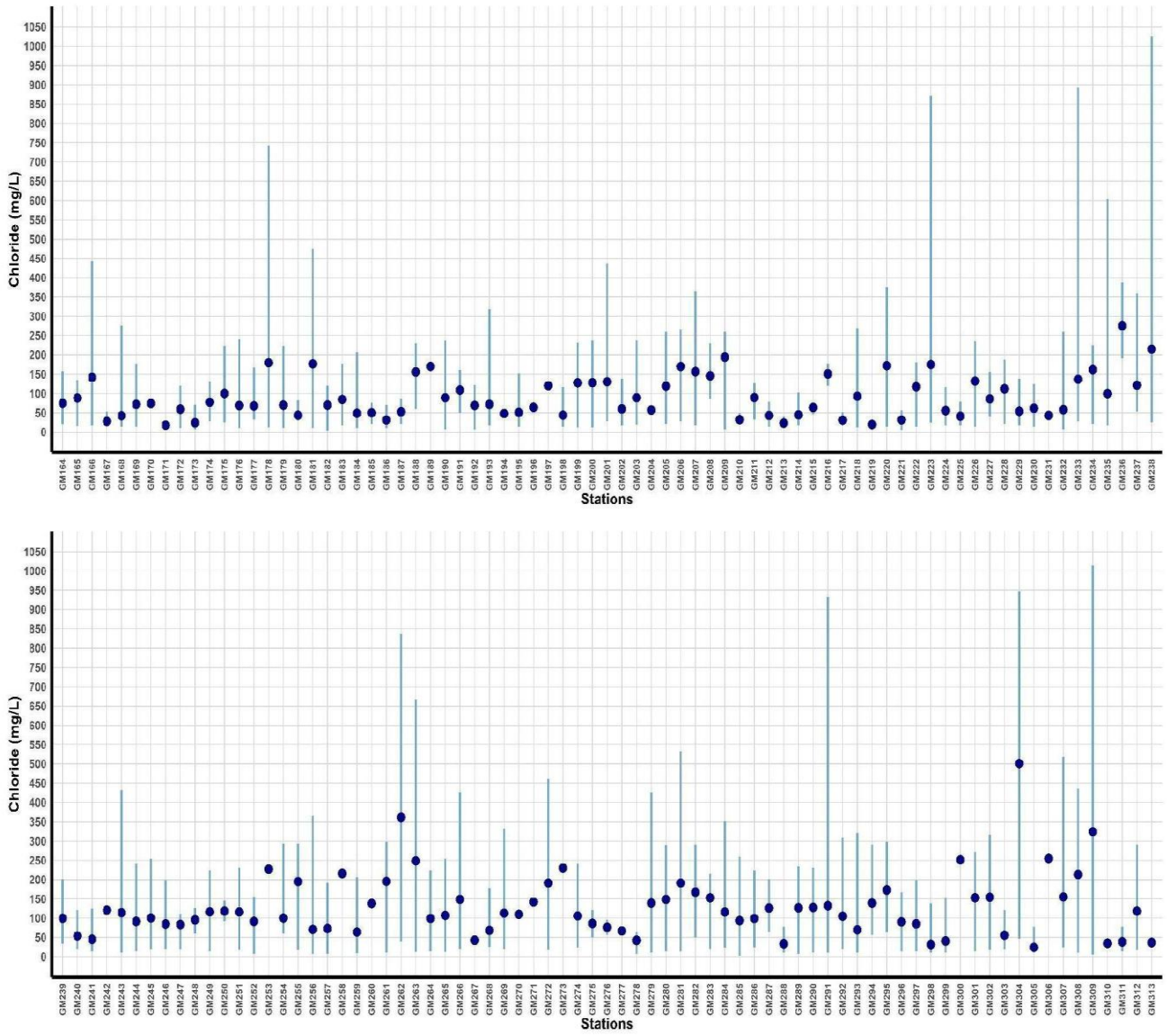


Figure 28: Average, Maximum, and Minimum values of 22 years of Chloride in groundwater across different sampling stations of the Middle Narmada Basin.

iii) Sulphate

The BIS standard limit for sulphate is 200-400mg/L. We have collected data and analyzed it from 2000 to 2022 across all sampling stations, showing that the highest average sulfate value

of all years, recorded at Station GM304 (Nanpur), was 183.9 mg/L. The maximum sulphate value observed was 650 mg/L at Station GM166 (Sanwalkhera) and 541 mg/L at Station GM243 (Ghatti). In all other years and stations, the sulphate average value of all years did not exceed the upper limit, as shown in Figure 29.

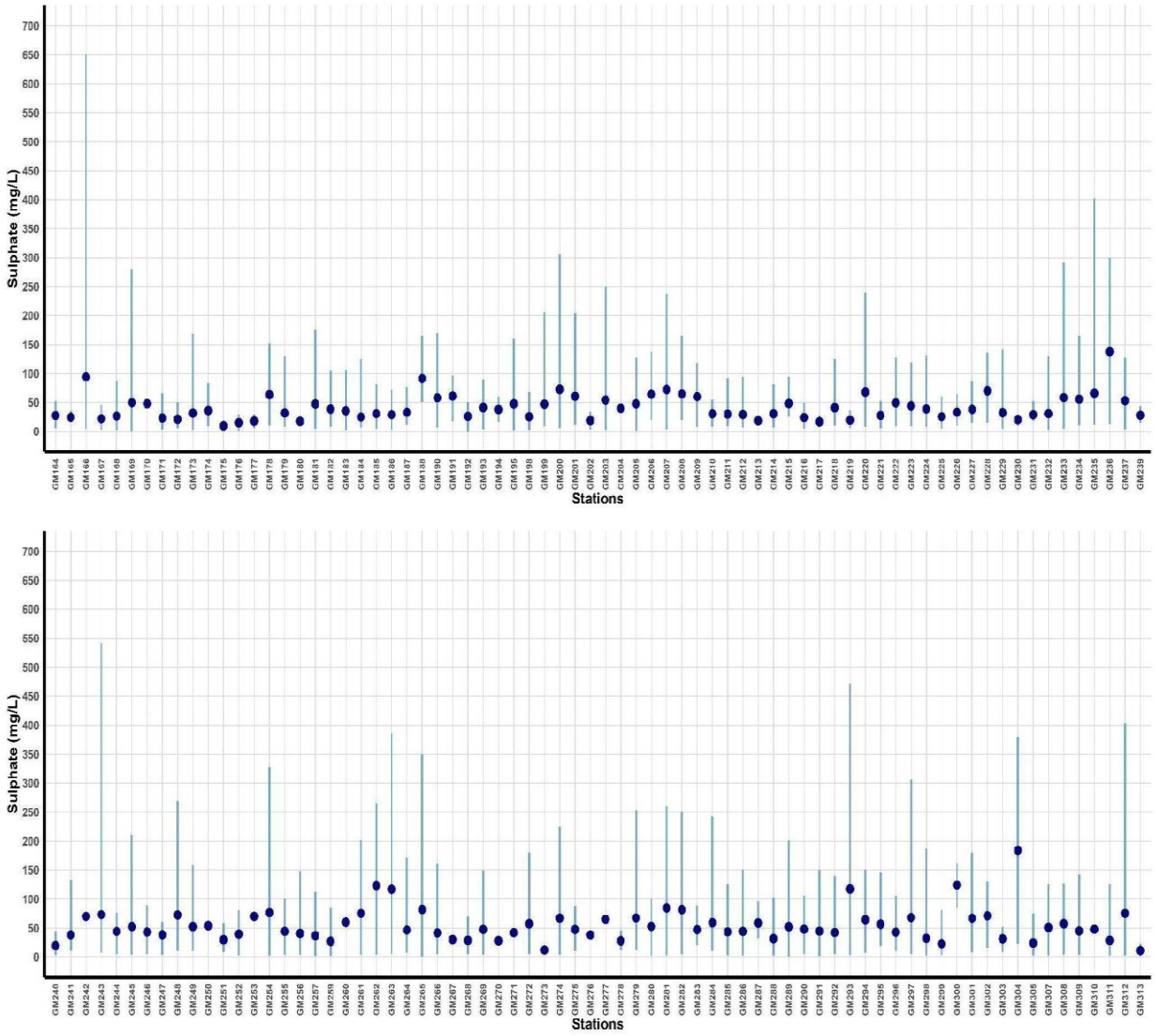


Figure 29: Average, Maximum, and Minimum values of all 20 years of Sulphate in groundwater across different sampling stations of the Middle Narmada Basin.

iv) Fluoride

The BIS standard limit for fluoride is 1-1.5 mg/L. The data for fluoride was collected over (2000-2022) for all sampling stations. The highest average fluoride value of all years was observed at 1.55 mg/L at Station GM193 (Handia), which falls within the standard limits. A maximum value of 2.8 mg/L was recorded at station GM249 (Bhikangaon). The average of all

years across all sampling stations is within the permissible limit, as shown in the Figure 30.

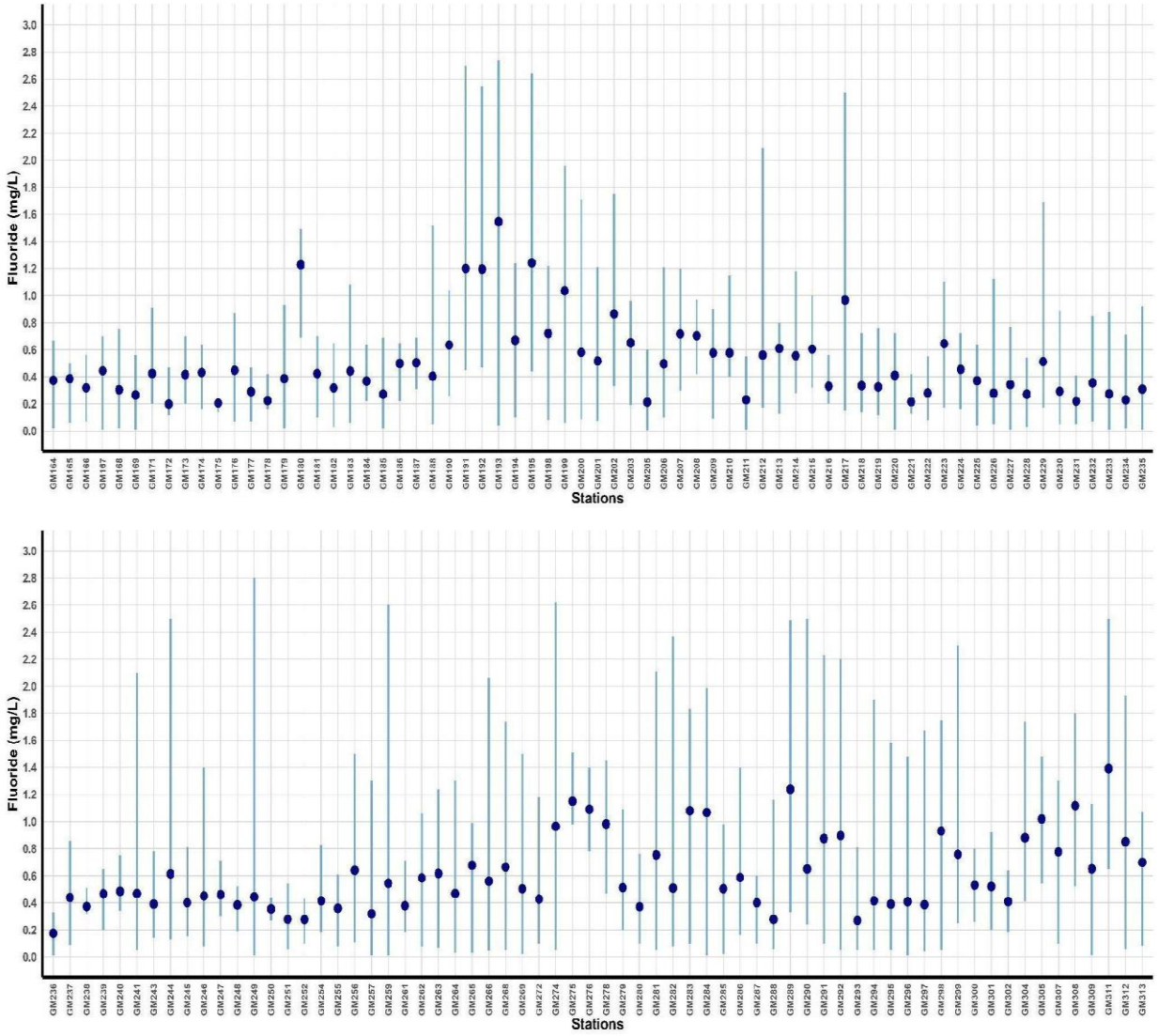


Figure 30: Average, Maximum, and Minimum values of all 23 years of fluoride in groundwater across different sampling stations of the Middle Narmada Basin.

v) Total Hardness

The BIS standard limit for Total hardness is between 200 and 600mg/L. We have collected and analysed data over the years 2000-2022) years for all stations, which shows that the highest average value of all years for total hardness is 768 mg/L, recorded at Station GM262 (Pipliyabuzrug), and also Stations GM236 (Pandhana), GM304 (Nanpur) also show high average values in the same range as shown in Figure 31.

The maximum total hardness value ranges between 1800 and 700 mg/L. The highest value of 1800 mg/L was observed at Station GM291 (Tawlal) in 2004 and at Station GM262 (Pipliyabuzrug) in 2010.

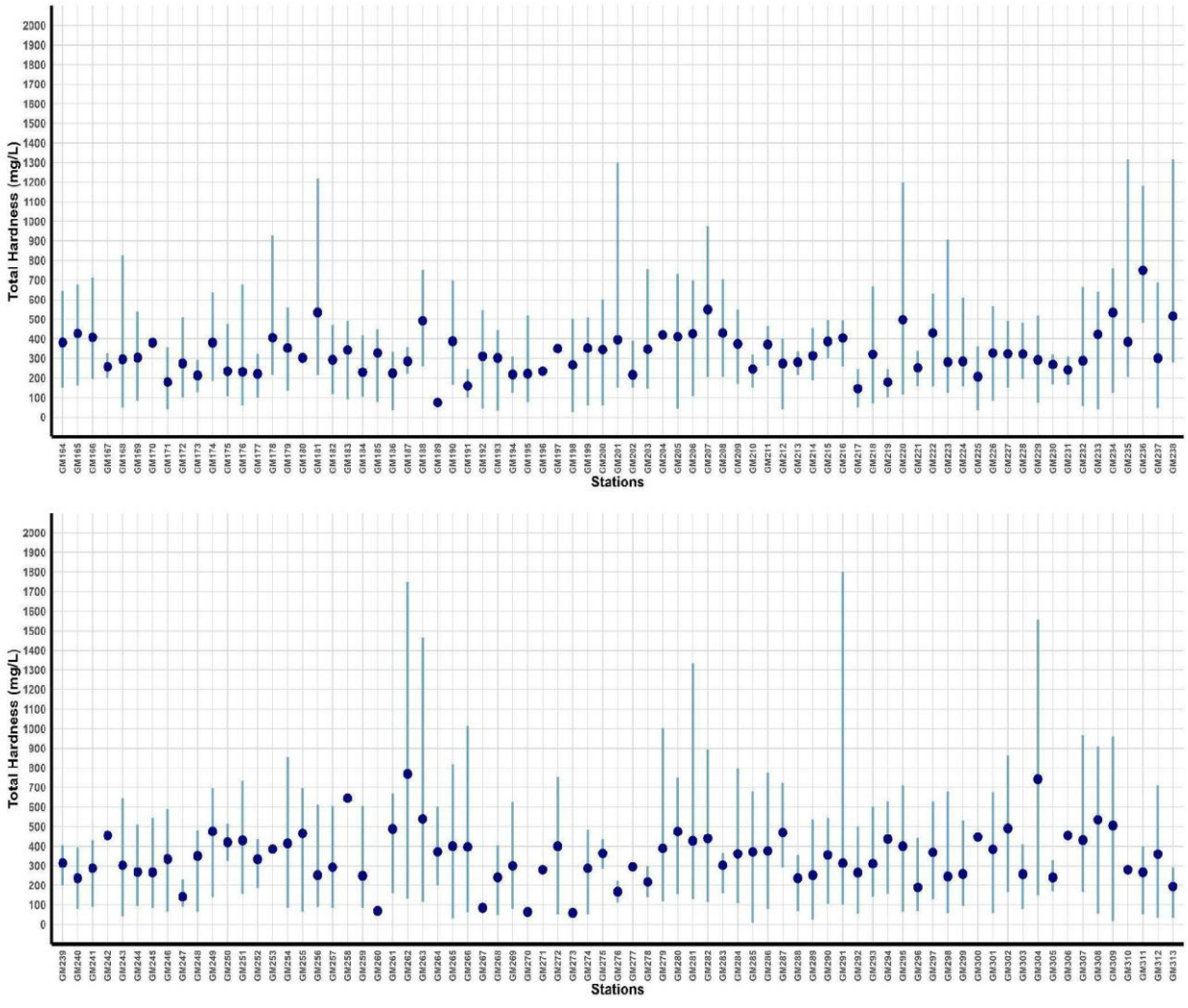


Figure 31: Average, Maximum, and Minimum values of all 21 years of Total Hardness in groundwater across different sampling stations of the Middle Narmada Basin.

vi) Total Alkalinity

The BIS standard limit for Total Alkalinity is between 200 and 600mg/L. We analyzed data from 2002 to 2015 for all stations, which shows that the highest average value for total alkalinity across all years was 457.87 mg/L, recorded at Station GM199 (Satwas New). The maximum total alkalinity value observed at Station GM166 (Sanwalkhera) was 600 mg/L and remained within the permissible limit during the entire period. The overall average values are shown in Figure 32.

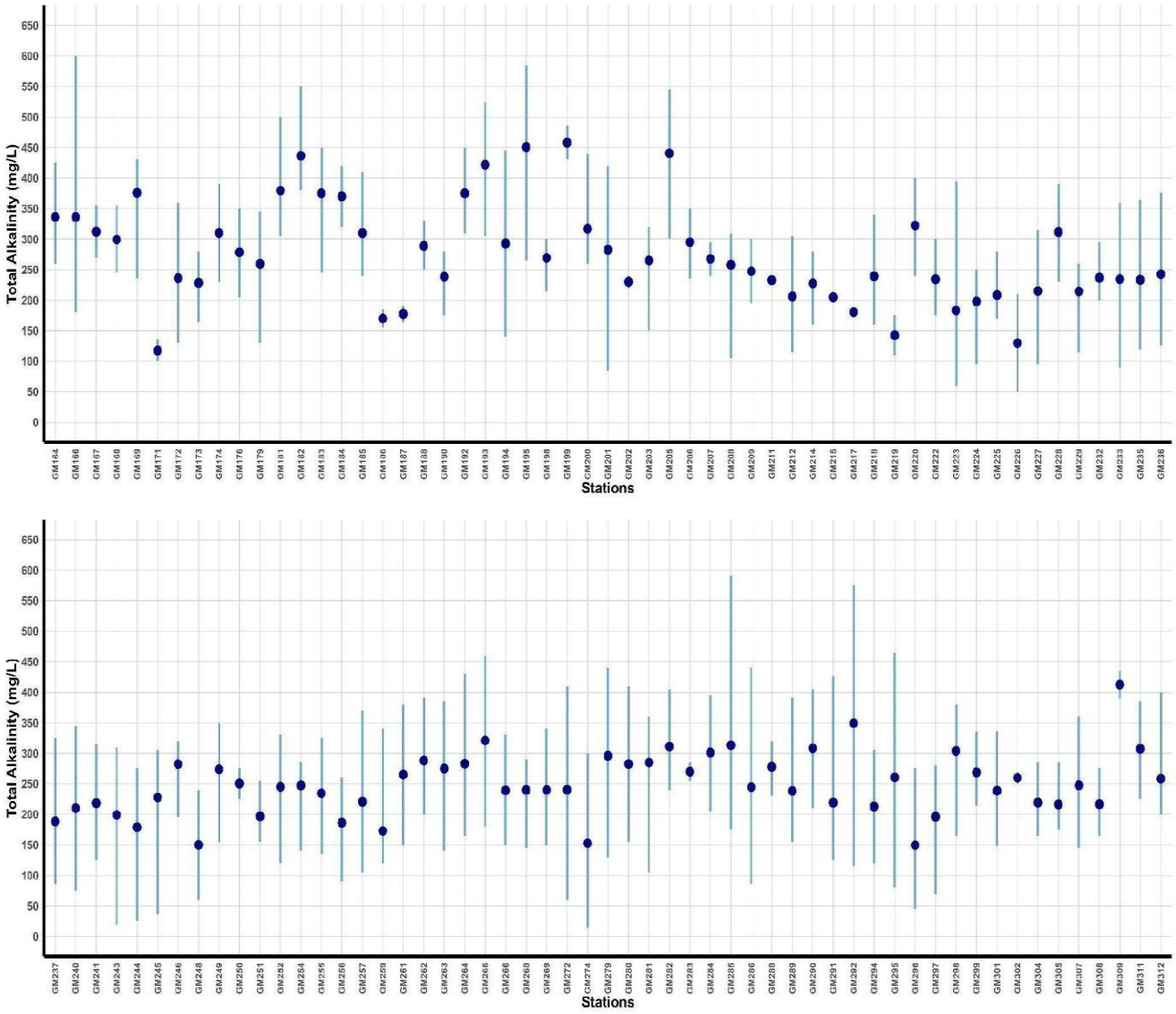


Figure 32: Average, Maximum, and Minimum values of all 12 years of Total Alkalinity in groundwater across different sampling stations of the Middle Narmada Basin.

vii) Silicon dioxide and Total Dissolved Solids

The SiO2 maximum value limit is 210 mg/L(Ref . 9), considered safe for drinking water, and the data collected from 2016 to 2022 for all stations lie within these limits. The maximum value

was observed at 62 mg/L at Station GM179 (Nadan).

Another parameter, Total Dissolved Solids, was also analyzed and shows that the highest value of the overall average across all years and stations was recorded at 1571 mg/L at Station GM178 (Bordi), which is within the range of the BIS standard, i.e., 500-2000 mg/L. The maximum value of 2471 was also observed at the same station. TDS values across all sampling stations for all years are within the limits, i.e. 500-2000mg/L.

viii) Nitrate

The BIS standard limit for Nitrate Nitrogen is set at < 45mg/L. The data collected and analysed over (2000-2022) across the stations. The highest average Nitrate value of all the years was observed at 198 mg/L at Station GM262 (Pipliyabuzrug), with a maximum value of 645 mg/L. Many of the stations located in the Middle Narmada Basin exhibited nitrate levels that are significantly elevated and exceeded the established limits.

Data Interpretation and Discussion

In this report, 9 years of data (2016-2025) have been analyzed, showing that most parameters are within the standard limits for drinking water.

After analysing the physical parameters, it can be considered that water is safe for drinking except for the stations which have extremely high turbidity values and most of them lies in the district of Sehore and Narmadapuram, which requires more ground- truthing and detail water sample analysis, so that exact reason can be identified for the high turbidity values during monsoon season in these areas.

All the chemical parameters of the Narmada River are within acceptable ranges, indicating good river health for both safe drinking water and aquatic life, which is essential for maintaining the overall health of the river. As we can see, there is a consistent concentration of dissolved oxygen, which further indicates the healthy river and thriving aquatic life, as adequate oxygen levels are essential for maintaining the ecological balance of the river. This indicates that the river water quality is safe and satisfactory, and is classified as A.

Overall, the basic condition of the river water is good and stable in the Upper and Middle Narmada Basin. However, two parameters stand out as major concerns: turbidity and total coliforms. The values for both are very high and exceed the permissible limits. These values

are from the maximum stations located in the districts of Sehore and Narmadapuram, which exhibit a seasonal pattern. Turbidity values shoot high in the monsoon season, and Total coliforms in the post-monsoon season. These seasonal peaks may be due to natural processes, such as heavy rainfall, sediment, and agricultural runoff, as well as human influences, especially in the case of Total coliforms, which affect the river's health.

To understand and to solve the issues, more ground-truthing, regular sampling and analysis, and field visits should be done to identify the source of pollution. These stations can be considered Hotspots and require attention to improve river water quality through proper management practices, community awareness, and regular monitoring.

The analysis of groundwater data for both the Upper and Middle stations of the Narmada Basin reveals that most of the assessed physico-chemical parameters, such as TDS, SiO2, chloride, sulphate, and fluoride, as well as pH, remain within the permissible limits across the periods. This indicated that the groundwater generally maintains stable water quality conditions for these parameters. contributions. Overall, the river remains within safe limits for most parameters.

However, the analysis also identifies that nitrate and hardness levels show an increase in the values at some sampling stations. These elevations raise concerns that there may be a potential influence from agricultural runoff, as well as organic and inorganic pollutants.

Additionally, natural geological factors could also play a role in altering these concentrations. Overall, despite increases in these values at some stations, groundwater quality remains within safe limits for most parameters. These analyses underscore the importance of regular monitoring and assessment in safeguarding groundwater resources in the Narmada Basin.
There are various factors influence the quality of groundwater in the Middle Narmada Basin, Dhar district, with particular concern over high levels of Nitrates, Total hardness, Iron, and pH in the region. Furthermore, it has been noted that approximately 70.51% of the area in Dhar district has been affected in terms of groundwater quality. (Ref .1)

Another study indicated that the groundwater in the region is significantly affected by elevated concentrations of nitrate and hardness, attributed to a rise in multivalent cations. The study also noted that the water is categorized from hard to very hard, and approximately 25.49% of the state’s groundwater contains nitrate levels that exceeded the acceptable limit. (Ref 2)

5. Soil Profile of the Narmada Basin

In the Upper Narmada basin, soils are predominantly shallow black soils formed through the erosion of trap basalts. These soils contain smectite-rich clay minerals with high water-holding capacity; their expandable clay structure swells upon wetting, which restricts drainage. Organic matter content is typically below 5%. The black soil here is primarily in-situ or colluvial deposits and is often interspersed with patches of red, sandy, or lateritic soils. The soil profile is generally shallow, covering hilltops and plateau surfaces.

Red soil in this region originates from the intense chemical weathering of basalts, during which most minerals are leached out, except for oxides of silica, iron, and aluminum. Due to this strong leaching, these soils exhibit good drainage but are deficient in plant-nutrients essential for growth.

In the Middle Narmada basin, particularly across the Vindhyan and Satpura plateau, soils range from shallow to medium black types. Near Hoshangabad, thick layers of recent alluvium occur, producing highly fertile soils that support crops such as cotton, jowar, and wheat.

Toward the Lower Narmada basin, valley areas, and the southern plateau are dominated by medium-deep black soils, while the northern plateau features mixed red and black soil types. At the Narmada estuary, Pliocene formations occur alongside recent alluvium. These alluvial soils are largely sandy loams, well-drained, highly fertile, and capable of sustaining productive agriculture. (Ref.3).

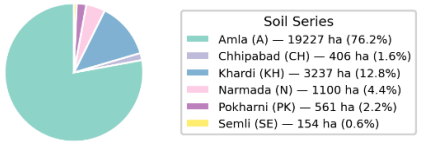
5.1 Soil series classification

The compiled datasets present a comprehensive, multi-decadal soil-series classification for the Narmada basin, retrived from detailed field surveys (1990–2025) done by Soil and Land Use Survey of India (SLUSI) across hundreds of micro- and sub-watersheds in the Narmada Basin (Ref.4). Surveys combined traditional topographic maps, cadastral plans, aerial/satellite imagery and high-resolution remote sensing to map soil types, extent and landscape features at scales suitable for watershed planning. The classifications capture the basin's major physiographic zones (western plateau, central plateau, and hill regions), diverse agro-climatic regimes, and fine-scale variability in series distribution information, which directly supports

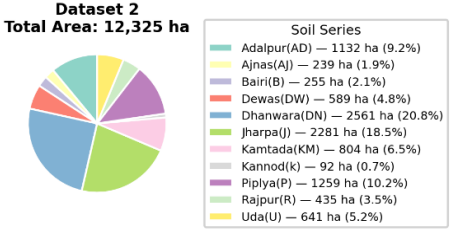
land-use planning, irrigation suitability, resource conservation, and targeted soil- and water-management interventions. Overall, the dataset forms a spatially explicit, policy-relevant foundation for integrated river-basin management and agricultural decision-making in the Narmada catchment. Annexure 5 A shows the dataset for each surveyed area, and Figure 33(Page1-4) shows the soil series classification of different surveyed areas.

Soil Series Distribution (Page 1/4)

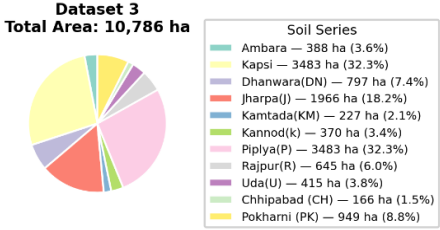
Dataset 1
Total Area: 25,245 ha



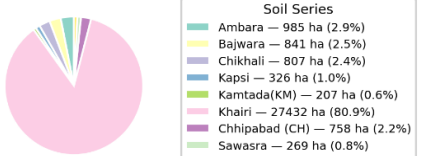
Dataset 2
Total Area: 12,325 ha



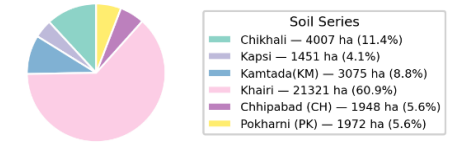
Dataset 3
Total Area: 10,786 ha



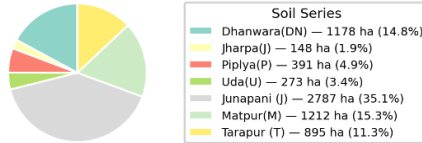
Dataset 4
Total Area: 33,918 ha



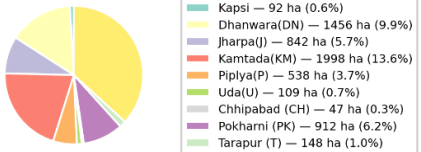
Dataset 5
Total Area: 35,005 ha



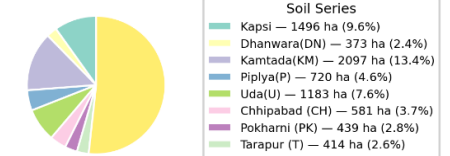
Dataset 6
Total Area: 7,940 ha



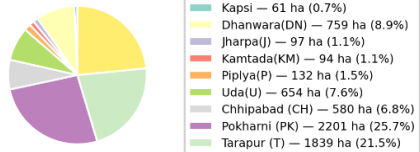
Dataset 7
Total Area: 14,738 ha



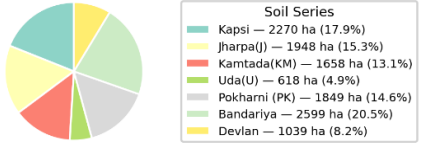
Dataset 8
Total Area: 15,663 ha



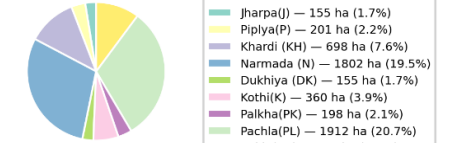
Dataset 9
Total Area: 8,555 ha



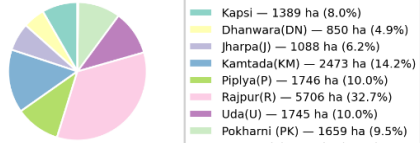
Dataset 10
Total Area: 12,697 ha



Dataset 11
Total Area: 9,231 ha

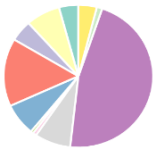


Dataset 12
Total Area: 17,449 ha



Soil Series Distribution (Page 2/4)

Dataset 13
Total Area: 10,695 ha



Soil Series
- Dhanwara(DN) — 423 ha (4.0%)
- Jharpa(J) — 750 ha (7.0%)
- Piplya(P) — 481 ha (4.5%)
- Khardi (KH) — 1551 ha (14.5%)
- Narmada (N) — 732 ha (6.8%)
- Bandariya — 60 ha (0.6%)
- Palkha(PK) — 78 ha (0.7%)
- Rekla(RK) — 785 ha (7.3%)
- Punsa(PN) — 4694 ha (43.9%)
- Sarnighat(S) — 119 ha (1.1%)
- Tikari(T) — 401 ha (3.7%)

Dataset 14
Total Area: 15,056 ha



Soil Series
- Kapsi — 3052 ha (20.3%)
- Dhanwara(DN) — 1143 ha (7.6%)
- Jharpa(J) — 1094 ha (7.3%)
- Kamtada(KM) — 1244 ha (8.3%)
- Piplya(P) — 174 ha (1.2%)
- Uda(U) — 627 ha (4.2%)
- Pokharni (PK) — 1305 ha (8.7%)
- Bandariya — 5472 ha (36.3%)

Dataset 15
Total Area: 12,879 ha



Soil Series
- Kapsi — 2200 ha (17.1%)
- Dhanwara(DN) — 497 ha (3.9%)
- Jharpa(J) — 355 ha (2.8%)
- Kamtada(KM) — 2599 ha (20.2%)
- Piplya(P) — 314 ha (2.4%)
- Uda(U) — 495 ha (3.8%)
- Pokharni (PK) — 1845 ha (14.3%)
- Bandariya — 4183 ha (32.5%)

Dataset 16
Total Area: 18,263 ha



Soil Series
- Kapsi — 2750 ha (15.1%)
- Dhanwara(DN) — 371 ha (2.0%)
- Jharpa(J) — 329 ha (1.8%)
- Kamtada(KM) — 1666 ha (9.1%)
- Piplya(P) — 882 ha (4.8%)
- Rajpur(R) — 1452 ha (8.0%)
- Uda(U) — 2205 ha (12.1%)
- Chhipabad (CH) — 288 ha (1.6%)
- Pokharni (PK) — 2401 ha (13.1%)
- Tarapur (T) — 377 ha (2.1%)
- Bandariya — 4995 ha (27.4%)

Dataset 17
Total Area: 983 ha



Soil Series
- Kapsi — 2235 ha (227.4%)
- Dhanwara(DN) — 569 ha (57.9%)
- Kamtada(KM) — 340 ha (34.6%)
- Piplya(P) — 1111 ha (113.0%)
- Rajpur(R) — 3340 ha (339.8%)
- Uda(U) — 199 ha (20.2%)
- Pokharni (PK) — 1731 ha (176.1%)

Dataset 18
Total Area: 13,469 ha



Soil Series
- Kapsi — 4332 ha (32.2%)
- Dhanwara(DN) — 425 ha (3.2%)
- Jharpa(J) — 101 ha (0.7%)
- Kamtada(KM) — 694 ha (5.2%)
- Piplya(P) — 492 ha (3.7%)
- Uda(U) — 659 ha (4.9%)
- Chhipabad (CH) — 89 ha (0.7%)
- Pokharni (PK) — 989 ha (7.3%)
- Tarapur (T) — 1000 ha (7.4%)
- Bandariya — 4223 ha (31.4%)

Dataset 19
Total Area: 17,270 ha



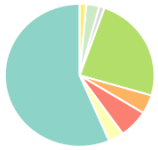
Soil Series
- Kapsi — 1308 ha (7.6%)
- Jharpa(J) — 13503 ha (78.2%)
- Uda(U) — 1909 ha (11.1%)

Dataset 20
Total Area: 8,871 ha



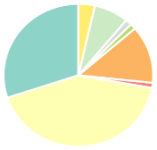
Soil Series
- Dhanwara(DN) — 757 ha (8.5%)
- Jharpa(J) — 3364 ha (37.9%)
- Kamtada(KM) — 661 ha (7.5%)
- Piplya(P) — 276 ha (3.1%)
- Uda(U) — 498 ha (5.6%)
- Tarapur (T) — 2529 ha (28.5%)

Dataset 21
Total Area: 7,591 ha



Soil Series
- Jharpa(J) — 3921 ha (51.7%)
- Kamtada(KM) — 230 ha (3.0%)
- Uda(U) — 444 ha (5.8%)
- Domanmau(D) — 298 ha (3.9%)
- Jamli(J) — 1679 ha (22.1%)
- Mandla(M) — 79 ha (1.0%)
- Pimpalghat(P) — 196 ha (2.6%)
- Rol gaon(R) — 98 ha (1.3%)

Dataset 22
Total Area: 17,717 ha



Soil Series
- Kapsi — 5139 ha (29.0%)
- Jharpa(J) — 7181 ha (40.5%)
- Kamtada(KM) — 193 ha (1.1%)
- Uda(U) — 2231 ha (12.6%)
- Pokharni (PK) — 214 ha (1.2%)
- Tarapur (T) — 232 ha (1.3%)
- Bandariya — 1274 ha (7.2%)
- Jamukhoh — 606 ha (3.4%)

Dataset 23
Total Area: 9,905 ha



Soil Series
- Kapsi — 1705 ha (17.2%)
- Dhanwara(DN) — 387 ha (3.9%)
- Jharpa(J) — 87 ha (0.9%)
- Kamtada(KM) — 182 ha (1.8%)
- Piplya(P) — 433 ha (4.4%)
- Uda(U) — 581 ha (5.9%)
- Chhipabad (CH) — 176 ha (1.8%)
- Pokharni (PK) — 2074 ha (20.9%)
- Tarapur (T) — 1469 ha (14.8%)
- Bandariya — 2391 ha (24.1%)

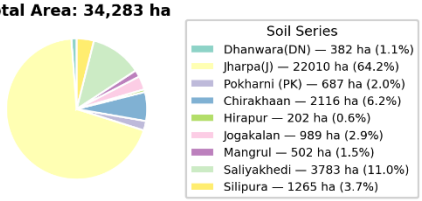
Dataset 24
Total Area: 12,232 ha



Soil Series
- Adalpur(AD) — 313 ha (2.6%)
- Ajnas(AJ) — 3062 ha (25.0%)
- Bairi(B) — 2509 ha (20.5%)
- Dewas(DW) — 921 ha (7.5%)
- Jharpa(J) — 3609 ha (29.5%)
- Uda(U) — 628 ha (5.1%)
- Relwan(RN) — 420 ha (3.4%)

Soil Series Distribution (Page 3/4)

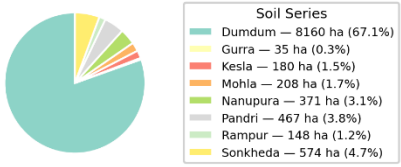
Dataset 25
Total Area: 34,283 ha



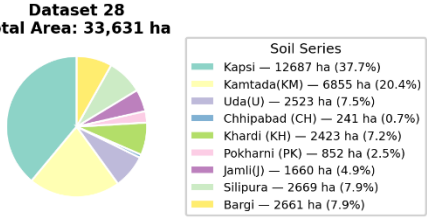
Dataset 26
Total Area: 10,553 ha



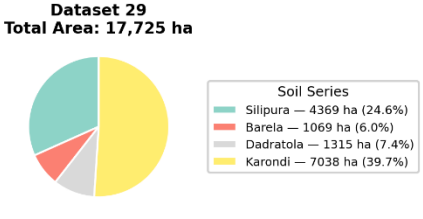
Dataset 27
Total Area: 12,155 ha



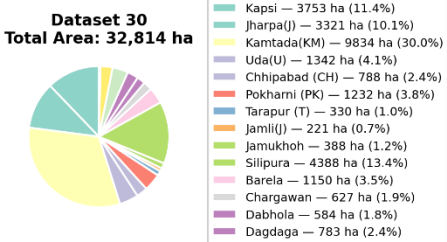
Dataset 28
Total Area: 33,631 ha



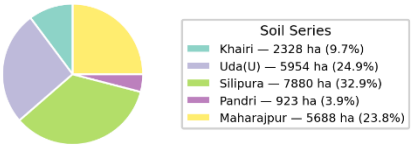
Dataset 29
Total Area: 17,725 ha



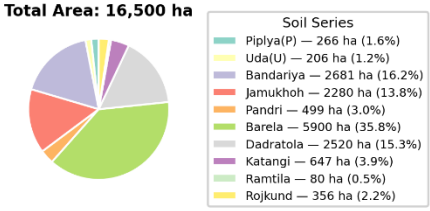
Dataset 30
Total Area: 32,814 ha



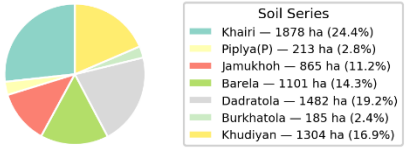
Dataset 31
Total Area: 23,947 ha



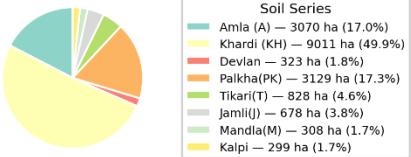
Dataset 32
Total Area: 16,500 ha



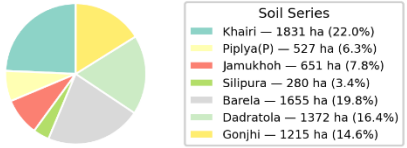
Dataset 33
Total Area: 7,704 ha



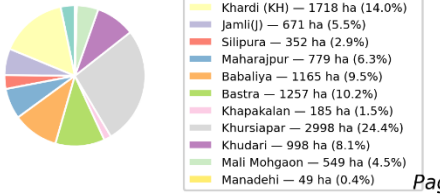
Dataset 34
Total Area: 18,062 ha



Dataset 35
Total Area: 8,341 ha



Dataset 36
Total Area: 12,270 ha



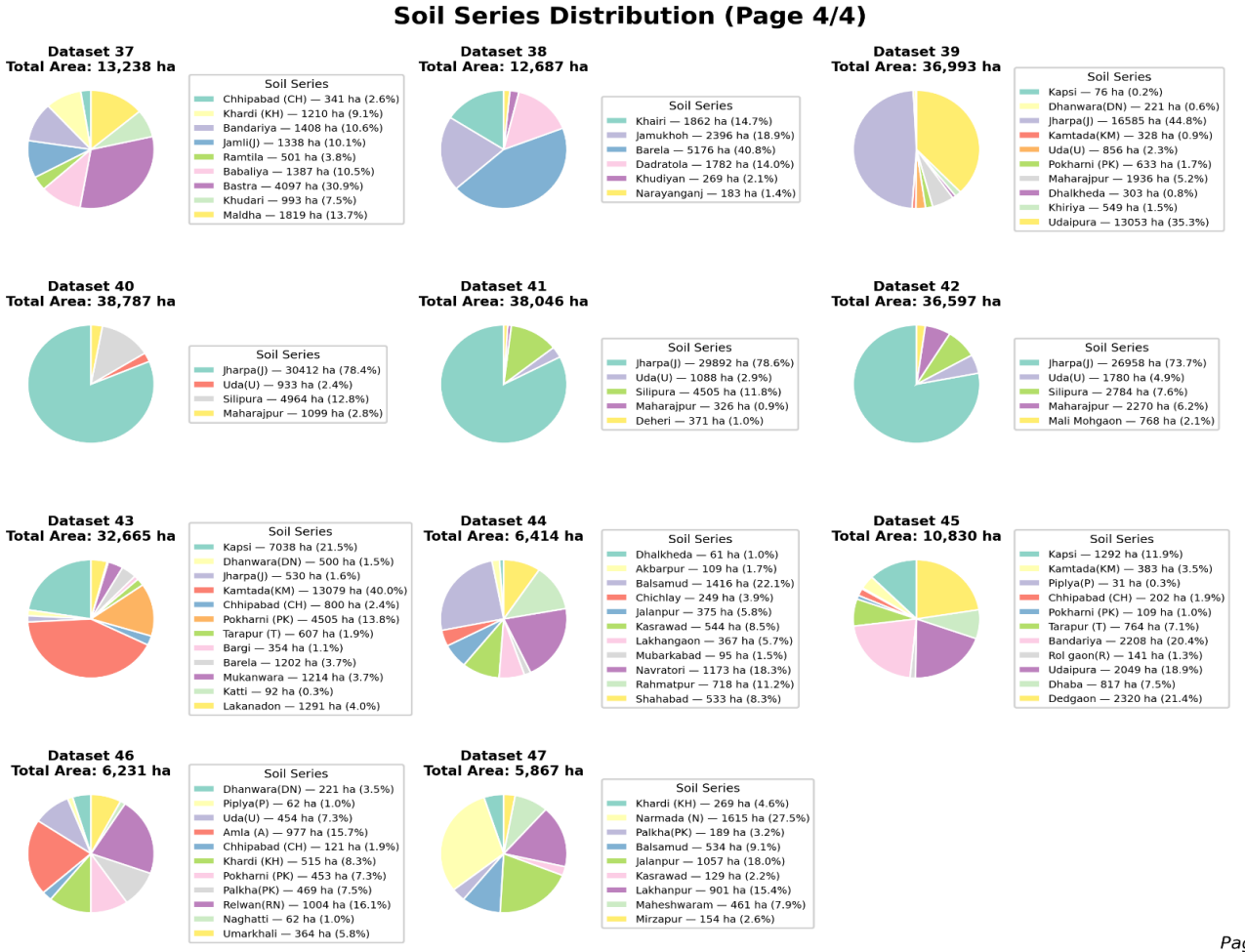


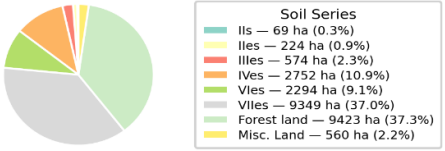
Figure 33 : Soil Series distribution for the different surveyed areas in the Basin.
Source - https://slusi.da.gov.in/dss/searchdss.html

5.2 Land Capability Classification

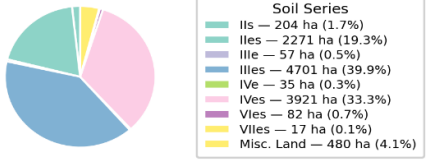
The land capability sub-class analysis of Datasets 1 to 47 as shown in Figure 34, which cover extensive micro- and sub-watersheds across the Narmada River basin, reveals a landscape shaped by diverse physiography, variations in soil depth, drainage conditions, and erosion intensities. Across the basin, Class III and Class IV lands dominate, especially the subclasses *IIIe, IIIs, IIIes, IVe,* and *IVes*, indicating widespread moderate to severe limitations due to erosion hazards, soil shallowness, and topographic constraints. These classes collectively represent the transitional agricultural zone of the basin, where cultivation is possible but requires significant soil and water conservation measures, contour farming, organic amendments, and erosion-control interventions. Class II soils, although present, are limited to relatively small pockets with deeper soil, gentler slopes, and lower erosion risks, reflecting more productive agro-zones in parts of Dewas, Harda, Sehore, and Barwani districts. Class V–VII soils appear intermittently, representing severely constrained lands that are often unsuitable for regular cropping. These areas are typically associated with stony outcrops, steep gradients, excessive runoff, or waterlogging tendencies, which tend to limit their utilization to forestry, grazing, watershed recharge interventions, or ecological restoration. The dataset also highlights substantial tracts of forest land and miscellaneous land, particularly in Mandla, Seoni, Jabalpur, Raisen, and Satpura-fringe regions, underscoring the basin's ecological richness and the presence of natural vegetation on fragile terrains. Together, the classification patterns across all 47 datasets illustrate a spatially heterogeneous basin in terms of agricultural capability, necessitating location-specific, conservation-oriented land-use planning. The results underscore the need to integrate land capability constraints into soil conservation strategies, watershed development programs, and sustainable river basin management frameworks for the Narmada catchment.

Land capability Sub-Class (Total Area) (Page 1/4)

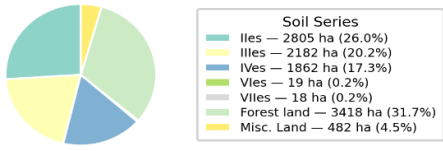
Dataset 1
Total Area: 25,245 ha



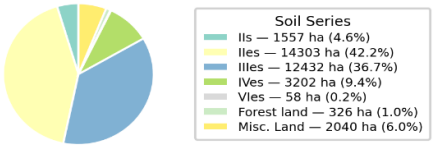
Dataset 2
Total Area: 11,768 ha



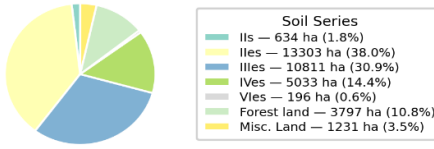
Dataset 3
Total Area: 10,786 ha



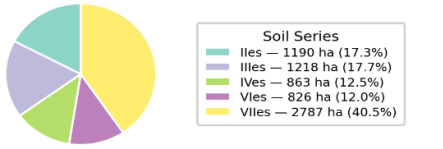
Dataset 4
Total Area: 33,918 ha



Dataset 5
Total Area: 35,005 ha



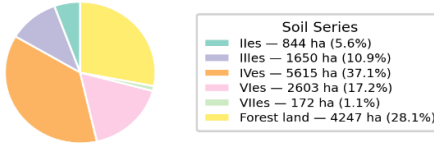
Dataset 6
Total Area: 6,884 ha



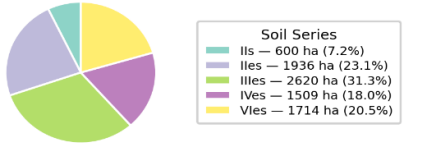
Dataset 7
Total Area: 14,738 ha



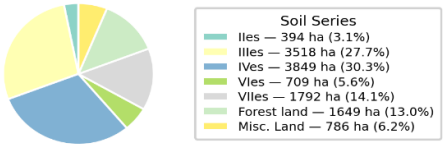
Dataset 8
Total Area: 15,131 ha



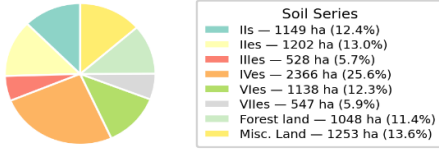
Dataset 9
Total Area: 8,379 ha



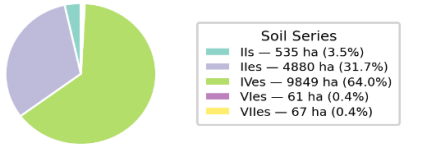
Dataset 10
Total Area: 12,697 ha



Dataset 11
Total Area: 9,231 ha



Dataset 12
Total Area: 15,392 ha

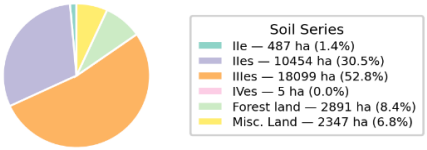


Land capability Sub-Class (Total Area) (Page 2/4)

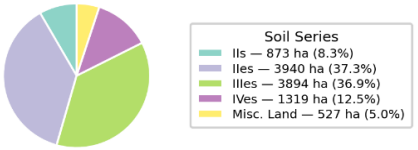
Dataset 13 Dataset 14 Dataset 15

Land capability Sub-Class (Total Area) (Page 3/4)

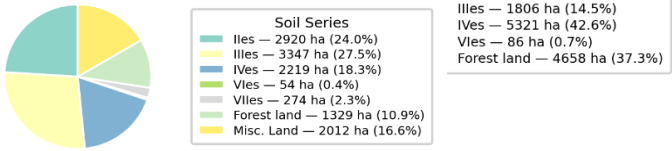
Dataset 25
Total Area: 34,283 ha



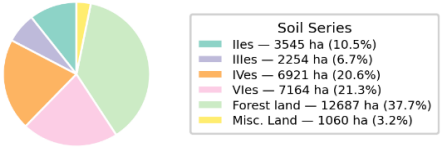
Dataset 26
Total Area: 10,553 ha



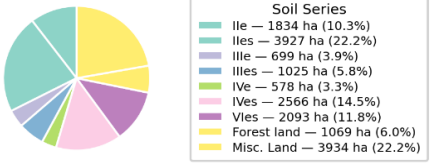
Dataset 27
Total Area: 12,155 ha



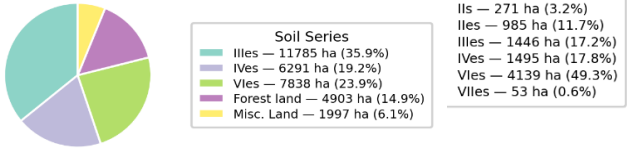
Dataset 28
Total Area: 33,631 ha



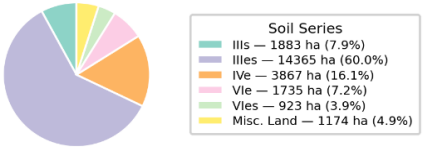
Dataset 29
Total Area: 17,725 ha



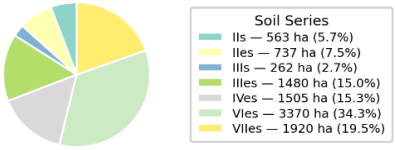
Dataset 30
Total Area: 32,814 ha



Dataset 31
Total Area: 23,947 ha



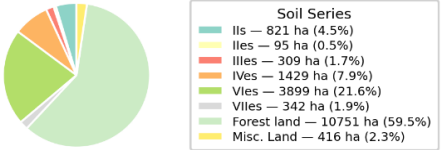
Dataset 32
Total Area: 9,837 ha



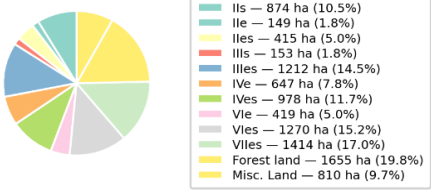
Dataset 33
Total Area: 7,704 ha



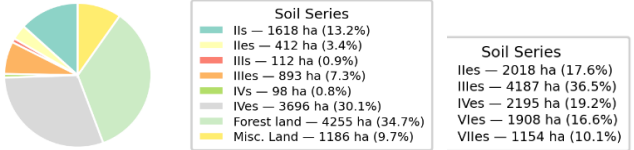
Dataset 34
Total Area: 18,062 ha



Dataset 35
Total Area: 8,341 ha



Dataset 36
Total Area: 12,270 ha



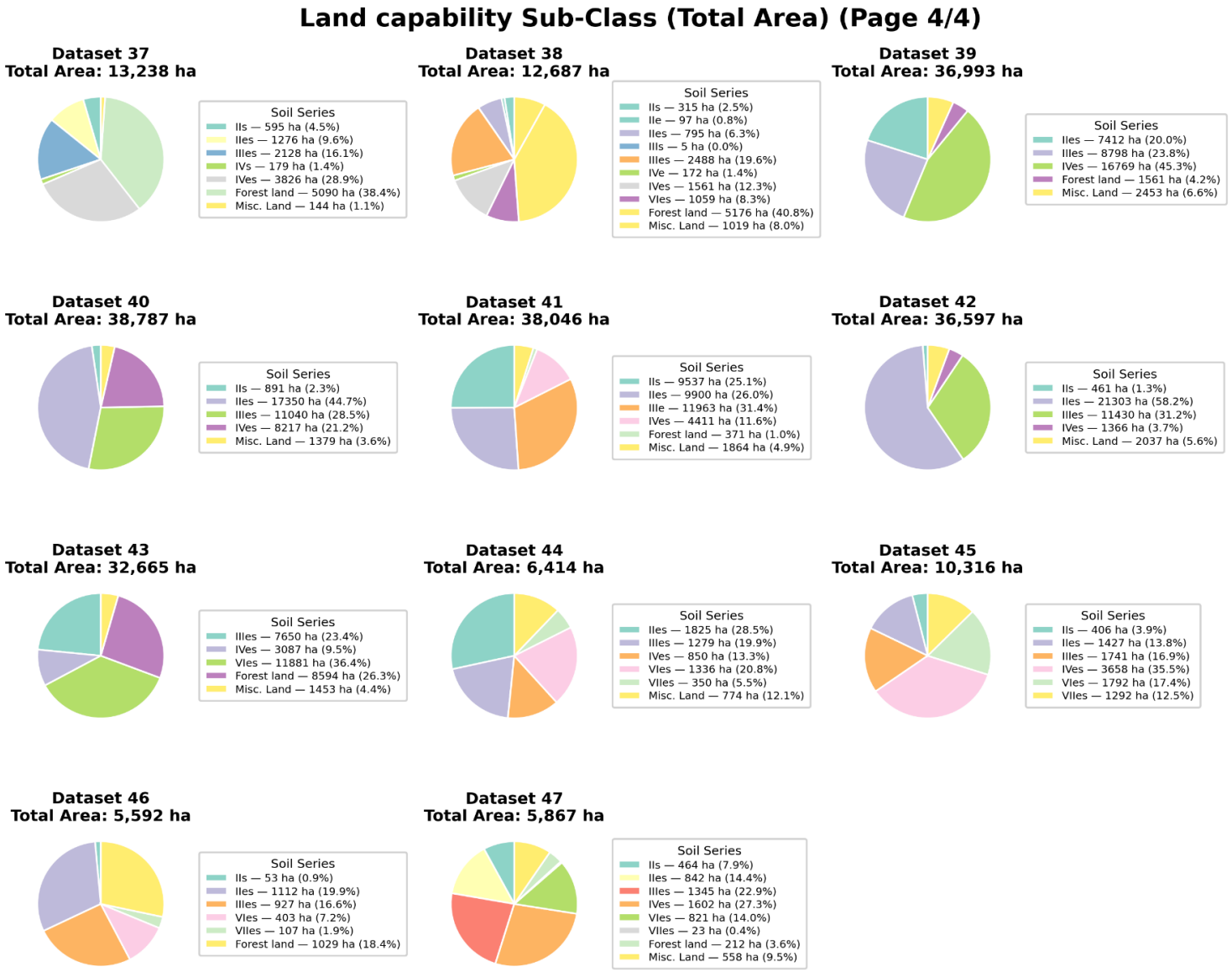


Figure 34: Land capability class of the surveyed area.

(Sourcehttps://slusi.da.gov.in/dss/searchdss.html)

5.3 Chemical Composition of Soils in the Narmada River Basin

The geochemical composition of soils across the Narmada River Basin reflects the dominance of Deccan basalt, with local contributions from alluvium and sedimentary formations. Data collected from The Geological Survey of India (GSI), OCBIS Portal (Ref.11) Four geochemical toposheets (55N/10, 55N/13, 54L/08, 55F/15–55J/03–55J/02) covering Narsinghpur, Seoni, Mandla, Jabalpur, Sagar, Vidisha, Betul, and Hoshangabad indicates consistent spatial trends in major oxides, trace metals, and rare earth elements (REEs). Data obtained from geochemical mapping sheets covering districts such as Narsinghpur, Seoni, Mandla, Jabalpur, Sagar, Vidisha, Betul, and Hoshangabad demonstrate consistent trends in major oxides, trace metals, and rare earth elements. These patterns reflect both the lithological diversity of the basin and the progressive weathering processes that shape its soil chemistry. The substantial variations in silica, iron, magnesium, and calcium oxides highlight the contrasting geological settings between the upper and middle basin regions, thereby providing crucial insight into the pedogenic evolution of Narmada soils.

i) Major Oxides

The major oxide composition of soils in the Narmada River Basin reflects the combined influence of Deccan basalt geology, weathering intensity, and d/s sediment dynamics. Silicon dioxide (SiO_2) shows substantial spatial variation, ranging from 40.13–50.31% in the upper basin districts of Narsinghpur, Seoni, Mandla, and Jabalpur, where basalt-derived soils dominate, and the silica content remains moderate due to limited quartz presence. Moving d/s into Betul and Hoshangabad, SiO_2 increases markedly to 57–67%, a trend that indicates stronger contributions from alluvial inputs, reworked sediments, and quartz-rich lithologies. This d/s enrichment in silica is typical of river basins where sediment sorting and mixing modify the original basaltic signature. Aluminium oxide (Al_2O_3) values consistently fall within 13.7–21.04%, reflecting the widespread formation of secondary clay minerals such as kaolinite and illite, produced through prolonged chemical weathering of feldspar-rich basalts.

A defining feature of the basin's soil chemistry is the abundance of iron oxide (Fe_2O_3), which occurs in high concentrations across all surveyed districts 14.69–21.22% in the upper basin and 5–9% in the Betul–Hoshangabad region. These elevated Fe levels strongly affirm the basaltic parent material, as iron-bearing minerals like magnetite and hematite are abundant in Deccan

trap rocks. The high Fe_2O_3 content is also responsible for the reddish to dark brown coloration commonly observed in these soils. Other oxides such as manganese (MnO: 0.20–0.34%), magnesium (MgO: 1.19–2.71%), and calcium (CaO: 1.26–5.20%) further underscore the mafic nature of the soil, indicating the presence of minerals like augite, olivine, and calcic plagioclase, which weather to release these oxides.

The concentrations of sodium oxide (Na_2O) and potassium oxide (K_2O), ranging from 0.15–2.07% and 0.19–1.31% respectively, signify the gradual breakdown of feldspar minerals, contributing important macro-nutrients that enhance soil fertility. Titanium oxide (TiO_2) occurs in notable quantities (1.52–4.25%), which is characteristic of Deccan basalts due to the presence of titanomagnetite and ilmenite. In contrast, phosphorus pentoxide (P_2O_5) remains very low (0.09–0.25%) across all locations, reflecting the limited availability of phosphate minerals within basaltic terrains and explaining the inherently low phosphorus fertility that often constrains agricultural productivity in these regions. Collectively, the major oxide profile of the Narmada Basin soils provides strong evidence of their basaltic heritage while also highlighting the d/smodifications introduced by sedimentary inputs and weathering processes.

ii) Trace Metals and Heavy Elements (ppm)

The distribution of trace metals and heavy elements across the Narmada River Basin provides strong evidence of the underlying basaltic and mafic geological framework, while also revealing subtle influences of sedimentary mixing in certain regions. Among the most diagnostic elements are chromium (Cr), nickel (Ni), and cobalt (Co), which occur in distinctly elevated concentrations Cr ranging from 266 to 699 ppm, Ni from 48 to 73 ppm, and Co from 6.1 to 26.83 ppm. Such elevated levels are typical of soils derived from ultramafic and mafic rock weathering, as these metals are commonly hosted in primary minerals like pyroxenes, olivine, and spinels. Their persistence in high concentrations across the basin reflects the chemical robustness of basalt-derived soils and confirms the strong geochemical imprint of the Deccan Traps, even in areas where weathering and sediment transport have modified the surface material.

Metals such as copper (Cu) and zinc (Zn) are present at concentrations typical of natural basalt-derived soils, with Cu ranging from 20 to 31 ppm and Zn from 160 to 322 ppm. These micronutrients play an essential role in soil fertility and plant nutrition, and their moderate to high presence indicates a healthy supply of geogenic micronutrients in most districts. Meanwhile, the concentrations of barium (Ba), strontium (Sr), and rubidium (Rb) show

significant spatial variability, with Ba ranging widely from 25 to 708 ppm, Sr from 59 to 196 ppm, and Rb from 30 to 40 ppm. The high values of Ba and Sr are indicative of plagioclase-rich lithology, as these elements commonly substitute for calcium in feldspar lattices. Their enhanced abundance in districts such as Sagar and Vidisha suggests the influence of sedimentary formations, particularly from the Vindhyan Supergroup, which contains carbonate-rich and feldspathic sequences. In contrast, the moderate Rb concentrations reflect minor inputs from felsic minerals, which may be introduced through alluvial processes or reworked sediments.

The basin also exhibits notable concentrations of High Field Strength Elements (HFSEs) such as yttrium (Y), zirconium (Zr), niobium (Nb), and hafnium (Hf). These elements occur at ranges typical of basaltic terrains, with Zr between 125 and 366 ppm, Nb between 214 and 414 ppm, Y between 14 and 31 ppm, and Hf between 4.86 and 8.29 ppm. HFSEs are highly resistant to weathering and are usually associated with accessory minerals like zircon, rutile, and titanite, which persist even in strongly weathered soils. Their distribution patterns across the Narmada basin indicate a dominant volcanic source, but the slightly elevated values in some middle-basin districts suggest a mixed volcanic–sedimentary signature, likely resulting from the incorporation of eroded sediments derived from non-basaltic formations. Overall, the trace metal and HFSE profiles reinforce the predominantly basaltic nature of the basin's soils while also highlighting localized lithological diversity arising from sedimentary contributions and alluvial reworking.

iii) Rare Earth Elements (REEs)

The Rare Earth Element (REE) composition of the Narmada River Basin soils provides an important geochemical fingerprint that further reinforces the dominant influence of Deccan basalt geology, while also revealing subtle variations associated with weathering intensity and sedimentary mixing. The concentrations of light REEs (LREEs) such as lanthanum (La: 10.54–24.96 ppm), cerium (Ce: 25.07–55.41 ppm), praseodymium (Pr: 3.12–7.02 ppm), and neodymium (Nd: 14.54–31.72 ppm) are consistently higher than those of heavy REEs (HREEs), which include elements such as ytterbium (Yb), lutetium (Lu), and trace quantities of samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), and erbium (Er). This LREE-enriched pattern is a defining characteristic of basalt-derived soils worldwide and is consistent with the behavior of REEs during magmatic differentiation, where LREEs tend to be concentrated in incompatible mineral phases. The presence of these elements

in stable concentrations across multiple districts underscores the uniform basaltic parentage of the basin's soils. Furthermore, the persistence of REEs in comparable proportions across the region indicates that these elements are largely housed within resistant accessory minerals such as zircon, apatite, allanite, and titanite, which do not readily break down during chemical weathering. Their stability allows them to retain the original geochemical signature of the parent rock even in areas where the soils have been subjected to intense tropical weathering or episodic alluvial reworking. Notably, the middle basin districts where mixed volcanic and sedimentary materials occur show only mild deviation in REE patterns, suggesting that sedimentary contributions, while present, do not significantly disturb the inherent basaltic REE profile. The preservation of Eu and Sm anomalies further supports the predominance of basalt-derived mineral phases, as these anomalies are characteristic of plagioclase-bearing mafic rocks.

Overall, the REE distribution across the Narmada Basin not only reaffirms the basaltic geochemical dominance but also highlights the capacity of REEs to serve as robust tracers of lithology and sediment provenance. Their relatively uniform concentration ranges limited anthropogenic influence, and enrichment in LREEs collectively validate the geological consistency of the basin while reflecting subtle spatial variations linked to weathering intensity and sedimentary admixture. This stable REE signature, when interpreted alongside major oxides and trace metals, provides a comprehensive understanding of the soil evolution and geochemical identity of the Narmada River Basin.

6. Lithological Profile of the Narmada River Basin

The lithological data is collected from the Geological Survey of India (GSI), OCBIS Portal for the Upper and Middle Narmada Basin. (Ref 11)

The Narmada River basin exhibits a highly diverse geological architecture shaped by multiple tectono-sedimentary cycles, making it one of India's most complex litho-physiographic regions. The upper and central basin is dominated by extensive exposures of the Gondwana Supergroup, represented by the Barakar, Motur, Bijori, Pachmarhi, Denwa, and Bagra formations. These formations consist primarily of medium- to coarse-grained feldspathic sandstones, conglomerates, red–green mudstones, clay lenses, and calcareous bands, reflecting repeated cycles of fluvial deposition within a braided to meandering river system developed in

a tectonically active pull-apart basin setting. The northern parts display thick, white, quartz-pebble-bearing sandstones of the Pachmarhi Formation, while the northwestern areas include the soft sandstones, clay-rich red beds, and calcareous mudstones of the Denwa Formation, overlain by the coarse, conglomeratic Bagra Formation interpreted to be alluvial-fan deposits formed due to rift-margin uplift and fault reactivation.

In the structural framework of the basin, the Narmada–Son Trough forms the principal tectonic lineament separating the Archaean and Bijawar crystalline basement from younger Vindhyan sedimentary sequences. The Vindhyan Supergroup contributes hard, horizontally bedded sandstones and shales, which form prominent escarpments and gorges along the valley, including the well-known marble gorge at Bheraghat. Further south, the basin is characterized by the Deccan Trap volcanic province, consisting of massive to vesicular basaltic lava flows with intertrappean limestone, sandstone, and clay horizons, commonly encountered in drilling records across Mandla, Betul, Hoshangabad, and Jhabua districts. These basalt flows form extensive flat-topped plateaus and ridges that dominate the Malwa and Satpura regions, frequently appearing at depths ranging from a few meters to over 200 m, depending on Quaternary cover thickness.

Large portions of the middle and lower basin are mantled by unconsolidated Quaternary alluvium, including sand, silt, clay, kankar, and gravel, deposited by the Narmada and its tributaries. Borehole lithologs consistently show alluvium extending to depths of 30–150 m before encountering underlying Gondwana, Vindhyan, Proterozoic, or Deccan Trap units. In several locations such as Mandla, Hoshangabad, Barwani, Jabalpur, and Narmadapuram the sub-alluvial lithology includes slates, phyllites, quartzites, dolomites, and granitic gneisses, indicating the presence of Archaean to Proterozoic crystalline basement beneath sedimentary and trap sequences. This diversity of lithologies reflects the long and complex evolutionary history of the basin, influenced by flexural subsidence, volcanic activity, basement faulting, and sustained fluvial processes, making the Narmada basin a geologically heterogeneous terrain with a distinct combination of volcanic, sedimentary, and crystalline rock assemblages.

6.1. Chemical composition of the Rocks of the Upper Narmada Basin

The chemical composition of rocks in the Upper Narmada Basin interpreted from NGCM geochemical datasets across geochemical toposheets 55N/10, 55N/13, 55N/01, 55N/11, 64B/07, and 64F/10 reveals a predominantly mafic to intermediate volcanic signature, shaped by the extensive spread of Deccan Trap basalts and interlayered Gondwana sedimentary

formations. These toposheets cover the key upper-basin districts of Narsinghpur, Seoni, Mandla, Jabalpur, Chhindwara, Balaghat, and Dindori, providing a comprehensive geochemical picture of the region. The dataset shows that these rocks are consistently enriched in iron, magnesium, calcium, titanium, chromium, nickel, and cobalt, highlighting their basaltic lineage. At the same time, the presence of silica-rich and alkali-bearing samples in specific locations reflects the influence of Gondwana sandstones, intertrappean sediments, and weathered Proterozoic units, producing a geochemically diverse yet volcanically dominated upper basin. High Field Strength Elements (HFSEs) and Rare Earth Element (REE) systematics further confirm a strong Deccan volcanic foundation with localized sedimentary overprints.

i) Major Oxides

The major oxide chemistry strongly supports the mafic volcanic origin of upper Narmada rocks. Silicon dioxide (SiO_2) varies widely from 39.67% to as high as 72.8% indicating both classic basaltic compositions (45–52%) and areas where weathered quartz-rich sediments or Gondwana formations elevate silica content above 60%. Mean SiO_2 values across Narsinghpur, Seoni, Mandla, and Jabalpur stabilize between 46–50%, while Mandla–Balaghat and Dindori include silica-rich samples up to 72%, representing sandstone intercalations or intense silica mobilisation. Aluminium oxide (Al_2O_3) ranges between 10–27%, with most values clustering around 14–17%, consistent with abundant feldspar and clay mineral presence. Iron oxide (Fe_2O_3) remains notably high 8–21% in most districts, with some samples exceeding 26%, reflecting the high Fe content of Deccan basalt and its derivative soils.

Magnesium oxide (MgO) exhibits broad variability, ranging from 0.7–8.8%, while Calcium oxide (CaO) spans 0.4–8.8%, highlighting the presence of augite, olivine, and plagioclase in fresh or moderately altered basalt flows. Even highly weathered rocks retain MgO–CaO signatures due to resistant mafic mineral remnants. Manganese oxide (MnO) remains low (0.2–0.6%) but stable across districts. Sodium (Na_2O, 0.13–1.89%) and Potassium oxides (K_2O, 0.17–1.89%) reflect feldspar weathering and are locally enriched in Mandla–Balaghat where Gondwana sandstones contribute to alkali release. Titanium dioxide (TiO_2) is strikingly high 1.5–4.5% a unique basaltic signature derived from titanomagnetite and ilmenite. Conversely, Phosphorus pentoxide (P_2O_5) remains consistently low (0.06–0.68%) across all districts, confirming the low natural apatite content typical of Deccan basalt terrains and explaining the poor intrinsic phosphorus fertility of these rocks.

ii) Trace Metals & Heavy Elements

Trace metal concentrations across upper Narmada rocks exhibit a deeply basaltic affinity. Chromium (Cr) displays extremely high levels 120–1281 ppm with Mandla, Balaghat, and Dindori presenting the highest values, often exceeding 1000 ppm, consistent with chromite-bearing basaltic minerals and less-altered mafic units. Nickel (Ni) ranges between 12–107 ppm, and Cobalt (Co) between 6–93 ppm, both diagnostic of olivine–pyroxene–rich basaltic rocks. These three transition metals together confirm the strong ultramafic–mafic influence in the upper basin geochemistry. Vanadium (V) also spans a wide range (14–169 ppm, occasionally up to 214 ppm), further characterising the mafic volcanic provenance.

Copper (Cu) (14–49 ppm) and Zinc (Zn) (60–322 ppm) fall within natural geogenic ranges, though some higher Zn values in Jabalpur and Dindori imply localized concentration due to secondary enrichment. Barium (Ba) shows substantial variability (25–1232 ppm), particularly high in Mandla–Balaghat and Dindori, suggesting inputs from feldspar-rich Gondwana sandstones or intertrappean calcareous beds. Strontium (Sr) (54–420 ppm) is persistently elevated, consistent with plagioclase-rich basaltic fabrics and carbonate-bearing intertrappean sediments. Rubidium (Rb) is moderate (12–106 ppm), indicating minor felsic or sedimentary input.

High Field Strength Elements (HFSEs) such as Zirconium (Zr) (75–712 ppm), Niobium (Nb) (33–167 ppm), Hafnium (Hf) (0.8–3.6 ppm), and Yttrium (Y) (8–76 ppm) confirm the presence of zircon, titanite, rutile, and other resistant accessory minerals, which persist even through deep tropical weathering. These HFSE distributions clearly reflect a mixed geological framework—dominantly volcanic, but with sedimentary and intertrappean contributions in regions like Mandla, Dindori, and Balaghat. Additionally, occasional detection of Lead (Pb) up to 54 ppm, Lithium (Li) up to 65 ppm, Arsenic (As) up to 11 ppm, and minor amounts of Antimony, Bismuth, Cadmium, and Silver show weak but detectable signatures of accessory mineralisation, though still within typical crustal limits.

iii) Rare Earth Elements (REEs)

The REE profile of upper Narmada rocks reinforces their strongly basaltic origin. Light REEs (LREEs) such as Lanthanum (La) (10–44 ppm), Cerium (Ce) (23–110 ppm), Praseodymium (Pr) (2–10 ppm), and Neodymium (Nd) (17–95 ppm) appear consistently enriched relative to Heavy REEs (HREEs) like Yb, Lu, Tb, and Dy. This LREE enrichment pattern is characteristic of Deccan Trap basalts and remains preserved even in strongly weathered terrains due to the

stability of REE-bearing accessory minerals. HREE concentrations remain low but stable (e.g., Yb 0.5–3.9 ppm, Lu 0.2–1.7 ppm), consistent with minimal heavy-REE host minerals.
Notably, the Mandla–Balaghat and Dindori regions show the highest REE variability, with Ce values occasionally exceeding 110 ppm and Nd reaching 94 ppm, suggesting localized enrichment due to retention in resistant phases like allanite, monazite, and zircon. Europium (Eu), Samarium (Sm), and Gadolinium (Gd) values show characteristic Deccan basalt anomalies, confirming the presence of plagioclase-rich mineral assemblages. Overall, the REE systematics clearly differentiate the upper basin as a dominantly basaltic geochemical province with modest sedimentary overprints and minimal anthropogenic disturbance.

6.2 Chemical composition of the rocks in the Middle Narmada basin

The chemical composition of the Middle Narmada Basin derived from NGCM datasets across Toposheet Nos. 54L/08, 46J/16, 46N/04, 46J/09, 55E/07–08, 46N/15, and 46N/16 reveals a complex but dominantly basaltic to basalt–sedimentary mixed geological environment. This zone covers the central districts of Sagar, Dhar, Barwani, Jhabua, Bhopal, Raisen, Sehore, Indore, and Khargone, all of which lie within the Deccan Volcanic Province but additionally include intertrappean limestone–sandstone formations, Vindhyan sediments, and alluvial patches. The chemical profiles from these districts illustrate strong signatures of mafic lithology marked by high Fe, Mg, Ca, Ti, Cr, Ni, and Co while elevated SiO_2 and alkali metals in some toposheets reflect the contribution of sandstone, siltstone, and calcareous intertrappean strata. This combination produces a geochemically hybrid middle basin where basaltic characteristics dominate but are periodically overprinted by sedimentary and alluvial geochemical inputs.

i) Major Oxides

The major oxide chemistry of the Middle Narmada Basin rocks reveals a broad spectrum arising from mixed volcanic and sedimentary parent materials. Silicon dioxide (SiO_2) ranges from 31–75%, with the lower values in Dhar–Barwani indicating fresher Deccan Trap basalts, while higher silica concentrations in Sagar, Vidisha, and Bhopal reflect quartz-rich Vindhyan sandstones and weathered claystones. Aluminium oxide (Al_2O_3) commonly falls between 7–20%, representing extensive feldspar weathering into clay minerals such as kaolinite, illite, and smectite, particularly in eastern middle basin districts. Iron oxide (Fe_2O_3) is characteristically high (9–24%), most pronounced in Jhabua and Dhar–Barwani where lateritic enrichment and basaltic minerals like magnetite–ilmenite dominate. Magnesium oxide (MgO) varies between

1.4–4.9% and Calcium oxide (CaO) between 1.3–16%, reflecting the presence of mafic minerals (augite, olivine) and calcic plagioclase in basaltic flows, as well as carbonate-rich intertrappean layers. Manganese oxide (MnO) stays low (0.07–0.3%), consistent with typical basalt-derived soils. Sodium oxide (Na_2O) ranges from 0.18–1.87% and Potassium oxide (K_2O) from 0.42–3.17%, marking zones of varying feldspar decomposition, with higher K_2O in shale–sandstone terrains and lower values in unaltered basaltic areas. Titanium oxide (TiO_2) shows consistent enrichment (1–5%), a diagnostic feature of Deccan basalts due to abundant titanomagnetite and ilmenite. Phosphorus pentoxide (P_2O_5) remains uniformly low (0.04–0.32%), indicating naturally low apatite content typical of both basaltic and sedimentary terrains of the basin.

ii) Trace & Heavy Metals

Trace metal distributions in the Middle Narmada Basin strongly reflect the mafic–intermediate volcanic foundation of the region. Chromium (Cr) levels range from 80 to over 1100 ppm, with Dhar, Barwani, and Jhabua recording the highest concentrations characteristic of ultramafic–mafic source rocks. Nickel (Ni) displays a similar trend, spanning 22–172 ppm, while cobalt (Co) ranges from 9–80 ppm, all confirming the dominance of basaltic material, particularly in the southwestern districts. Moderate concentrations of copper (Cu) (22–61 ppm) and zinc (Zn) (70–390 ppm) reflect natural mafic sources with localized sedimentary enrichment. Barium (Ba) values vary widely (75–1232 ppm), especially high in Sagar–Vidisha–Bhopal where felsic sandstone and shale components contribute. Strontium (Sr) spans 60–420 ppm, highlighting plagioclase weathering, while rubidium (Rb) (6–66 ppm) is elevated in sedimentary terrains, signifying higher K-bearing mineral content. High Field Strength Elements including zirconium (Zr) (160–700+ ppm), niobium (Nb) (30–160 ppm), hafnium (Hf) (0.8–3.6 ppm), and yttrium (Y) (6–70 ppm) appear significantly enriched due to the presence of resistant accessory minerals such as zircon, titanite, rutile, and monazite. Additionally, trace occurrences of lead (Pb) (up to ~97 ppm), lithium (Li) (~65 ppm), arsenic (As) (~10 ppm), tungsten (W) (up to 377 ppm), molybdenum, silver, cadmium, and bismuth appear in geogenic background levels, with no indication of anthropogenic contamination in these bedrock-derived samples. The overall trace metal pattern firmly indicates a composite landscape shaped primarily by Deccan Trap volcanism modified by sedimentary intrusions.

iii) Rare Earth Elements

The rare earth element (REE) geochemistry of the Middle Narmada Basin reveals clear enrichment in light rare earth elements (LREEs) and moderate but consistent levels of heavy rare earth elements (HREEs) a pattern characteristic of Deccan Trap basalts and associated weathering crusts. Lanthanum (La) values typically range from 10–44 ppm, cerium (Ce) from 35–110 ppm, praseodymium (Pr) between 2–10 ppm, and neodymium (Nd) between 18–95 ppm, showing strong LREE dominance driven by the presence of monazite, allanite, and LREE-enriched basaltic minerals. Europium (Eu) concentrations remain low (0.5–1.7 ppm), reflecting mild to moderate negative Eu anomalies caused by plagioclase fractionation in parent lava flows. Among HREEs, elements such as samarium (Sm) (1.8–10 ppm), gadolinium (Gd) (3.9–10.6 ppm), dysprosium (Dy) (4–10 ppm), erbium (Er) (1.7–5 ppm), ytterbium (Yb) (1–4 ppm), and lutetium (Lu) (0.3–1.4 ppm) show moderately enriched but tightly clustered values that point to the persistence of zircon, titanite, and xenotime in both volcanic and sedimentary settings. Collectively, the REE distribution highlights the mixed provenance of Middle Basin rocks, where basaltic LREE enrichment intersects with sedimentary contributions from Vindhyan and Gondwana formations, producing a balanced but distinctly mafic-biased REE signature.

7. Recommendations

- Higher values of some specific water parameters highlight zones of concern, and attention is required because this affects water quality and overall health.
- Soil and lithological profile are also very important parameters to explore, as they affect the overall river ecosystem.
- There is a need for further ground-truthing and continued monitoring of water and soil quality in the Narmada River Basin. Mapping of all point and non-point sources affecting river health is required to be done.
- Continuous monitoring and assessment would lead to decision-making for effective management plans and strategies to control overall river health.

8. References

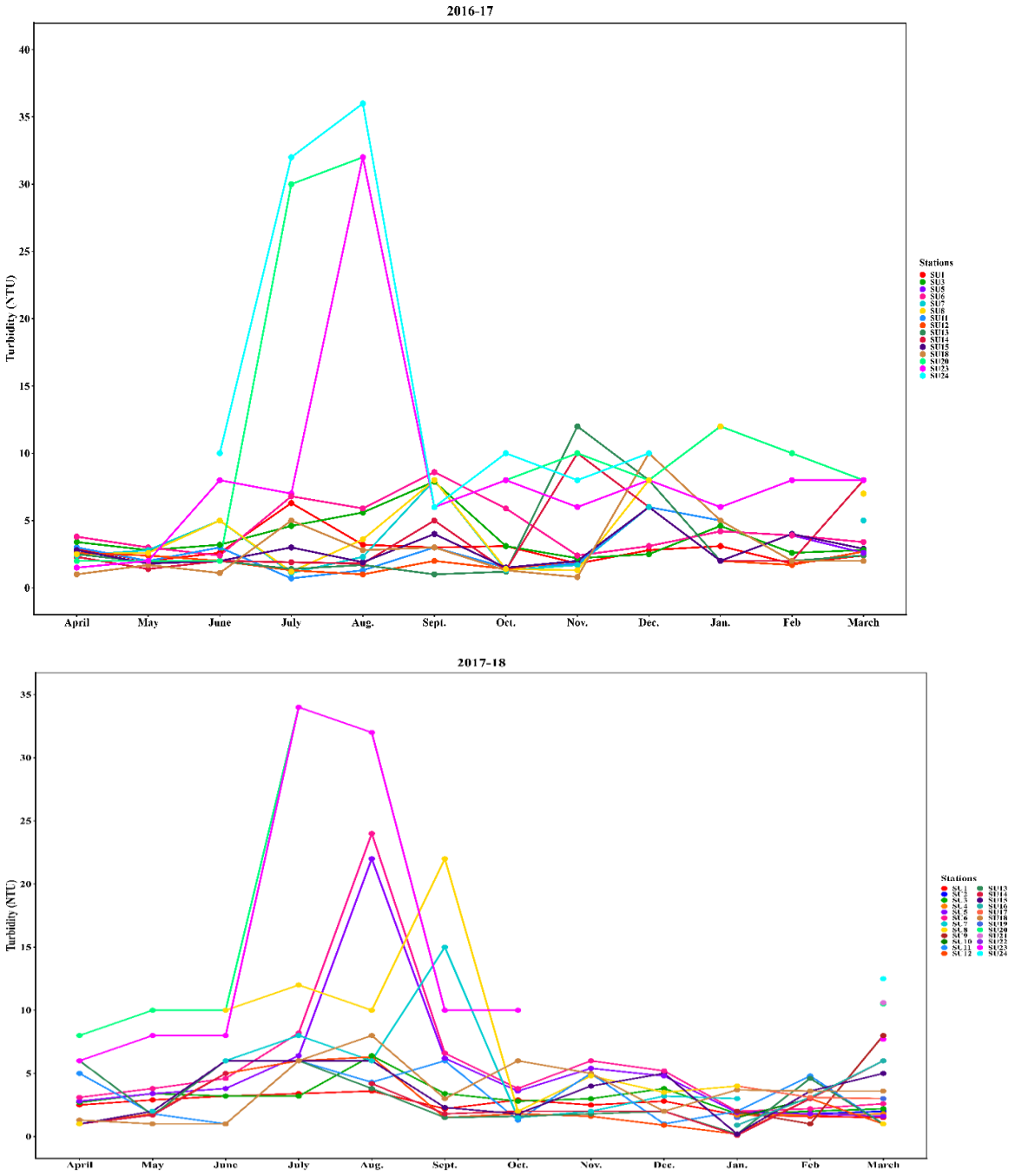
1) Umak, D. K., Rastogi, S. K., & Das, I. C. (2022). Spatial data analysis for groundwater

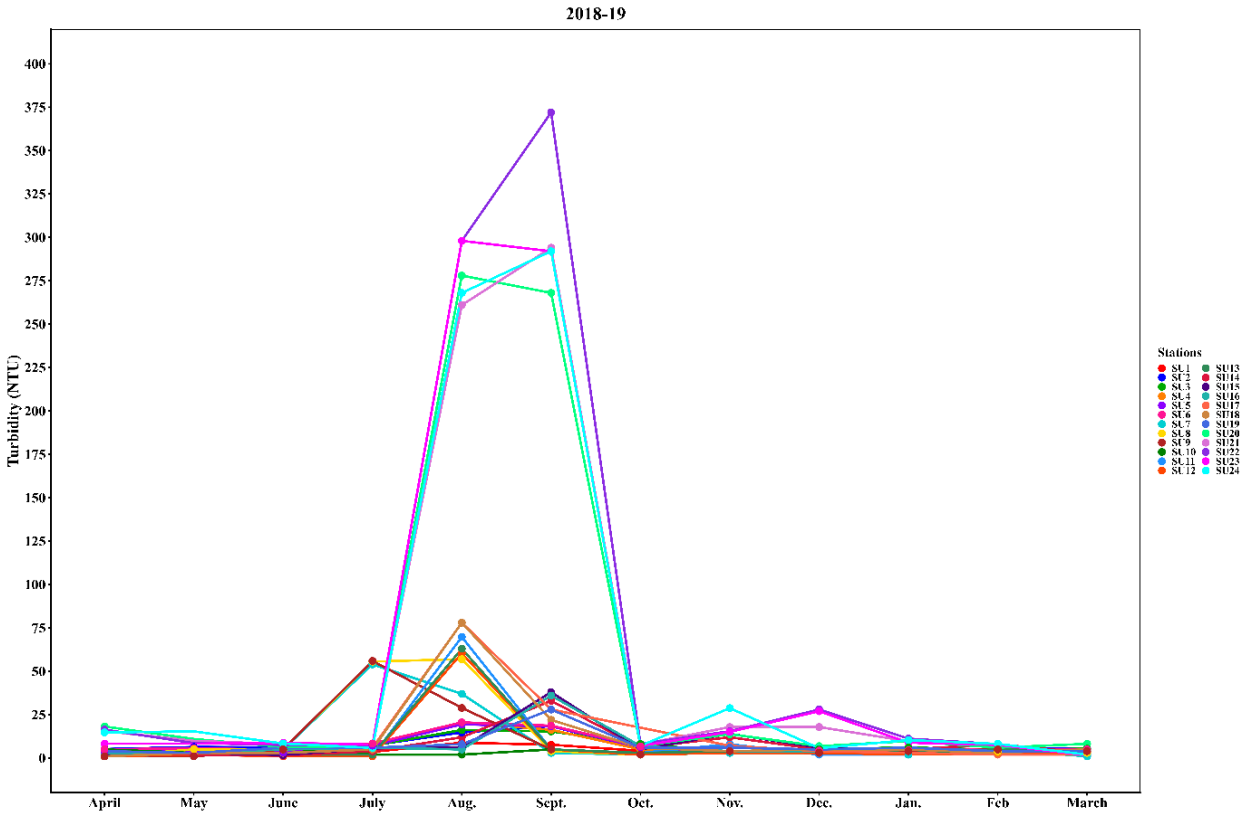
quality assessment with special reference to fluoride: A case study of Dhar District, Madhya Pradesh, India. International Journal of Chemistry, 14(1), 8. https://doi.org/10.5539/ijc.v14n1p8

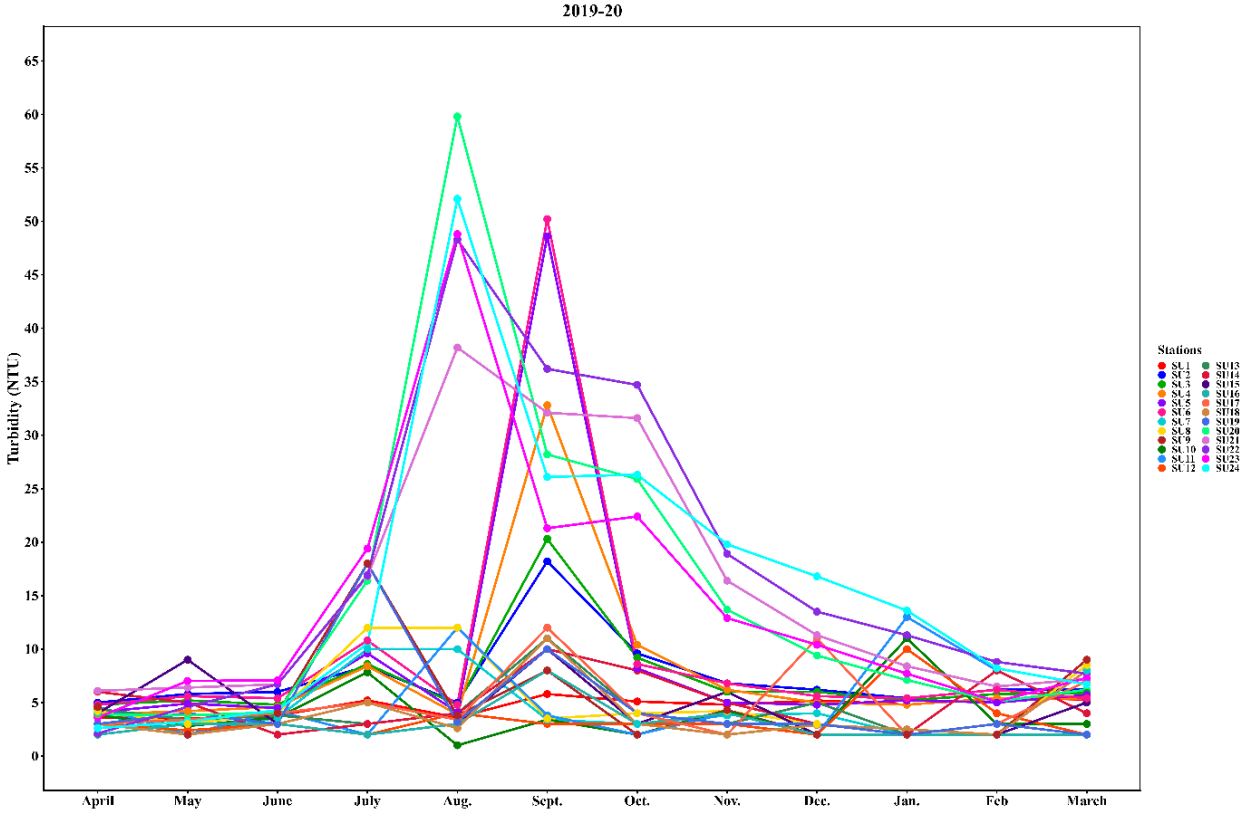
2) Mishra, S., Chauhan, M. S., & Sundaramurthy, S. (2024). *Assessing groundwater quality dynamics in Madhya Pradesh: Chemical contaminants and their temporal patterns*. **Environmental Research, 252**(Part2), 118887. https://doi.org/10.1016/j.envres.2024.118887
3) India-WRIS, NRSC/ISRO & Central Water Commission. (2014). *Narmada Basin Report*. National Remote Sensing Centre (NRSC), Indian Space Research Organisation (ISRO) & Central Water Commission (CWC), Government of India.
4) Soil and Land Use Survey of India (SLUSI). (1990-2025). *Detailed Soil Survey Reports*. Department of Agriculture & Farmers Welfare, Government of India. Retrieved from https://slusi.da.gov.in/dss/searchdss.html
5) https://www.mppcb.mp.gov.in/narmada-report16-17-17-18eng.aspx
6) https://sites.iitgn.ac.in/cnarmada/assets/files/Report.pdf
7) *https://cpcb.nic.in/water-quality-criteria/*
8) https://cdn.who.int/media/docs/default-source/wash-documents/wash-chemicals/potassium-background.pdf?sfvrsn=4542eda3_4
9) https://indiawris.gov.in/wris/
10) https://guidelines.nhmrc.gov.au/australian-drinking-water-guidelines/part-5/physical-chemical-characteristics/silica
11) The Geological Survey of India, reports retrieved from OCBIS Portal, website link https://www.gsi.gov.in/webcenter/portal/OCBIS

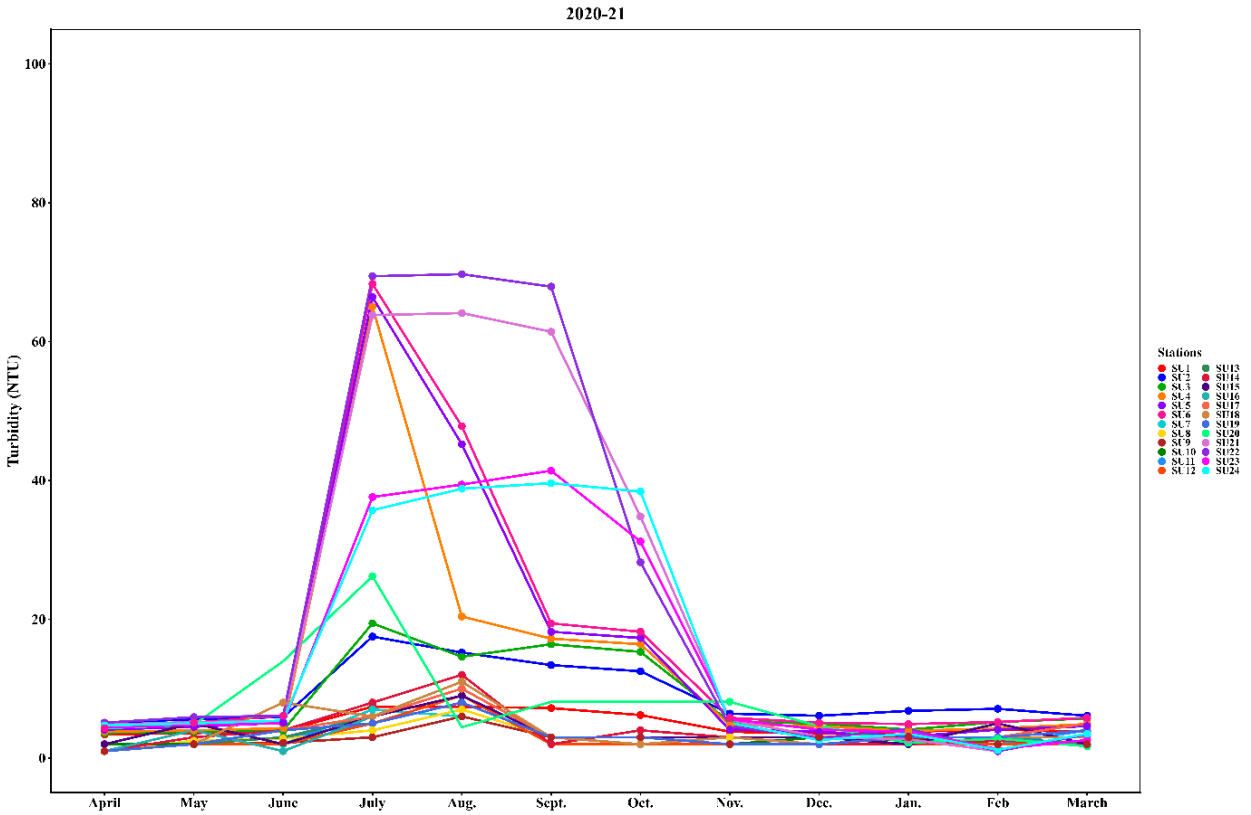
9. ANNEXURES

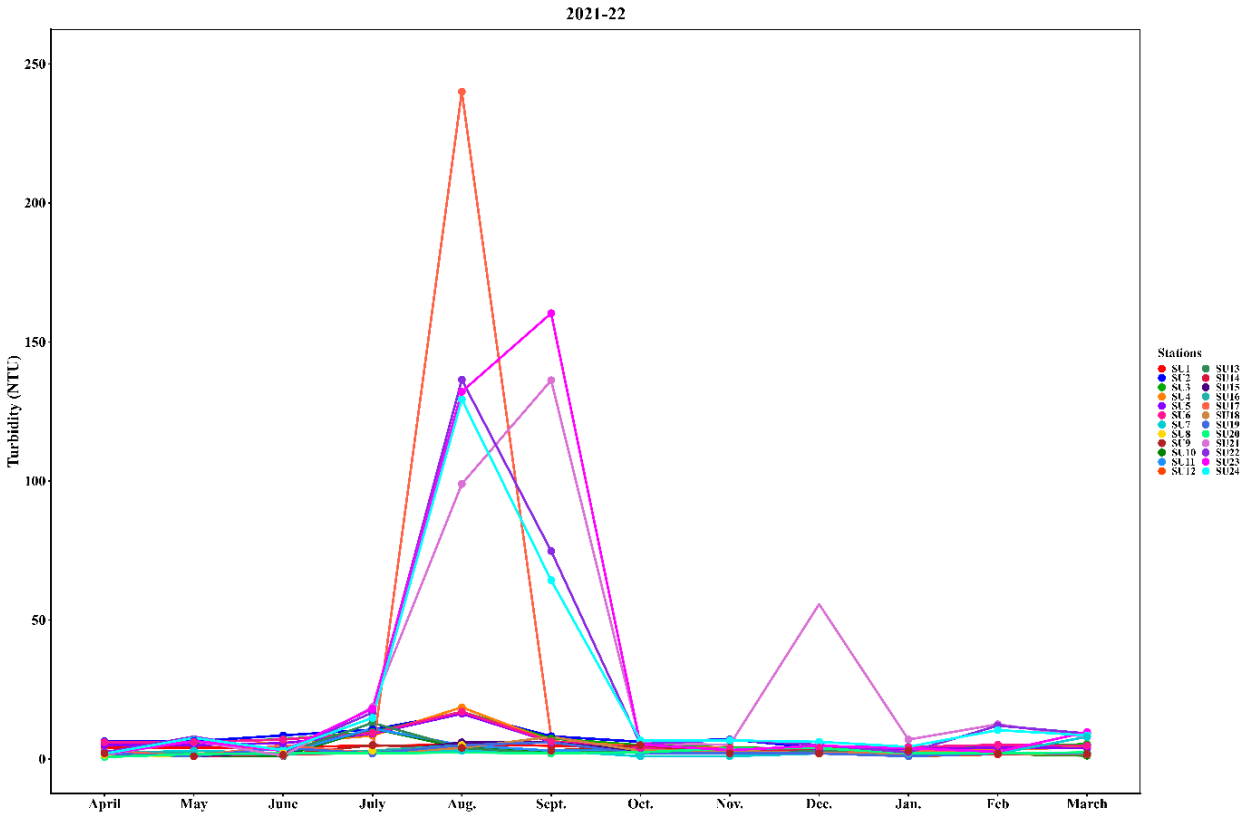
Annexure 3A: Monthly variation of turbidity for 9 years (2016-2025) at different sampling stations of the Upper Narmada Basin

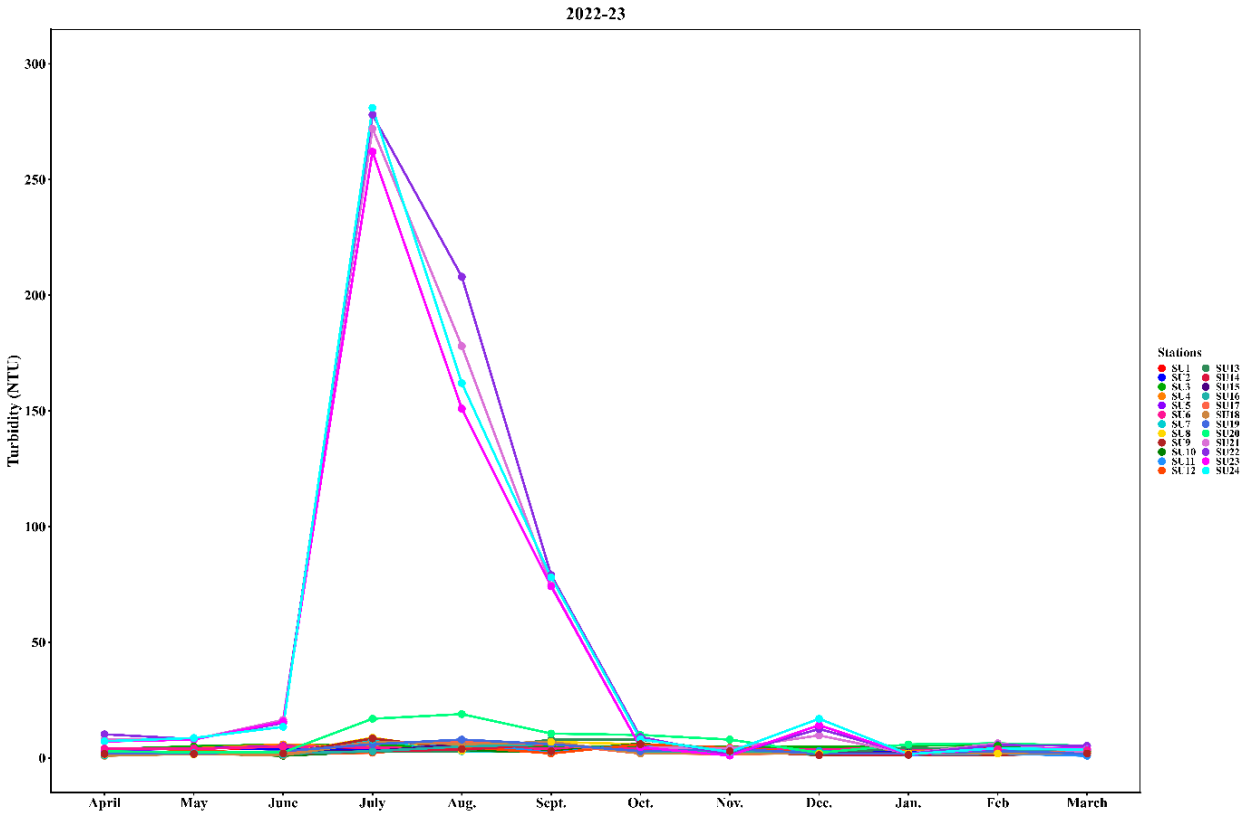


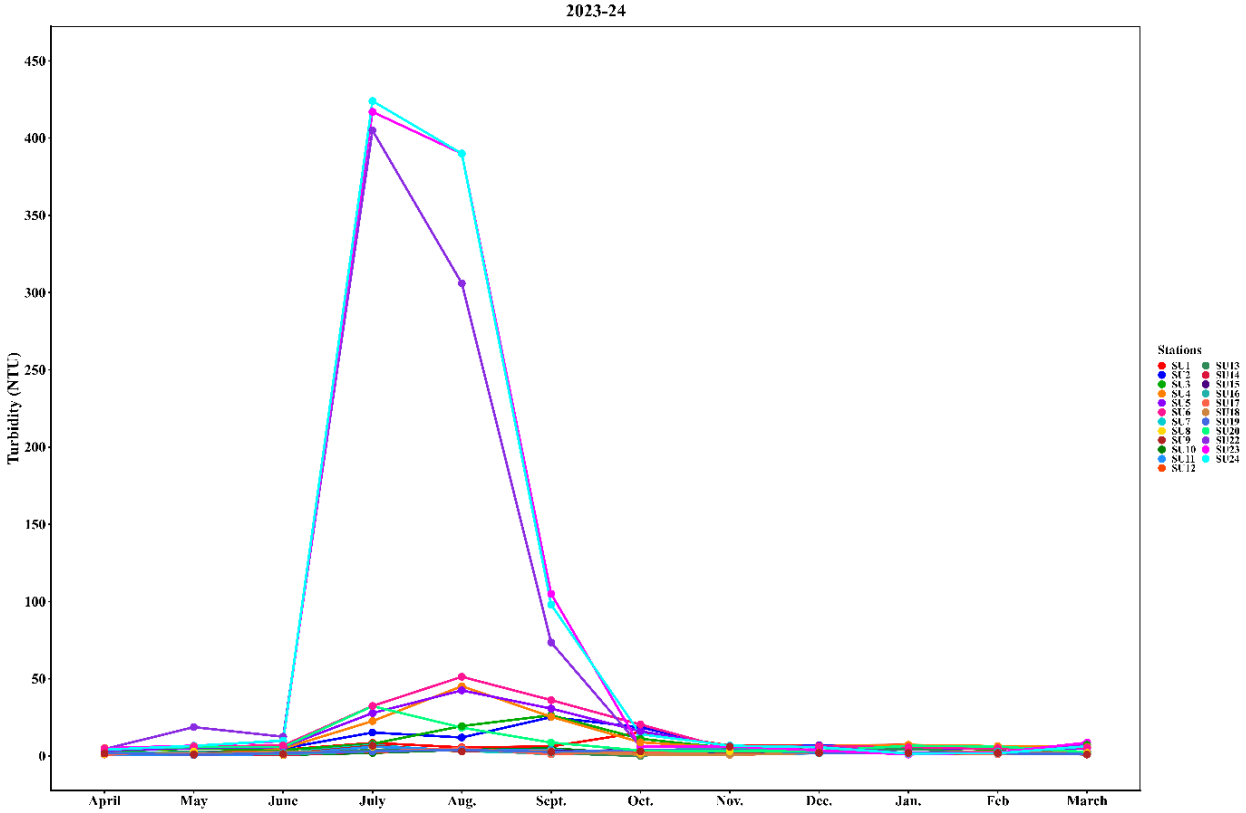

2018-19
Turbidity (NTU)
400
375
350
325
300
275
250
225
200
175
150
125
100
75
50
25
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU21
SU22
SU23
SU24

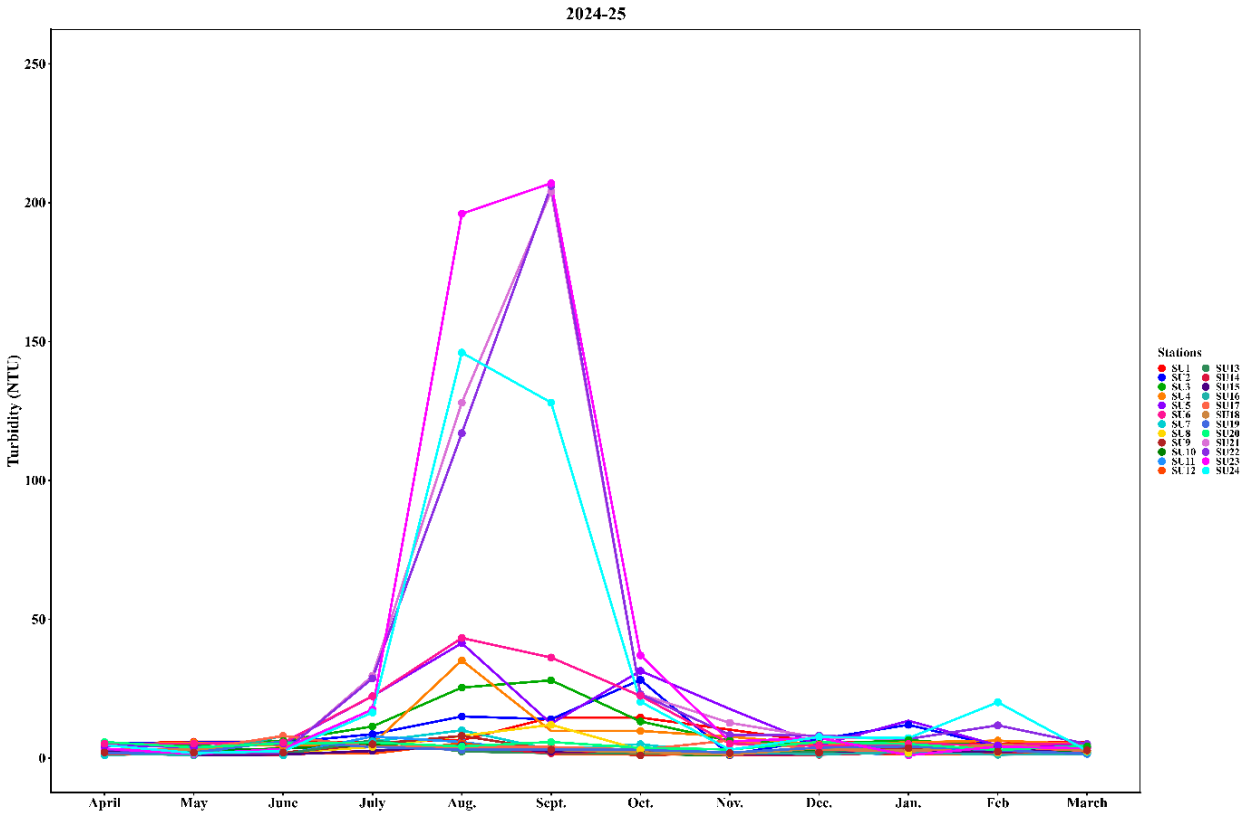

2019-20
Turbidity (NTU)
65
60
55
50
45
40
35
30
25
20
15
10
5
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU21
SU22
SU23
SU24


2020-21
Turbidity (NTU)
100
80
60
40
20
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU21
SU22
SU23
SU24

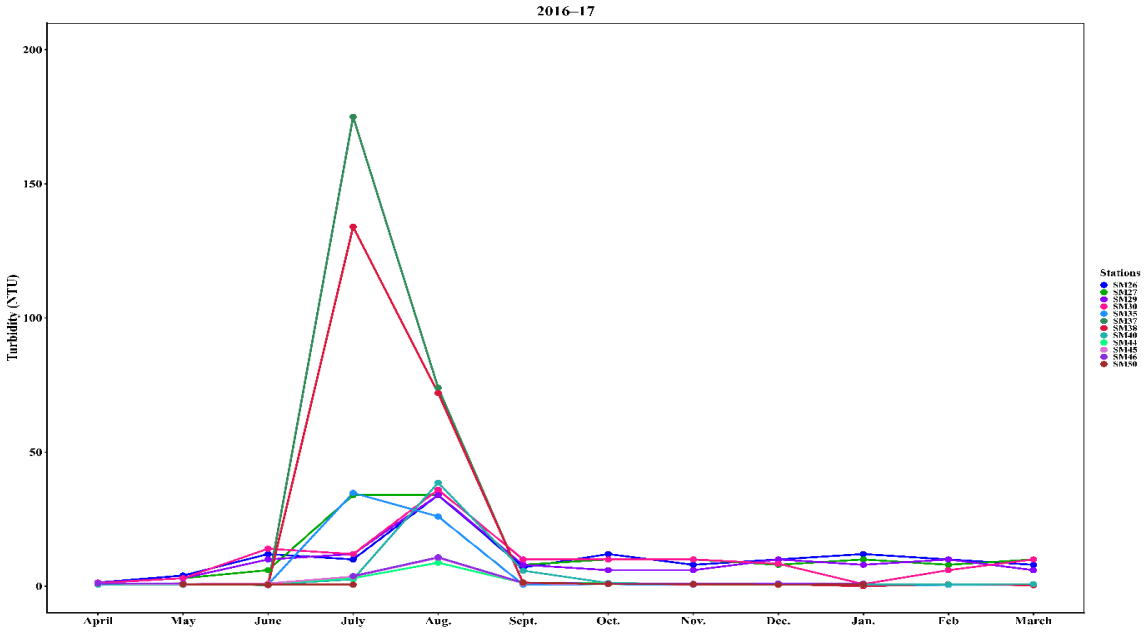

2021-22
Turbidity (NTU)
250
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU21
SU22
SU23
SU24

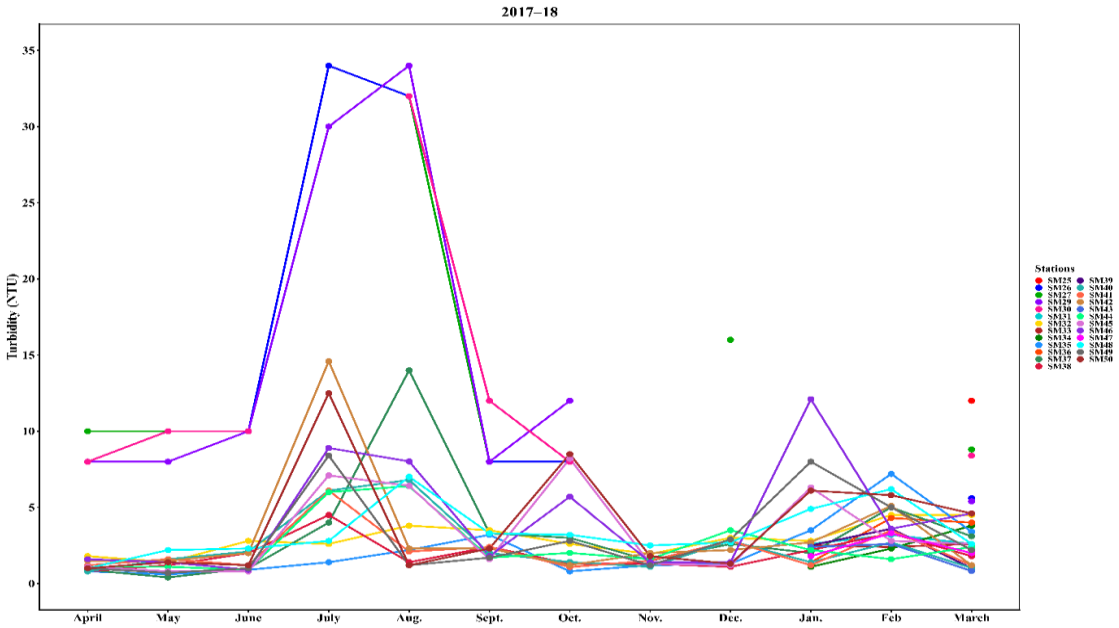

2022-23
Turbidity (NTU)
300
250
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU21
SU22
SU23
SU24

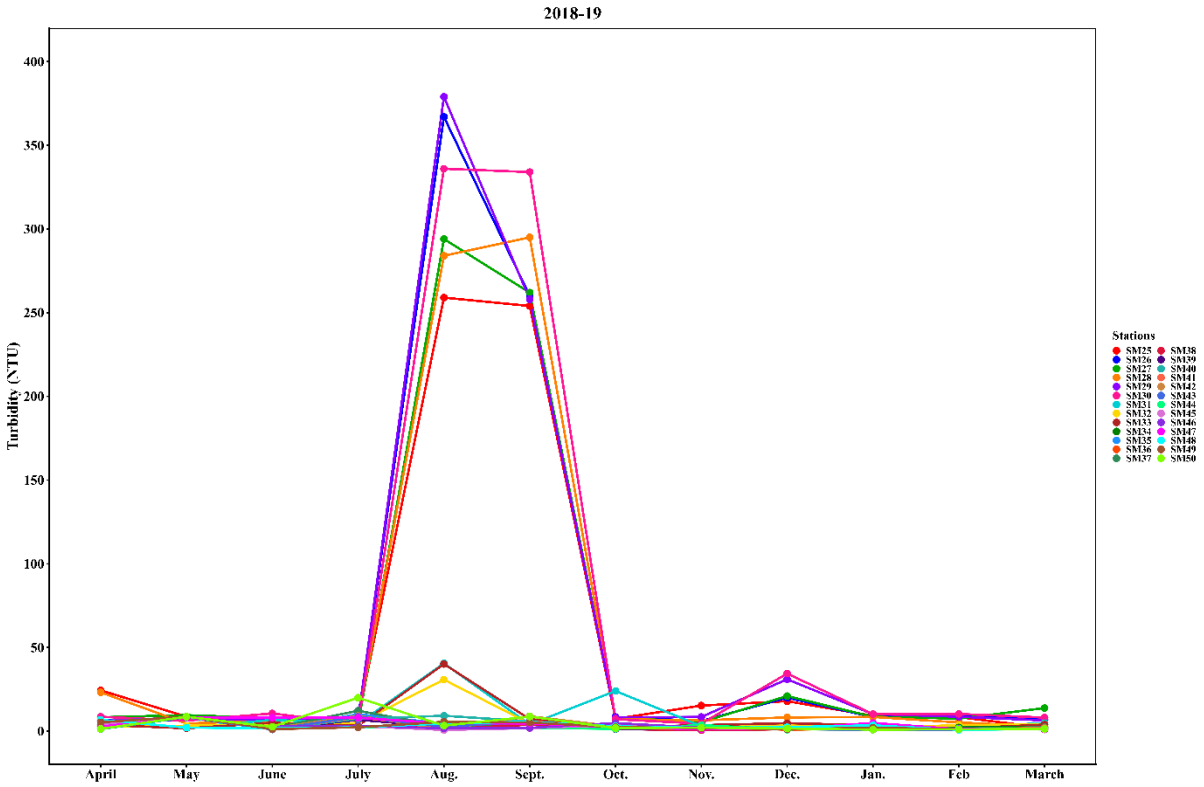

2023-24
Turbidity (NTU)
450
400
350
300
250
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU22
SU23
SU24

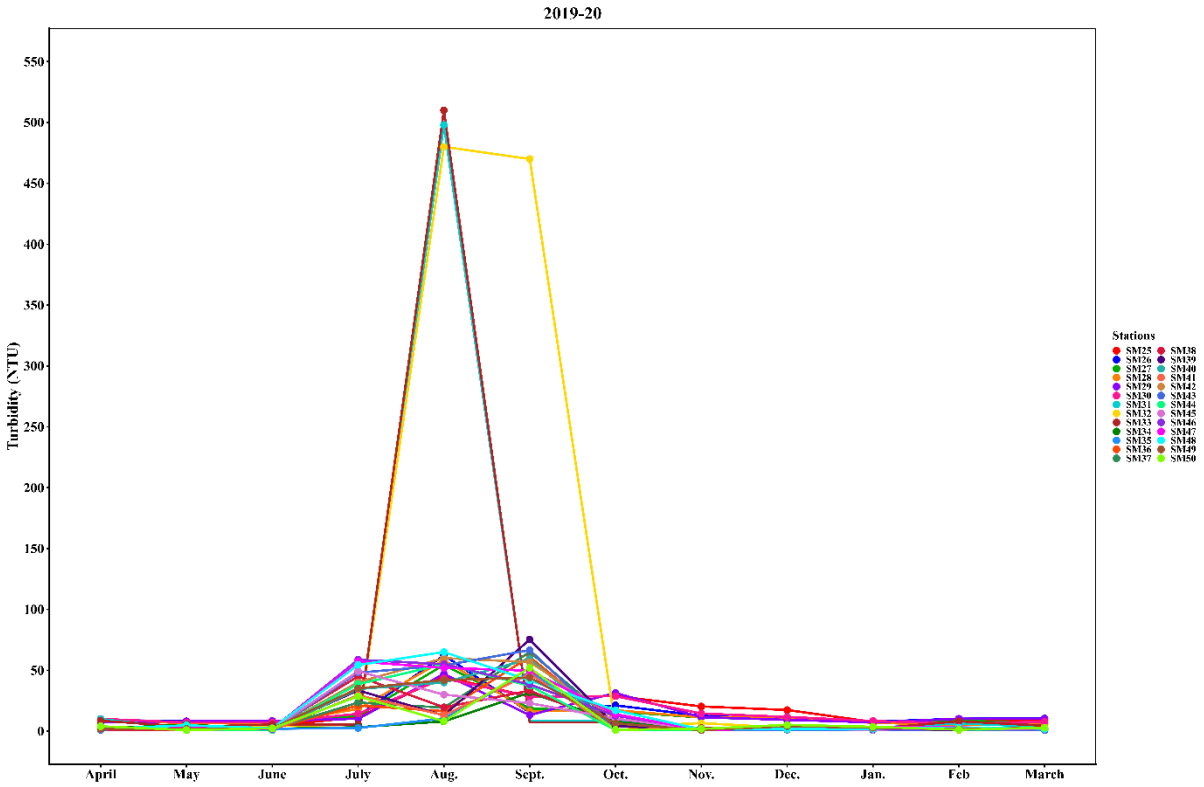

2024-25
Turbidity (NTU)
250
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SU1
SU2
SU3
SU4
SU5
SU6
SU7
SU8
SU9
SU10
SU11
SU12
SU13
SU14
SU15
SU16
SU17
SU18
SU19
SU20
SU21
SU22
SU23
SU24

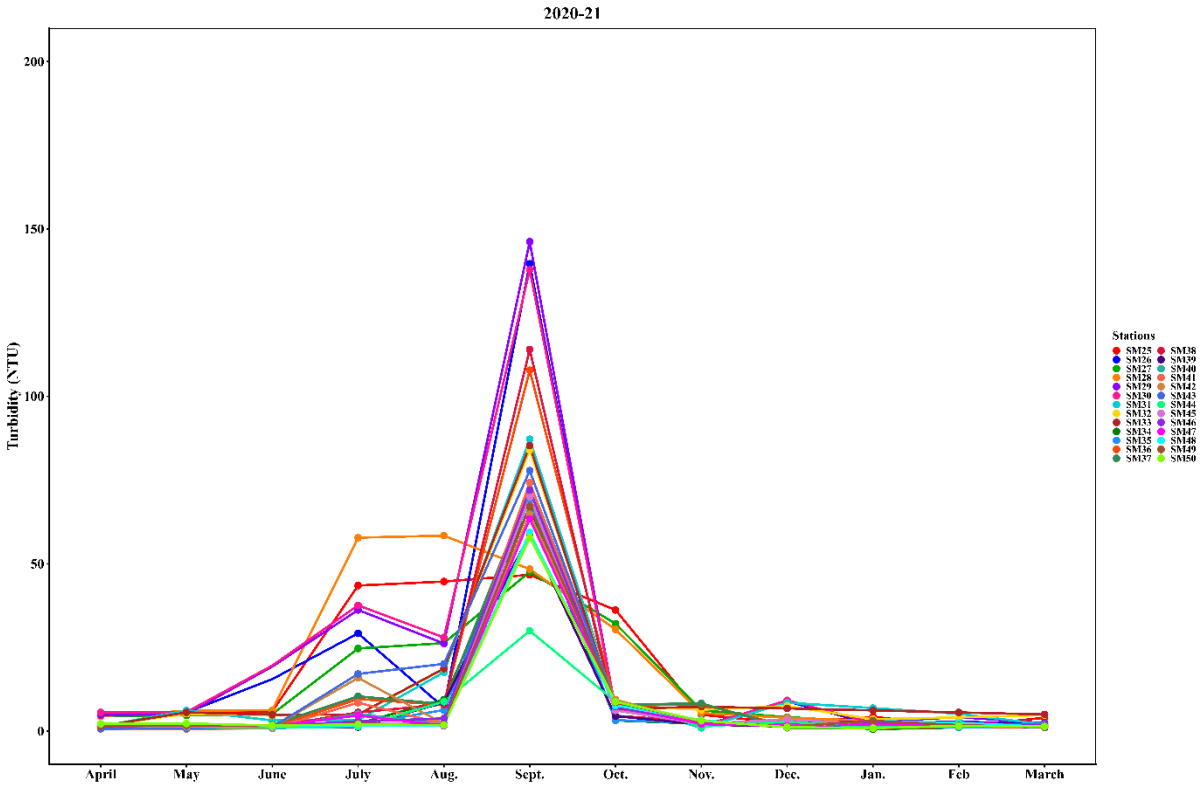
Annexure 3 B: Monthly data variation of turbidity for 9 years (2016-2025) at different sampling stations of the Middle Narmada Basin

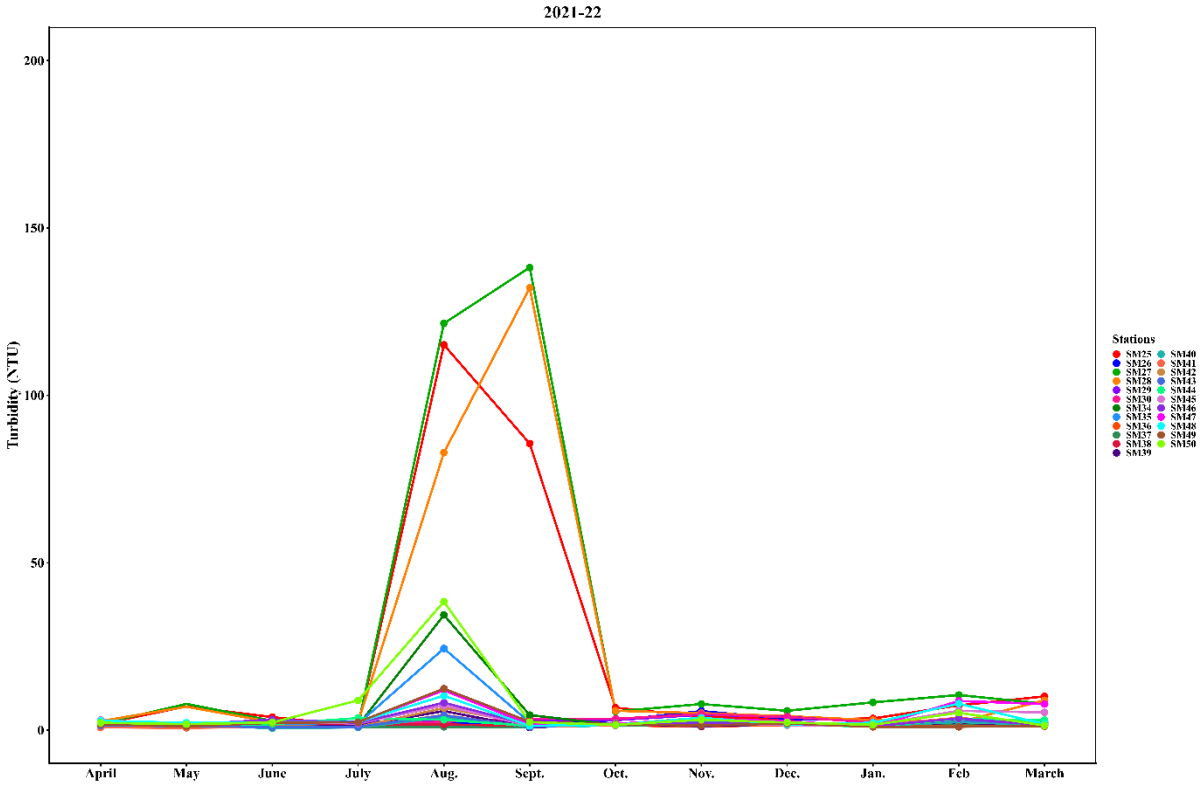


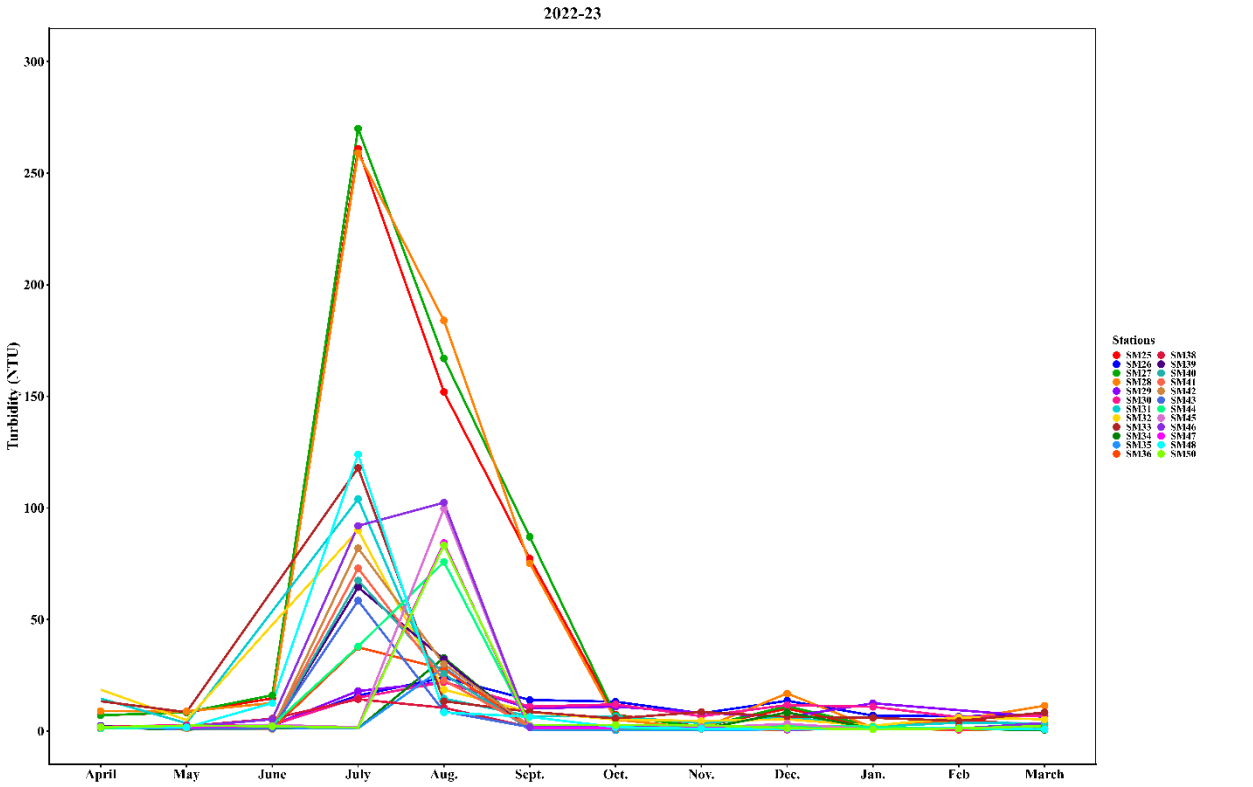


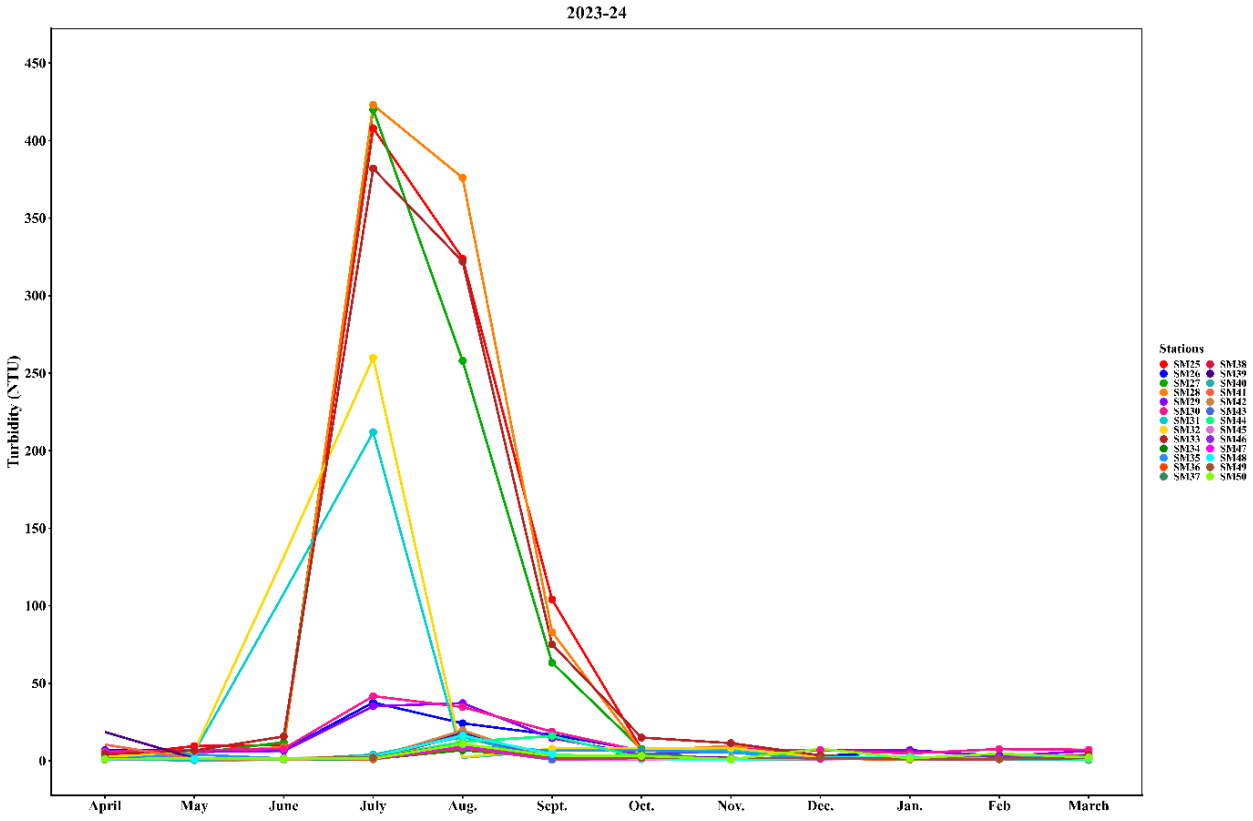

2018-19
Turbidity (NTU)
0
50
100
150
200
250
300
350
400
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SM25 SM38
SM26 SM39
SM27 SM40
SM28 SM41
SM29 SM42
SM30 SM43
SM31 SM44
SM32 SM45
SM33 SM46
SM34 SM47
SM35 SM48
SM36 SM49
SM37 SM50

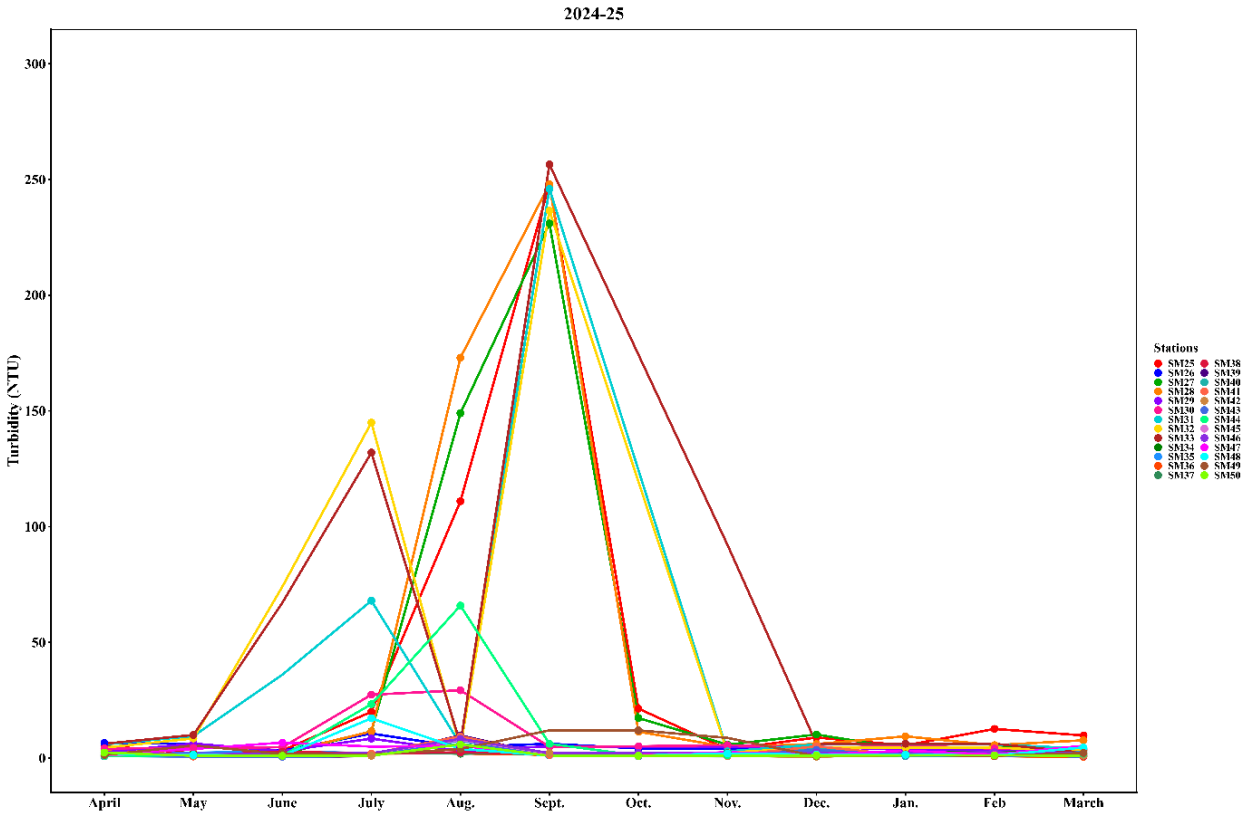

2019-20
Turbidity (NTU)
0
50
100
150
200
250
300
350
400
450
500
550
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SM25 SM38
SM26 SM39
SM27 SM40
SM28 SM41
SM29 SM42
SM30 SM43
SM31 SM44
SM32 SM45
SM33 SM46
SM34 SM47
SM35 SM48
SM36 SM49
SM37 SM50


2020-21
Turbidity (NTU)
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SM25 SM38
SM26 SM39
SM27 SM40
SM28 SM41
SM29 SM42
SM30 SM43
SM31 SM44
SM32 SM45
SM33 SM46
SM34 SM47
SM35 SM48
SM36 SM49
SM37 SM50


2021-22
Turbidity (NTU)
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SM25 SM40
SM26 SM41
SM27 SM42
SM28 SM43
SM29 SM44
SM30 SM45
SM34 SM46
SM35 SM47
SM36 SM48
SM37 SM49
SM38 SM50
SM39


2022-23
Turbidity (NTU)
300
250
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SM25 SM38
SM26 SM39
SM27 SM40
SM28 SM41
SM29 SM42
SM30 SM43
SM31 SM44
SM32 SM45
SM33 SM46
SM34 SM47
SM35 SM48
SM36 SM50


2023-24
Turbidity (NTU)
450
400
350
300
250
200
150
100
50
0
April
May
June
July
Aug.
Sept.
Oct.
Nov.
Dec.
Jan.
Feb
March
Stations
SM25 SM38
SM26 SM39
SM27 SM40
SM28 SM41
SM29 SM42
SM30 SM43
SM31 SM44
SM32 SM45
SM33 SM46
SM34 SM47
SM35 SM48
SM36 SM49
SM37 SM50



Annexure 5A: Detailed soil Survey report, Survey areas in Narmada River Basin (Source - https://slusi.da.gov.in/dss/searchdss.html)

S No	**Dataset Number**	**Surveyed Area**	**Total area reported & mapped** (hectares)
1	**Dataset 1**	Subwatersheds Ndlf, Nd1g, Nd1h, Nd1j, Nd1k, Nd1m, Nd1n, Nd1p, Nd1t and Nd1u of Narmada Catchment; Tahsil & District: Barwani, Madhya Pradesh.	25,245 ha
2	**Dataset 2**	Surveyed area - Nm2b, Nm2c, Nm3b, Nm3f, Nm3j, Nm5f and Nm5g. Narmada catchment Madhya Pradesh.	12325 ha
3	**Dataset 3**	Survey area: Nm4a, Nm4f, Nm4g, Nm4q and Nm4t Subwatersheds of Narmada Catchment, Madhya Pradesh	10,786 ha
4	**Dataset 4**	"5D2D1f1-5, D2a1-8, D2b1-5, D2c1-2, D3a1-6, D3b1-2, D3c1-2, D3d1-2, D3f1-6 and D3h1-2" micro watersheds of Narmada Catchment (RVP), Taluka. - Khategaon, Dist.- Dewas and Taluka-Nasarulaganj, Dist.- Sehore, Madhya Pradesh State	33918 ha
5	**Dataset 5**	"5D2D3g1 to g5, D4a1 to a4, D4b1 to b4, D4c1 to c3, D4h1 to h9, D4j1 to j8, D4k1 to k3, D4m1, D5j1 to j2 and D6h1 to h2" micro watersheds of Narmada Catchment (5D2),RVP, Taluka-Khategaon, Dist.-	35,005 ha

		Dewas, and Taluka-Nasarulaganj, Dist.- Sehore, Madhya Pradesh State	
6	**Dataset 6**	Surveyed area : Subwatersheds - Nc7w, Nd8h, Nd8j, Nf4v and Ng3h, Narmada catchment, Tahsil Manawar and Dhar of Dhar District and, Maheshwar, Thikari of Khargone District, Madhya Pradesh.	7940 ha
7	**Dataset 7**	Subwatersheds - Nn8n, Nn8p Nn8q, Nn8r and Nn8t Narmada Catchment Tahsil - Harsud , Dist - Khandwa (Madhya Pradesh)	14738 ha
8	**Dataset 8**	Survey area - Nk3v, Nk3x, Nk3y, Nk4d, Nk4m and Nk4n Subwatersheds of Narmada R.V.P. Catchment, Madhya Pradesh.	15,663 ha
9	**Dataset 9**	Surveyed area: Subwatersheds - Nn8a, Nn8b, Nn8c, Nn8d, Narmada catchment, Tahsil - Khirkıya of Hoshangabad district and Harsud tahsil of Khandwa district, Madhya Pradesh	8555 ha
10	**Dataset 10**	Subwatersheds: Nk1h, Nk1j, Nn5k, Nn5m & Nn5p- Narmada Catchment, Madhya Pradesh	12697 hectares
11	**Dataset 11**	Surveyed Area: Nh5a, Nh5b, Nh5c, Nh5g, Nh6c, Nh6k and Nh6m Subwatersheds, Narmada Catchment (Madhya Pradesh)	9231 ha
12	**Dataset 12**	Subwatersheds – Nk6a, Nk6∂, Nk6c, Nk6d, Nk9a, Nk9d, Nk9i, Nk9m, Nk9r Narmada catchment. Tahsil–Pandhana and Khandwa of Khandwa District and Jhirnya Tahsil of Khargone District, Madhya Pradesh	17,449 ha
13	**Dataset 13**	Survey area: - Nh7b, Nh7d, Nh7h, Nh8a, Nh8b, Nh8c, Nh8h, and Nh8k Sub watersheds of Narmada Catchment, Madhya Pradesh.	10,695 ha
14	**Dataset 14**	Surveyed area: Subwatersheds - Nn9g, Nn9f, Nn9g, Nn9h, Nn9i, Nn9m and Nn9n Narmada catchment, Madhya Pradesh	15056 ha
15	**Dataset 15**	Subwatersheds Nk1a, Nk1n, Nk1p, Nk2a and Nk2b, Narmada Catchment, Madhya Pradesh	12,879 ha
16	**Dataset 16**	Surveyed area: Nk3b, Nk3c, Nk3f, Nk3k, Nk3m and Nk3q Subwatersheds, Narmada Catchment (Madhya Pradesh)	18263 ha

17	**Dataset 17**	Subwatersheds - Nk5p, Nk5s, Nk5t, Nk5u and Nk5w. Narmada catchment Tahsil Harsud and Khandwa of Khandwa district, Madhya Pradesh	983 ha
18	**Dataset 18**	Subwatersheds - Nn5n, Nn8f, Nn8g, Nn8j, Nn8m and Nn9q; Narmada catchment, Tahsil - Khirkhiya of Hoshangabad district and Harsud tahsil of Khandwa district, Madhya Pradesh	13469 ha
19	**Dataset 19**	Subwatersheds- Nn2b, Nn2c, Nn2f, Nn2g, Nn2h, Nn2j, Nn2k and Nn2r, Narmada Catchment, Madhya Pradesh.	17270 ha
20	**Dataset 20**	Subwatersheds–Nn5d .Nn5f, Nr5h & Nn5j Narmada Catchment, Madhya Pradesh	8,871 ha
21	**Dataset 21**	Survey area - Nn4a, Nn4h and Nn4j Subwatersheds of Narmada R.V.P. catchment M.P.	7591 ha
22	**Dataset 22**	Subwatersheds -Nn4k, Nn4m, Nn4q,Nn4r, Nn4s and Nn4t, Narmada catchment, Madhya Pradesh	17,717 ha
23	**Dataset 23**	Subwatershed - Nn6m, Nn7c, Nn7f, Nn7g, Nn7j and Nn7k Narmada catchment, Madhya Pradesh	9905 ha
24	**Dataset 24**	Surveyed area : Nn1c, Nn1d, Nn1f, Nn1g and Nn1j subwatersheds, Narmada Catchment, Madhya Pradesh	12232 ha
25	**Dataset 25**	Survey Area: “5D3D1a1, b1 to b9, c4 to c8, d1 to d9, f1 to f4, g1 to g4 and h1 to h3”, micro-watersheds, Narmada Catchments, Taluka- Handia, Harda and Timarni, District-Harda, Madhya Pradesh State	34,283 ha
26	**Dataset 26**	Surveyed area: Priority subwatersheds Nq6j, Nq6k, Nq6p and Nq6s; Narmada catchment	10,553 ha
27	**Dataset 27**	Surveyed Area: Nr1d, Nr1g, Nr1h, Nr1j, Nr1k, Nr1m & Nr1n Subwatersheds of Narmada RVP Catchment (MP)	12,155 ha
28	**Dataset 28**	Survey area: Detailed Soil Survey and Land Use Plan of “5D5C6m6, m7, n4, p2 to p7, r1 to r3, t1 to t5; 5D6A1a1 to a5, b2, b4, b7, c1, c2, d1 to d6, f1, f2, g1 to g3 & k1” micro-watersheds of Narmada Catchments, Taluka- Jabalpur and Shahpura dist.- Jabalpur & Taluka- Lakhnadou Dist.- Seoni, Madhya Pradesh using Remote Sensing	33631 ha
29	**Dataset 29**	Survey Area: 5D4B6f1-5, 5D4B7g1-3, j1-2, n1-2, r1-2, 5D4B8a1-3, c4-6, j1, m2 and z1-4 micro-watersheds, Narmada Catchment District-Jabalpur, Madhya Pradesh State	17,725 ha
30	**Dataset 30**	Survey Area: Detailed Soil Survey and Land Use Plan of “5D5C6m3,m5, n1-2, n3, n5; 5D5C7c2, c4, f1-f3, g1-2, h1-h4, j1-7, k1-3, m1-2, n1-n4, p1-p4, q1-q3, 5D6A1b1, b3, b5-6” micro-watersheds of Narmada Catchments, Taluka-Jabalpur and Shahpura dist.- Jabalpur, Madhya Pradesh using Remote Sensing	32, 814 ha

31	**Dataset 31**	Report on Detailed Soil and Land Use Survey of "5D4A8g1, j-1-2, 5D4B1a1-3, b1-3, d1-5, g1-4, k3-4, 5D4B2f1-2, 5D4B7a1-2, b1-2, c1-2, d1-2" Micro-watersheds of Narmada catchment, Taluka- Shahpur& Patan, Distt- Jabalpur & Taluka- Kardi, Dist- Narsinghpur, Madhya Pradesh State	23,947 ha
32	**Dataset 32**	SURVEY AREA: Subwatersheds Nv1d, Nv1g, Nv1m, Nv1n, Nv1p, Nv1q, Nv1r, Nv1s, Nv1t, & Nv1v, Narmada catchment, Tah. -Niwas, Dist. - Mandla (MP)	16,500 ha
33	**Dataset 33**	Subwatersheds Ny1b, Ny1c, Ny1f and Ny1h of Narmada Catchment. Tehsil and District: - Mandla, Madhya Pradesh	7,704 ha
34	**Dataset 34**	Subwatersheds Nv1x, Nv5b, Nv5d, Nv5f, Nv5g, Nv5h, Nv5j, Nv5k, & Nv5m; Narmada Catchment, Tahsil-Niwas, District-Mandla, Madhya Pradesh	18,062 ha
35	**Dataset 35**	Subwatersheds NV6k, NV6p, NV6r, & NV6u of Narmada Catchment, Tehsil and District: Mandla, Madhya Pradesh State.	8,341 ha
36	**Dataset 36**	Subwatersheds Nv4g, h, j, k, m, n and p of Narmada Catchment, Tah. and Distt- Mandla, Madhya Pradesh	12,270 ha
37	**Dataset 37**	Report on Detailed Soil Survey and Land Use ofNv5n, Nv5q, Nv5r, Nv5s, Nv5t, Nv5u, and Nv5y Sub watersheds of Narmada RVP Catchments, Tahsil: Niwas, District: Mandla, Madhya Pradesh	13,238 ha
38	**Dataset 38**	Sub watersheds Nv6a, Nv6b, Nv6c, Nv6d, Nv6f, Nv6g & Nv6j of Narmada Catchment, Tehsil and District: Mandla, Madhya Pradesh	12,687 ha.
39	**Dataset 39**	"5D4A4p2, p3 , q1,q2 & q3, 5D4A6a3,b2, b4, c1, c2, c4, f3,f4, h3, m3, t1, t3, u2, v1,v2, v3 & w4, 5D4A7a1, b1, b2, c1, c2, f1 & f3" micro watersheds of Narmada Catchment (RVP), Taluka- Bareli & Udaipura, Dist. – Raisen and Taluka- Tendukhera, Dist. – Narsimhapur, Madhya Pradesh	36,993 ha
40	**Dataset 40**	5D4A2v2 to v4, 5D4A4a1 to a7, b1 to b4, c1 to c4, d1 to d3, f1 to f8 and h1 to h9" micro watersheds of Narmada Catchment (RVP), Taluka- Bareli, Silwani and Udaipura, Dist. -Raisen, Madhya Pradesh	38,787 ha
41	**Dataset 41**	"5D4A1j1 to j5, p1 to p3, r1 to r5, s1 to s3, t1 to t4, u1 to u3, x1 to x3, 5D4A2a1 to a4, b1 to b3, c1 to c3, d1 to d6, q1 to q3 & v1" micro watersheds of Narmada Catchment (RVP), Taluka- Budhani, , Dist. – Sehore and Talukas- Bareli & Bari, , Dist. - Raisen, Madhya Pradesh State	38,046 ha

42	**Dataset 42**	“5D2D5a1 to a6, b1 to b5, c1 to c4, d1 to d5, f1 to f3, g1 to g2, m1 to m3, 5D2D7a1 to a2, b1 to b5 and c1 to c4 ”micro-watersheds, Narmada Catchments taluka- Nasrullaganj, and Rehti, District-Sehore, Madhya Pradesh	36,597 ha
43	**Dataset 43**	“5D6A1g4,h1 to h3, j1 to j3, k2 to k4, m1 ,m2, n1 to n4, p1 to p3, q1 to q3, r1 to r3, s1 to s3, t1 to t7, y1,\ y3, y6 & y7 ” micro watersheds of Narmada Catchment (RVP), Taluka- Lakhandaun &Ghansaur, Dist. – Seoni, Taluka+ Dist.- Jabalpur, Madhya Pradesh	32,665 ha
44	**Dataset 44**	Nf2j, Nf2k subwatersheds, Narmada catchment	6414 ha
45	**Dataset 45**	Subwatershds Nf3a, Nf3b, Nf3g and Nf3r, Narmada catchment, Tahsil: Bhikangaon, Kasarawad, District: Khargone, Madhya Pradesh	10830 ha
46	**Dataset 46**	Subwatershds Nf4q and Nf6f. Narmada catchment, Tahsil - Bhagwanpura and Khargone of Khargone District, Madhya Pradesh	6231 ha
47	**Dataset 47**	Subwatershds Ng3a, Ng3j, Ng3k, Ng3n and Ng5a Narmada Catchment, Tahsil – Maheshwar of Khargone – District, Madhya-Pradesh. 5867 ha	5867 ha